

SingleRAN

RF Unit and Topology Management Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

Change Description	Parameter Change
Added the pRRU5755GR (1800+2100+2300+2600+3500). For details, see 4.1.2.4 Basic RRU Information .	None
Added the AAU5368md (2300+3600). For details, see 4.1.2.3 Basic AAU Information .	None

Editorial Changes

Added incompatibility of BTS5929R with Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(eNodeB). For details, see [4.4.3 Hardware](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue includes the following changes.

Technical Changes

Change Description	Parameter Change
Added the RRU5522t (700+800+900) and pRRU5551JR (26000). For details, see 4.1.2.4 Basic RRU Information .	None
Added the AAU5785t (700+800+900) and AAU5685md (2300+2600). For details, see 4.1.2.3 Basic AAU Information .	None
Added support for MN (FDD) by the RRU5910 (800). For details, see 4.1.2.4 Basic RRU Information .	None

Editorial Changes

None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE, and NR. This feature works the same way with each of these RATs.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the RF Unit and Topology Management feature, including its technical principles and engineering guidelines. [Table 2-1](#) lists related features.

Table 2-1 Features related to RF Unit and Topology Management

RAT	Feature ID	Feature Name	Chapter/Section
GSM	MRFD-210002	Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(GBTS)	4 Topologies
UMTS	MRFD-220002	Multi-mode BS RRU/RFU star-connection with separate CPRI interface(NodeB)	4 Topologies
LTE FDD	MRFD-230002	Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(eNodeB)	4 Topologies
NR	MRFD-151169	Multi-mode BS RRU/RFU star-connection with separate CPRI interface (NR)	4 Topologies

2.4 Differences

Differences Between NR FDD and NR TDD

Function Name	Difference
RF Unit and Topology Management	Depends on the differences between RF modules.

Differences Between NSA and SA

Function Name	Difference
RF Unit and Topology Management	None

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high

frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference
RF Unit and Topology Management	Depends on the differences between RF modules.

3 Overview

This document does not involve the CPRI trunk chain and description of the trunk chain topology with CPRI MUX. For details, see *RMU in 3900 & 5900 Series Base Station Product Documentation*.

This document applies to macro base stations (including BTS3900, BTS3900L, BTS3900A, BTS3900AL, BTS3900C, DBS3900, BTS5900, BTS5900L, DBS5900, DBS5900E, and DBS5900A) as well as DBS3900 LampSite and DBS5900 LampSite (referred to as LampSite base stations in this document).

Any parameters, alarms, counters, or managed objects (MOs) described herein apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB.

Unless otherwise specified, in this document, NR and gNodeB always include FDD and TDD. In scenarios where they need to be distinguished, NR FDD and NR TDD are used. The same rules apply to gNodeB.

The "G", "C", "U", "L", "T", "M", "N", "N (TDD)", and "N (FDD)" in RAT acronyms refer to GSM, CDMA, UMTS, LTE FDD, LTE TDD, LTE NB-IoT, NR, NR TDD, and NR FDD, respectively.

Based on the main control board type, an eGBTS can be classified into the following types:

- eGBTS with the GTMUc functioning as the main control board: The main control board does not support multi-RAT concurrency and is referred to as eGBTS (GTMUc) in this document.
- eGBTS with the UMPT functioning as the main control board: The main control board supports multi-RAT concurrency and is referred to as eGBTS (UMPT) or co-MPT base station in this document.

3.1 Introduction to RF Modules

RF modules include remote radio units (RRUs), radio frequency units (RFUs), and radio units (RUs) of active antenna units (AAUs).

- RRU: applies to outdoor macro base stations and LampSite base stations. It is used to transmit and receive baseband signals, modulate and demodulate RF signals, process data, and amplify the power of signals.
- RFU: applies to indoor macro base stations. It is used to transmit and receive baseband signals, modulate and demodulate RF signals, process data, and amplify the power of signals.
- AAU: applies to distributed base stations and macro base stations. It integrates the functions of an RU and an antenna.

 NOTE

Only the pico remote radio unit (pRRU) applies to LampSite base stations.

3.2 Introduction to Topologies

RRU/RFU/AAU topologies are networking between the baseband unit (BBU) and RRU/RFU/AAUs over common public radio interface (CPRI) ports.

CPRI may refer to the fronthaul interface between the BBU and RF module, or the CPRI protocol. The fronthaul interface may run the CPRI or enhanced common public radio interface (eCPRI) protocol. However, eCPRI only represents the eCPRI protocol and it still uses the CPRI port. Therefore, CPRI in CPRI ports and links indicates the fronthaul interface, but both CPRI and eCPRI protocols are involved.

The transmission link between an RRU/RFU/AAU and a BBU is called a CPRI link, and is used to transmit control- and user-plane data.

The eCPRI protocol is used for the connection between a BBU and an AAU in a wireless base station. Based on this protocol, a link between the BBU and AAU is set up to transmit control-, user-, and synchronization-plane data. If eCPRI is used, certain processing units at the physical layer are deployed on the AAU instead of the BBU, and Ethernet is used instead of time division multiplexing (TDM) for data transmission.

Topologies include the star topology, chain topology, ring topology, load sharing topology, dual-star topology, branch chain (single-link) topology, and branch load sharing topology. For details about the support for CPRI and eCPRI, see [4 Topologies](#).

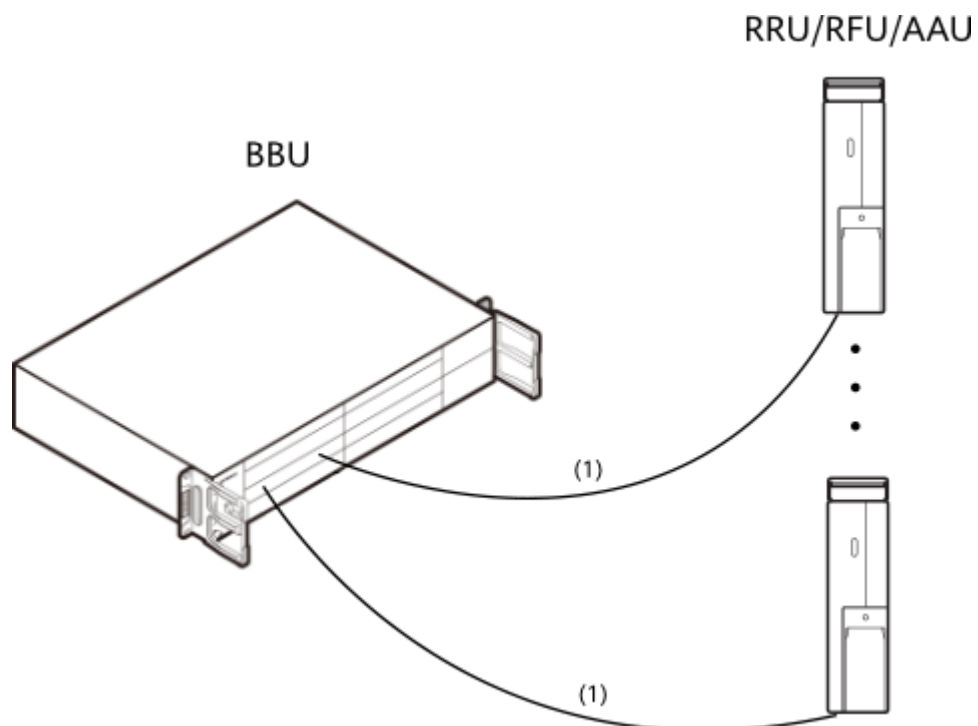
4 Topologies

4.1 Principles

4.1.1 Topologies

4.1.1.1 Overview

In a macro base station, the RRU/RFU/AAU is directly connected to the baseband processing unit (BBP) or baseband radio interface unit (BRI) in the BBU. The topology is shown in the following figure.

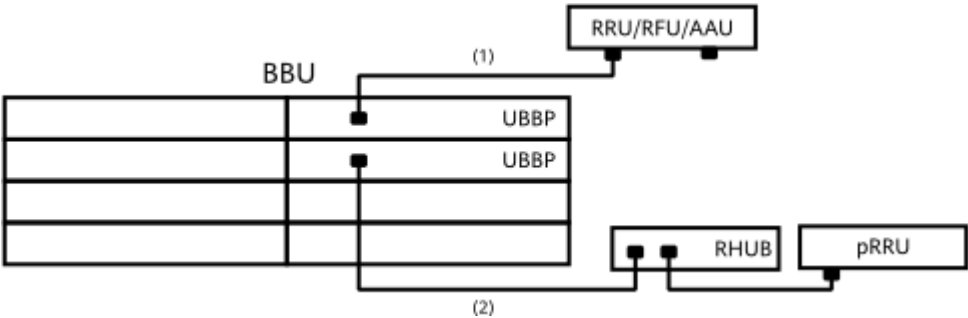


(1) Link between the BBU and RRU/RFU/AAU

 NOTE

- If intermediate equipment, such as the optical transport network (OTN) or radio transmission node (RTN), exists between the BBU and RF modules, see the topologies of the intermediate equipment in the related documents.
- Link switchover of intermediate equipment affects services on the links.

In a macro base station, both the RRU/RFU/AAU and pRRU are used. The topology is shown in the following figure.



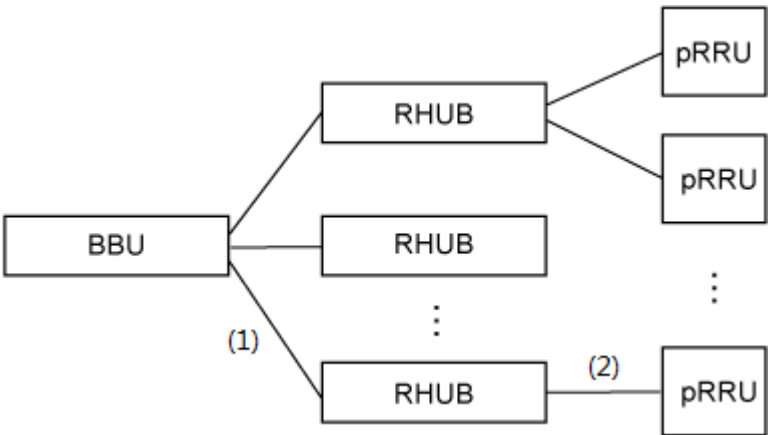
(1) Link between the BBU and RRU/RFU/AAU	(2) Link between the BBU and RRU HUB (RHUB)
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NOTICE

This scenario has the following restrictions:

- The link between the BBU and RRU/RFU/AAU and the link between the BBU and RHUB do not share a BBP.
- The RRU/RFU/AAU and pRRU do not share a CPRI link.

In a LampSite base station, the pRRU is connected to the RHUB. The RHUB is then connected to the BBP/BRI in the BBU. The topology is shown in the following figure.



(1) Link between the BBU and RHUB (main chain/ring)	(2) Link between the RHUB and pRRU (branch chain/ring)
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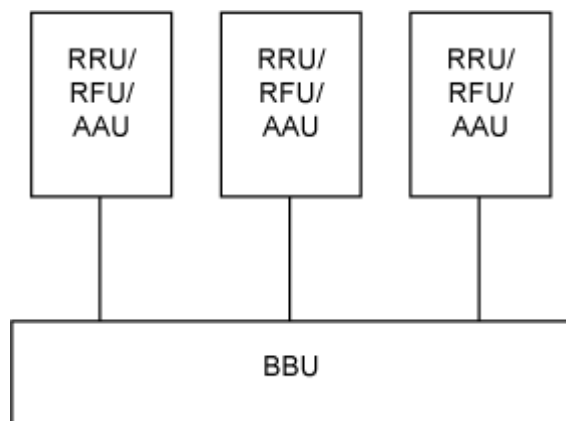
4.1.1.2 Star Topology

The star topology applies to most areas, especially densely populated areas, such as downtown. [Figure 4-1](#) shows the star topology.

NOTE

When the star topology is used, the interface principles of the eCPRI and CPRI protocols are the same.

Figure 4-1 Star topology



In the star topology, different RRUs/RFUs/AAUs are directly connected to different CPRI ports on the BBP(s)/BRI(s) in the BBU. The eCPRI protocol does not support the ring topology.

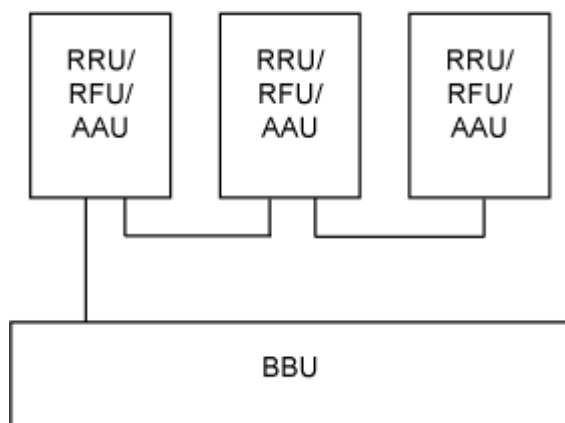
An operation and maintenance link (OML) is set up between each RRU/RFU/AAU and the BBP/BRI in the BBU. The control- and user-plane data is transmitted on the CPRI link.

In the star topology, if a CPRI port or CPRI link is faulty, only services carried on the RRU/RFU/AAU that communicates with the BBU over the faulty CPRI port or CPRI link are affected. Therefore, the star topology is relatively reliable, but it requires more CPRI cables.

4.1.1.3 Chain Topology

The chain topology applies to long and narrow, and sparsely populated areas, such as expressways and railways. [Figure 4-2](#) shows the chain topology.

Figure 4-2 Chain topology



For each chain in the chain topology, only the first-level RRU/RFU/AAU is directly connected to the BBP/BRI in the BBU through the CPRI port, and other RRUs/RFUs/AAUs are connected to their upper-level RRUs/RFUs/AAUs.

An OML is set up between each RRU/RFU/AAU on the chain and the BBP/BRI in the BBU. The control- and user-plane data is transmitted on the CPRI link. Data of a lower-level RRU/RFU/AAU is forwarded by its upper-level RRU/RFU/AAU. The total physical bandwidth occupied by all RRUs/RFUs/AAUs on the chain cannot exceed the physical bandwidth capacity of the connected CPRI port on the BBP/BRI.

In the chain topology, if a CPRI port or CPRI link is faulty, all services carried on the lower-level RRU/RFU/AAU that communicates with the BBU over the faulty CPRI port or CPRI link are affected. Therefore, the reliability of the chain topology is relatively low. However, compared with other topologies, the chain topology requires fewer CPRI cables.

NOTE

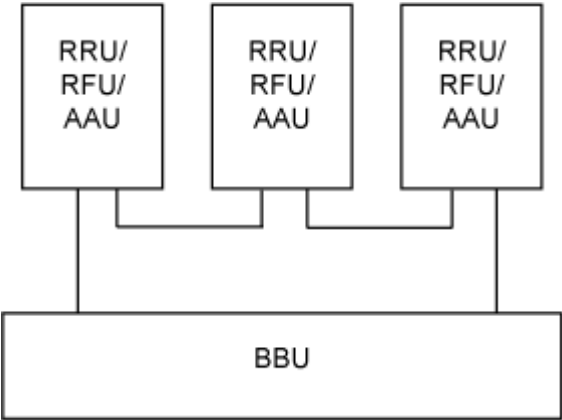
All RF modules on a chain must work at the same rate.

4.1.1.4 Ring Topology

In the ring topology, RRUs/RFUs/AAUs are connected to the BBP(s)/BRI(s) in the BBU to form a ring. One CPRI port functions as the ring head, and the other as the ring tail.

[Figure 4-3](#) shows the ring topology.

Figure 4-3 Ring topology



When the protocol type is CPRI, different ring topology types are supported by different RATs, as described in the following table. The eCPRI protocol does not support the ring topology.

Table 4-1 Ring topology types supported by different RATs (CPRI protocol)

Type of Ring Topology	GSM Only	UMTS Only	LTE Only	NR Only	Multimod e ^a
Intra-board cold backup ring topology	Yes	Yes	Yes	Yes	Yes
Inter-board cold backup ring topology	Yes	Yes	Yes	Yes	Yes
Hot backup ring topology	No	Yes	Yes	Yes	No
a: Multimode indicates any combination of GSM only, UMTS only, LTE only, and NR only.					

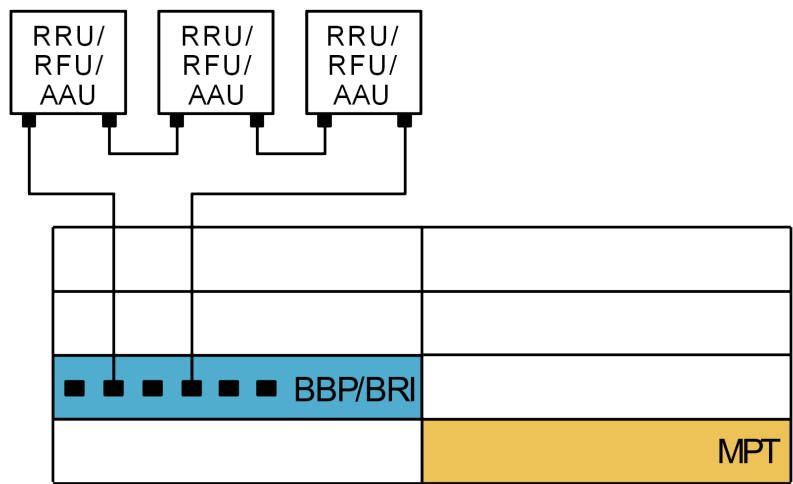
NOTE

- Base stations configured with a BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a do not involve the inter-board cold backup ring topology or hot backup ring topology.
- In multimode scenarios, the intra-board and inter-board cold backup ring topologies are supported only by co-MPT multimode base stations.
- All RF modules on a cascading link must work at the same rate.
- In a ring topology, the CPRI rates of the head port and the tail port on the ring must meet the bandwidth requirements for cells established on all RF modules on the ring. Otherwise, when a CPRI port or CPRI link transmitting services on the ring is faulty and the CPRI link on the other port is used for service transmission, some cells may become faulty due to insufficient CPRI bandwidth.

4.1.1.4.1 Intra-Board Cold Backup Ring Topology

Figure 4-4 shows the intra-board cold backup ring topology.

Figure 4-4 Intra-board cold backup ring topology



NOTE

MPT indicates the main processing and transmission unit, BBP indicates the baseband processing unit, and BRI indicates the baseband radio interface unit.

In the intra-board cold backup ring topology, the ring head and ring tail are connected to different CPRI ports on the same BBP/BRI.

RRUs/RFUs/AAUs on the ring set up OMLs with the BBU over one CPRI port on the BBP/BRI. The control- and user-plane data is transmitted on the CPRI link.

In the intra-board cold backup ring topology:

- If the BBP/BRI is faulty, services carried on all RRUs/RFUs/AAUs on the ring are interrupted and cannot be restored.
- If the CPRI port or CPRI link that transmits services on the ring is faulty, services carried on the RRUs/RFUs/AAUs and transmitted over the CPRI port or CPRI link are interrupted and then return to normal after the services are switched to the CPRI cable on the other port side. If the CPRI port or CPRI link

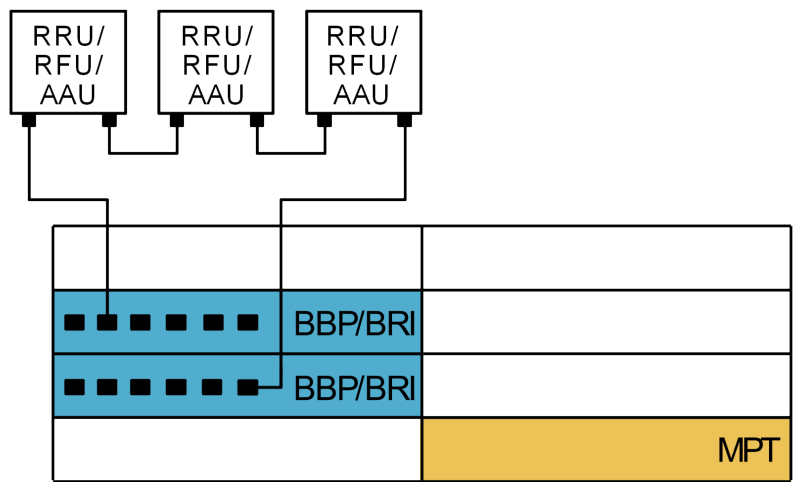
that does not transmit services is faulty, services carried on the RRUs/RFUs/AAUs are not affected.

- In scenarios where both the baseband extension mode and non-baseband extension mode are used on a ring, services of carriers in baseband extension mode are also affected when the services of carriers in non-baseband extension mode are switched to the other port side because of a port fault.

4.1.1.4.2 Inter-Board Cold Backup Ring Topology

Figure 4-5 shows the inter-board cold backup ring topology.

Figure 4-5 Inter-board cold backup ring topology



In the inter-board cold backup ring topology, the ring head and ring tail are connected to different BBPs/BRIs.

RRUs/RFUs/AAUs on the ring set up OMLs with one BBP/BRI in the BBU. The control- and user-plane data is transmitted on the CPRI link.

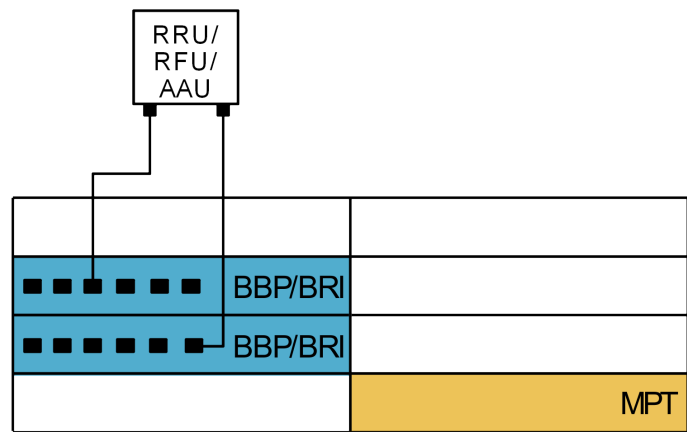
In the inter-board cold backup ring topology:

- If the BBP/BRI that transmits services is faulty, all services carried on RRUs/RFUs/AAUs on the ring are interrupted and then return to normal after the RRUs/RFUs/AAUs successfully communicate with the other BBP/BRI. If the BBP/BRI that does not transmit services is faulty, services carried on the RRUs/RFUs/AAUs are not affected.
- If the CPRI port or CPRI link that transmits services on the ring is faulty, services carried on the RRUs/RFUs/AAUs and transmitted over the CPRI port or CPRI link are interrupted and then return to normal after the services are switched to the CPRI cable on the other port side. If the CPRI port or CPRI link that does not transmit services is faulty, services carried on the RRUs/RFUs/AAUs are not affected.

4.1.1.4.3 Hot Backup Ring Topology

Figure 4-6 shows the hot backup ring topology.

Figure 4-6 Hot backup ring topology



In the hot backup ring topology, each RRU/RFU/AAU is connected to two BBPs/BRIs, and there is only one RRU/RFU/AAU on the ring.

An OML is set up between an RRU/RFU/AAU and one BBP/BRI in the BBU. The same user-plane data is simultaneously transmitted on two CPRI links.

In the hot backup ring topology:

- If the BBP/BRI that sets up the OML with the RRU/RFU/AAU on the ring is faulty, services carried on the RRU/RFU/AAU are interrupted and then return to normal after the RRU/RFU/AAU successfully communicates with the other BBP/BRI. If the BBP/BRI that does not set up the OML with the RRU/RFU/AAU is faulty, services carried on the RRU/RFU/AAU are not affected.
- If the CPRI port or CPRI link that carries OM services is faulty, services carried on the RRU/RFU/AAU are quickly switched to the CPRI cable on the other port and then return to normal. If the CPRI port or CPRI link that does not carry OM services is faulty, services carried on the RRU/RFU/AAU are not affected.

The hot backup ring topology is more reliable than the cold backup ring topology.

4.1.1.5 Load Sharing Topology

NOTE

- Base stations configured with a BBU3910A/BBU3910C/BookBBU5901/BookBBU5901a do not involve the inter-board load sharing topology.
- Base stations configured with a BBU5900C do not involve the load sharing topology.
- When the load sharing topology is used, the interface principles of the eCPRI and CPRI protocols are the same.

When the protocol type is CPRI or eCPRI, different load sharing topology types are supported by different RATs, as described in Table 4-2 and Table 4-3. "Yes" indicates supported, and "No" indicates not supported.

Table 4-2 Load sharing topology types supported by different RATs (CPRI protocol)

Type of Load Sharing Topology	GSM Only	UMTS Only	LTE Only	NR Only	Multimode ^a
Intra-board load sharing topology	No	No	Yes	Yes	No
Inter-board load sharing topology	No	No	Yes	Yes	Yes
a: Multimode indicates any combination of GSM only, UMTS only, LTE only, and NR only.					

Table 4-3 Load sharing topology types supported by different RATs (eCPRI protocol)

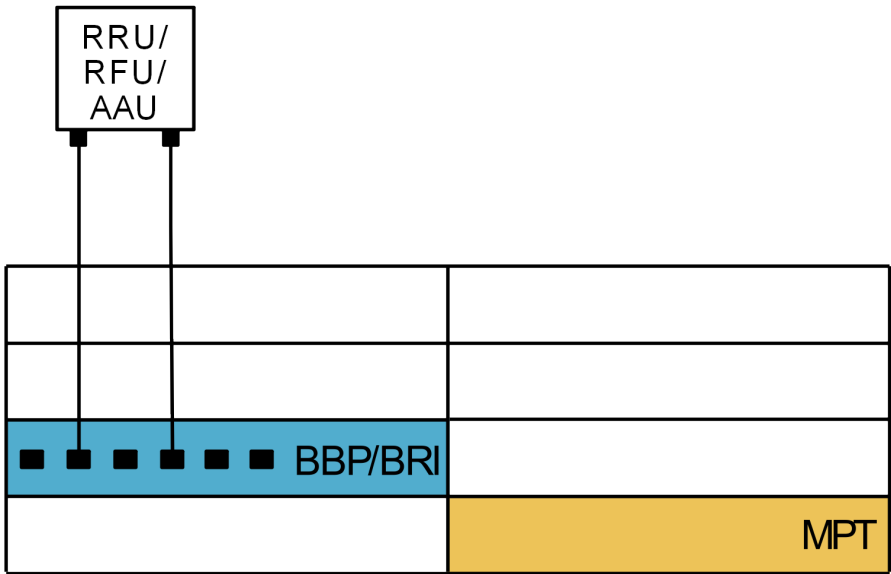
Type of Load Sharing Topology	LTE Only	NR Only	Multimode ^a
Intra-board load sharing topology	Yes	Yes	Yes
Inter-board load sharing topology	Yes	Yes	Yes
a: "Multimode" indicates the combination of LTE only and NR only.			

4.1.1.5.1 Intra-Board Load Sharing Topology

Intra-Board Load Sharing Topology

If the physical bandwidth of a CPRI port is insufficient to support a cell, the intra-board load sharing topology can be used. Figure 4-7 shows the intra-board load sharing topology.

Figure 4-7 Intra-board load sharing topology



In the intra-board load sharing topology, one RRU/RFU/AAU is connected to two CPRI ports on one BBP/BRI. This ensures that the physical bandwidths of the two ports meet the cell bandwidth requirements.

An OML is set up for transmitting OM data between a BBU and an RRU/RFU/AAU through either of the CPRI ports on the BBP/BRI. The user-plane data is transmitted simultaneously on the two CPRI links.

In the intra-board load sharing topology, if the CPRI port or CPRI link carrying OM services is faulty, the OML is automatically switched over to the other CPRI link. If one of the CPRI ports is faulty and the bandwidth of the CPRI ports fails to meet the cell bandwidth requirements, the cell may be unavailable or the cell service capability may deteriorate.

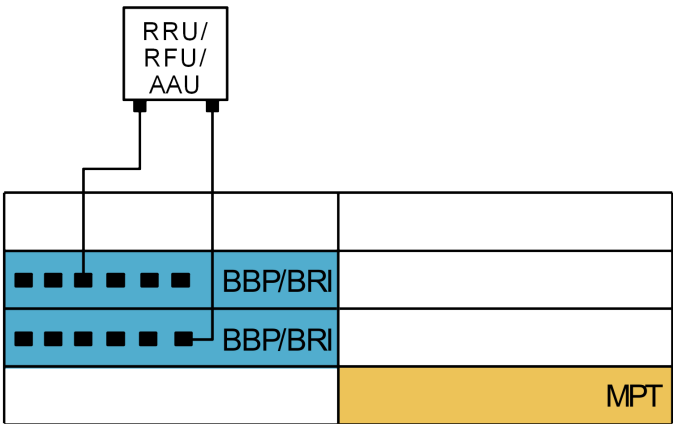
4.1.1.5.2 Inter-Board Load Sharing Topology

Inter-board load-sharing topologies are classified into inter-board load-sharing topology (single mode) and inter-board load-sharing topology (multimode).

Inter-Board Load Sharing Topology (Single Mode)

The inter-board load sharing topology applies to multi-carrier RRUs/RFUs/AAUs. [Figure 4-8](#) shows the inter-board load sharing topology.

Figure 4-8 Inter-board load sharing topology (single mode)



In the inter-board load sharing topology, one RRU/RFU/AAU is connected to the CPRI ports on two BBPs/BRIs.

An OML is set up for transmitting OM data between a BBU and an RRU/RFU/AAU through one BBP/BRI. The user-plane data is transmitted simultaneously on the two CPRI links.

In the inter-board load sharing topology, if the CPRI port or CPRI link carrying OM services is faulty, the OML is automatically switched over to the other CPRI link. If one of the CPRI ports is faulty and the bandwidth of the CPRI ports fails to meet the cell bandwidth requirements, the cell may be unavailable or the cell service capability may deteriorate.

Inter-Board Load Sharing Topology (Multimode)

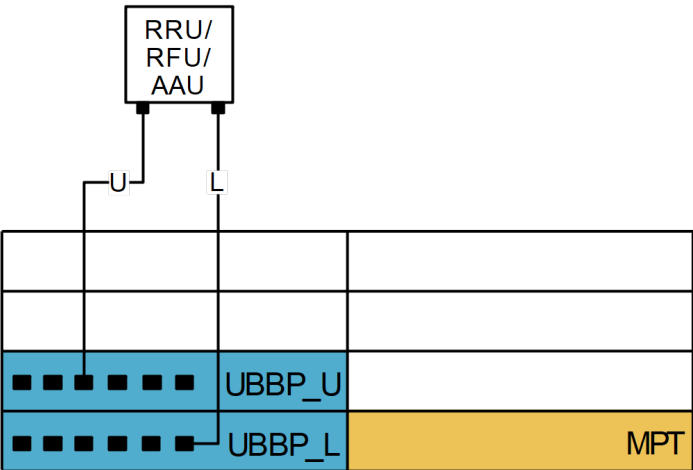
The inter-board load sharing topology (multimode) applies to co-MPT multimode base stations.

In the inter-board load sharing topology (multimode), two CPRI links of a multimode RF module carry services of two or more RATs, and each of them carries services of different RATs.

- NOTE
- If a CPRI link of a multimode RF module carries services of multiple RATs, the CPRI MUX feature must be enabled. For details, see *CPRI MUX*.
 - If a UBBP is used in the inter-board load sharing topology (multimode), the UBBP must have been configured with baseband resources of the target RAT. Otherwise, carriers cannot be successfully allocated to the CPRI ports on this UBBP.

For multimode RF modules, the inter-board load sharing topology can be used. The UL load sharing topology is used as an example, as shown in [Figure 4-9](#). UBBP_U/L indicates the working mode. For example, if a UBBP supports GSM, UMTS, and LTE but works in UMTS mode, then UBBP_U is used.

Figure 4-9 Inter-board load sharing topology (multimode)



In the inter-board load sharing topology (multimode), the RRU/RFU/AAU is connected to two BBPs/BRIs of different RATs (the RATs supported by the two BBPs/BRIs are not identical). For example, when a BBU3900 is used in the UL inter-board load sharing topology (multimode), the two CPRI ports on the RRU/RFU/AAU are connected to a UMTS BBP and an LTE BBP, respectively.

An OML is set up for transmitting OM data between the BBU and the RRU/RFU/AAU through either of the one-RAT BBPs/BRIs. The user-plane data of two RATs is transmitted separately on two different CPRI links.

If the BBP/BRI that is used for OML establishment is faulty, the services carried on the RRU/RFU/AAU will be interrupted and then return to normal after the RRU/RFU/AAU successfully communicates with the other BBP/BRI. In the process, all the services carried on the BBP/BRI will be affected. After the communication is reestablished for the RF module, services carried on the normal port will recover and those carried on the faulty port will not recover. If the BBP/BRI that is not

used for OML establishment is faulty, the services carried on the port of this board will be affected and those carried on the port of the other board will not be affected.

NOTE

If the BBU3900 is used in a load sharing topology involving UMTS and two or more BBPs are configured for UMTS, a WBBPf or UBBP_Ux must be installed in slot 2 or 3. UBBP_Ux indicates a UBBP configured with UMTS baseband resources.

4.1.1.6 Dual-Star Topology

The dual-star topology applies to separate-MPT multimode base stations.

This section describes the following features of multimode base stations:

- MRFD-210002 Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(GBTS)
- MRFD-220002 Multi-mode BS RRU/RFU star-connection with separate CPRI interface(NodeB)
- MRFD-230002 Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(eNodeB)
- MRFD-151169 Multi-mode BS RRU/RFU star-connection with separate CPRI interface (NR)

In the dual-star topology, an RRU/RFU/AAU is connected to two BBPs/BRIs of different RATs. Taking a GU base station using the BBU3900 as an example, [Figure 4-10](#) and [Figure 4-11](#) show the intra-subrack dual-star topology and inter-subrack dual-star topology, respectively.

Figure 4-10 Intra-subrack dual-star topology

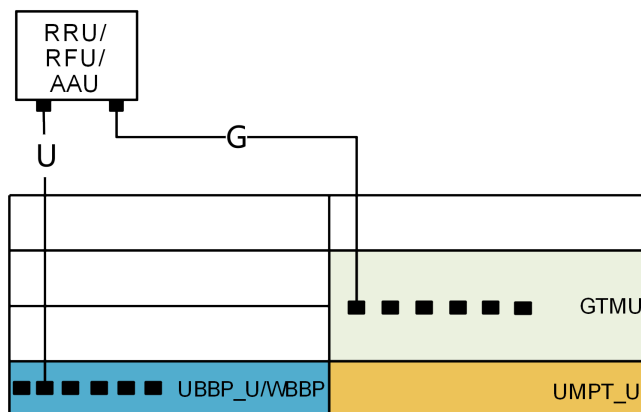
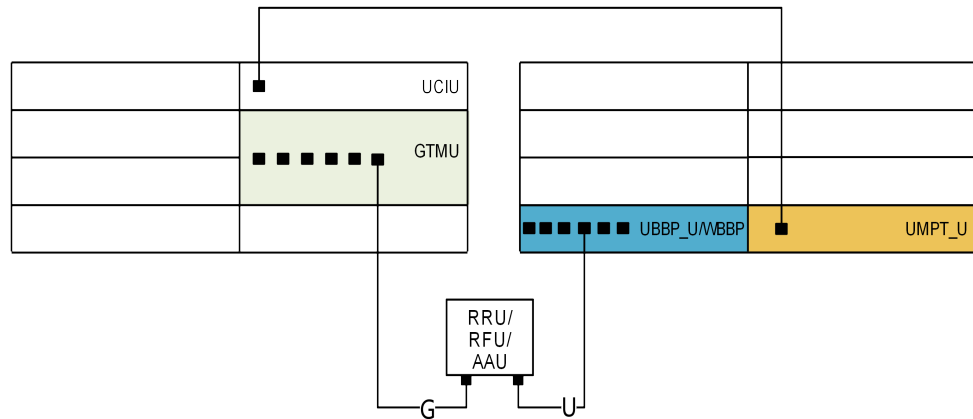


Figure 4-11 Inter-subrack dual-star topology



NOTE

- Base stations configured with a BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a do not involve the intra-subrack dual-star topology or inter-subrack dual-star topology.
- The inter-subrack dual-star topology requires that the physical connections and data configuration for BBU interconnection have been completed.
- When the dual-star topology is used, RF modules cannot be cascaded.
- The cabinet number, subrack number, and slot number of the RF module shared by two BBUs must be consistent between the two BBUs.

In the dual-star topology, two OMLs are set up between the BBU(s) and an RRU/RFU/AAU through two BBPs/BRIs of different RATs. Control- and user-plane data of each RAT is transmitted on the CPRI link of the corresponding RAT.

In the dual-star topology, if one CPRI port or CPRI link for one RAT is faulty due to the reset of the main control board, the services of the other RAT are affected.

4.1.1.7 LampSite Branch Chain/Ring Topology

In the LampSite solution, the NodeB or eNodeB consists of the BBU, pRRU, and RHUB. The BBU connects to RHUBs using optical fibers, and RHUBs connect to pRRUs using network cables or optical fibers. Typical scenarios include branch chain (single-link) and branch load sharing.

In the LampSite solution, the CPRI networking between the RHUB and BBU can be the star topology or chain topology. For details on specifications (including the maximum number of RHUBs supported by a BBU, the maximum number of RHUBs supported by a CPRI link, and the maximum number of pRRUs supported by an RHUB), see *LampSite Solution Description* in *3900 & 5900 Series Base Station Product Documentation*.

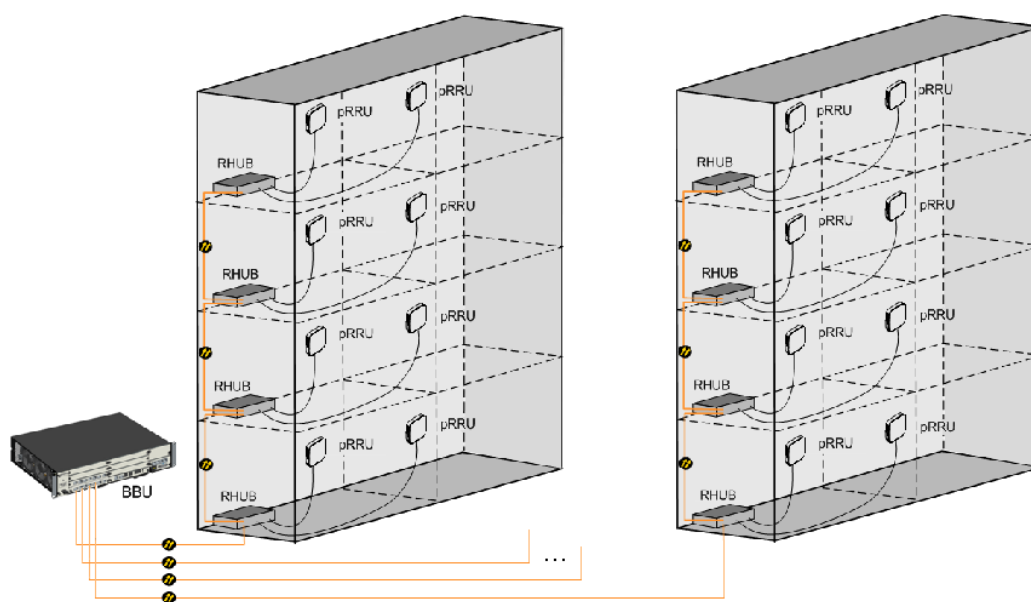
 NOTE

- When connected by the network cable, the maximum stable rate supported by a LampSite branch chain depends on the type and length of the cable. They are as follows: 1.25 Gbit/s for Cat5e non-shielded cable; 2.5 Gbit/s for Cat5e shielded cable; 10.1 Gbit/s for Cat6a shielded cable.
- For the reliability, a LampSite branch chain only supports auto-negotiation of the 1.25 Gbit/s rate. Before using any other rate, check the network cable type, pRRU model, and RHUB model and then specify the rate of the LampSite branch chain in fixed rate mode.

4.1.1.7.1 Branch Chain (Single-Link) Topology

Figure 4-12 shows the branch chain (single-link) topology.

Figure 4-12 Branch chain (single-link) topology



The branch chain (single-link) topology is classified into the star topology and chain topology. In the branch chain (single-link) topology, a pRRU connects to an RHUB over one network cable or optical fiber. pRRU cascading is not supported.

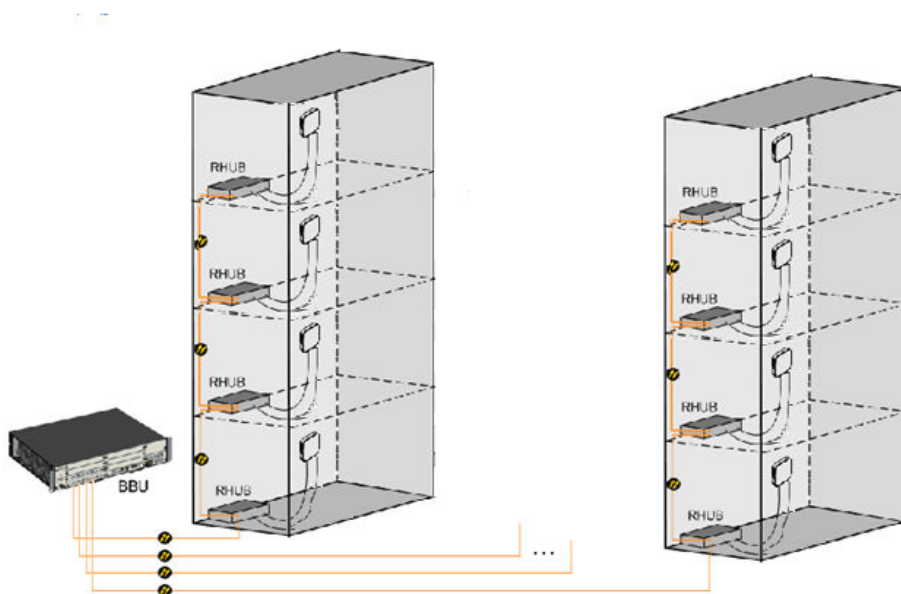
 NOTE

- Before changing the CPRI rate of a pRRU on a cascading chain or adding a pRRU to or deleting a pRRU from this chain, run the **BLK CPRI PORT** command to block the CPRI port on the BBU side of this chain. After the change, unblock this CPRI port at least 30 seconds later.
- When a cascading chain with RF modules of different rates is changed into one with RF modules of the same rate by activating the configuration or CB, if link setup fails on the BBP on the cascading chain or the BBP cannot work at the highest line rate after the main control board is restarted, block the CPRI port on the BBU side of this chain for at least 30 seconds using the **BLK CPRI PORT** command, and then unblock the CPRI port.

4.1.1.7.2 Branch Load Sharing Topology

Figure 4-13 shows the branch load sharing topology.

Figure 4-13 Branch load sharing topology



In the branch load sharing topology, a pRRU connects to an RHUB over two network cables. Application scenarios of the branch load sharing topology are as follows:

- The physical bandwidth of a CPRI port on a pRRU is insufficient to support cells served by the pRRU.
- A pRRU is configured with two or more RATs.

NOTE

The pRRU565x4B modules (x is a digit; A and B stand for different letters) do not support the branch load sharing topology.

In branch load sharing scenarios, CPRI resources cannot be automatically defragmented. If the cell service capability or TRP service capability degrades due to operations such as board reset, cell addition or removal, and bandwidth modification, you can attempt to block or unblock the corresponding RHUB port to restore service capability.

4.1.1.8 LampSite Trunk Chain Topology

NOTE

- RHUB39xx series and RHUB592x series RHUBs do not support the LampSite trunk chain topology.
- In LampSite networking scenarios, only the CPRI protocol is involved.
- In LampSite networking scenarios, the CPRI ports on all main chains in the trunk chain topology must use the same data rate.
- In LampSite networking scenarios, if the CPRI link or CPRI port that carries OM services is faulty, the OML is faulty and the corresponding alarm is reported. If the CPRI link or CPRI port over which no OML is set up with the RHUB is faulty, only the services carried on the faulty port will be affected and those carried on the other port will not be affected.

When the protocol type is CPRI, different trunk chain topology types are supported by different RATs, as described in the following table.

Table 4-4 Trunk chain topology types supported by different RATs (CPRI protocol)

Trunk Chain Topology	GSM Only	UMTS Only	LTE Only	NR Only	RFA Only	Multimode ^a
Single-mode trunk chain topology (intra-board)	No	Yes	Yes	Yes	No	No
Single-mode trunk chain topology (inter-board)	No	No	No	Yes	No	No
Multimode trunk chain topology (intra-board)	No	No	No	No	No	Yes

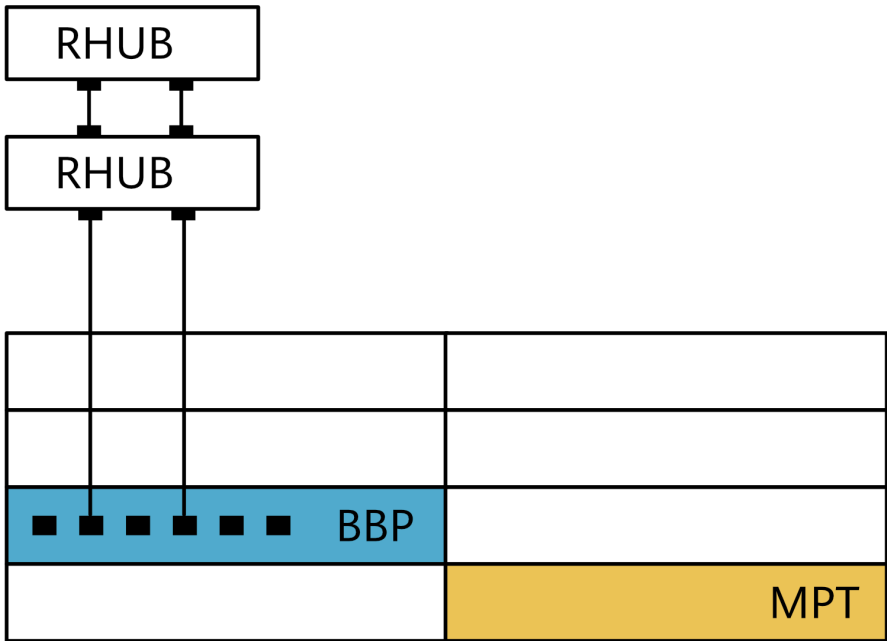
a: In the multimode trunk chain topology, any combination of UMTS, LTE, NR, and RFA single modes can be used. RFA must share the same main control board with LTE or NR before being combined with other modes.

4.1.1.8.1 Single-Mode Trunk Chain Topology

If the physical bandwidth of a CPRI port is insufficient to support a cell, the trunk chain topology can be used.

Figure 4-14 shows the single-mode intra-board trunk chain topology.

Figure 4-14 Single-mode intra-board trunk chain topology

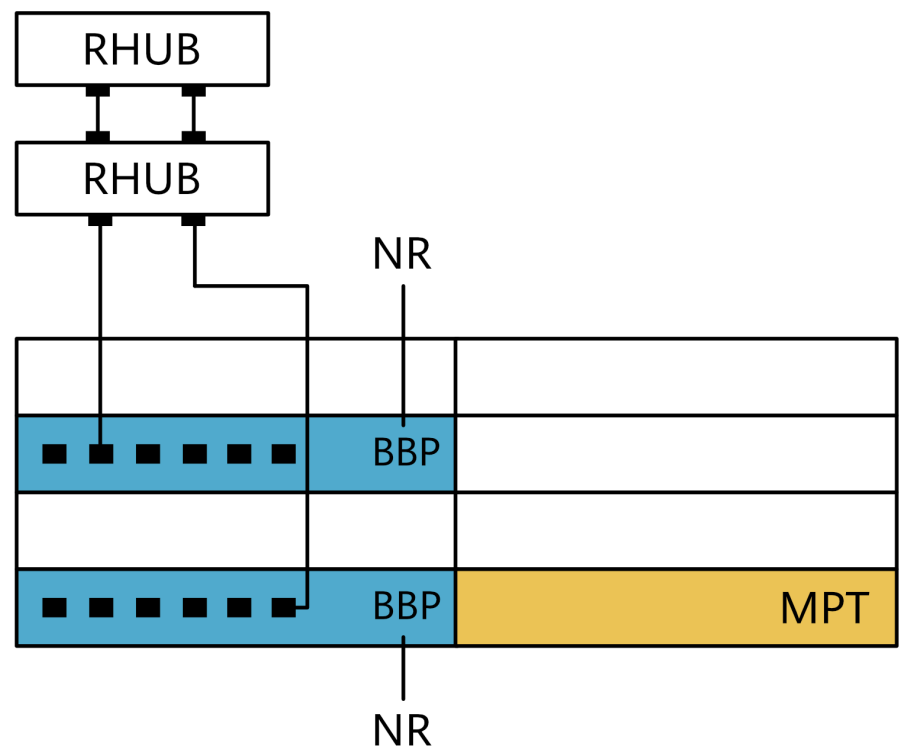


In the single-mode trunk chain topology, two CPRI ports on an RHUB connect to two CPRI ports on a BBP, respectively. This ensures that the physical bandwidth of two CPRI ports meets the cell bandwidth requirements.

An OML is set up for transmitting OM data between a BBU and an RHUB through a CPRI port on the BBP. The user-plane data is transmitted simultaneously on two CPRI links.

Figure 4-15 shows the single-mode inter-board trunk chain topology.

Figure 4-15 Single-mode inter-board trunk chain topology



When the single-mode inter-board trunk chain topology is used, two CPRI ports on an RHUB connect to two CPRI ports on two BBPs, respectively.

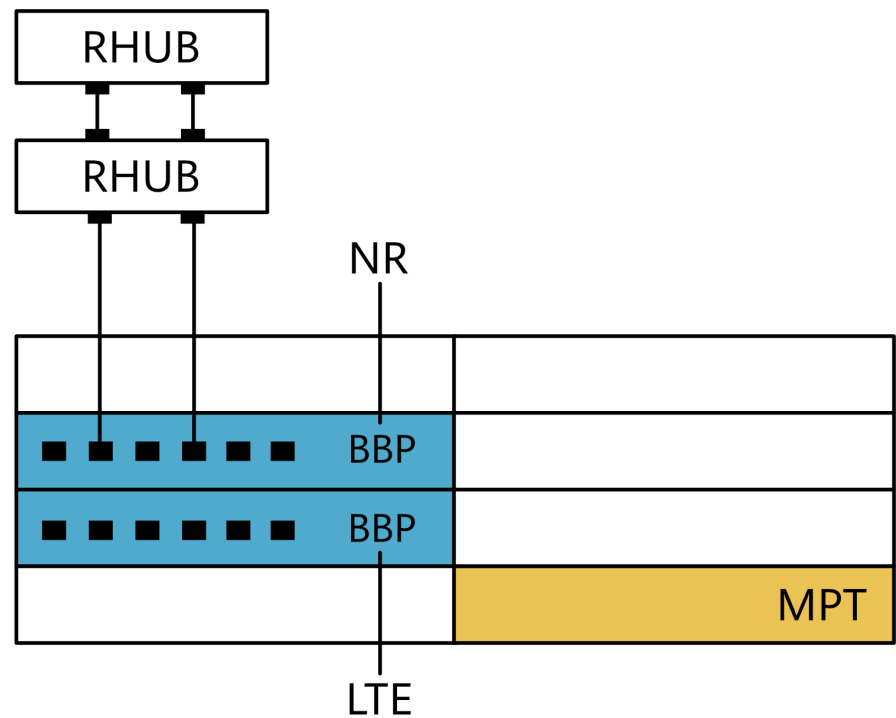
An OML is set up for transmitting OM data between a BBU and an RHUB through a CPRI port on the BBP. The user-plane data is transmitted simultaneously on two CPRI links. If the CPRI OML becomes faulty, data is automatically switched over to the other CPRI link. During the OML switchover, services are interrupted for a short period of time.

4.1.1.8.2 Multimode Trunk Chain Topology

The multimode trunk chain topology applies to co-MPT, separate-MPT, and hybrid-MPT multimode base stations. If two CPRI links of a multimode RF module support services of two or more RATs, this type of topology is called the multimode trunk chain topology.

This document describes only the multimode trunk chain scenario where each CPRI link of a multimode RF module carries services of two RATs. Figure 4-16 uses the LNR trunk chain as an example.

Figure 4-16 Multimode intra-board trunk chain topology



In the multimode trunk chain topology, two CPRI ports on an RHUB connect to two CPRI ports on a BBP. Taking the BBU5900 as an example, in the LNR trunk chain topology, two CPRI ports on the RHUB connect to two ports on the NR BBP. An OML is set up for transmitting OM data between each BBP in the BBU and the RHUB through a physical CPRI link. The user-plane data of two RATs is transmitted on two CPRI links.

4.1.2 RF Modules

4.1.2.1 Overview

- The following table lists the topologies supported by RF modules of different RATs in macro base stations. "CPRI" indicates that the topology is supported only in the case of the CPRI protocol. "eCPRI" indicates that the topology is supported only in the case of the eCPRI protocol. "CPRI/eCPRI" indicates that the topology is supported either in the case of the CPRI or eCPRI protocol. "/" Indicates that the topology is supported neither in the case of the CPRI or eCPRI protocol.

Table 4-5 Topologies supported by RF modules of each RAT in macro base stations

Topology Type	GSM Only	UMTS Only	LTE Only	NR Only	Multimod e ^a
Star	CPRI	CPRI	CPRI/eCPRI	CPRI/eCPRI	CPRI/eCPRI
Chain	CPRI	CPRI	CPRI	CPRI	CPRI

Topology Type		GSM Only	UMTS Only	LTE Only	NR Only	Multimode ^a
Ring	Intra-board cold backup	CPRI	CPRI	CPRI	CPRI	CPRI
	Inter-board cold backup	CPRI	CPRI	CPRI	CPRI	CPRI
	Hot backup	/	CPRI	CPRI	CPRI	/
Load sharing	Intra-board	/	/	CPRI/eCPRI	CPRI/eCPRI	eCPRI
	Inter-board	/	/	CPRI/eCPRI	CPRI/eCPRI	CPRI/eCPRI
Dual-star		/	/	/	/	CPRI/eCPRI

a: In the case of the CPRI protocol, "multimode" indicates any combination of GSM only, UMTS only, LTE only, and NR only. In the case of the eCPRI protocol, "multimode" indicates the combination of LTE only and NR only.

- The topologies supported by RF modules in LampSite base stations are as follows:
 - Between the BBU and RHUB: star, chain, load sharing, and trunk chain
-  **NOTE**
- The main chain supports the load sharing topology only when the RHUB3918/RHUB5921/RHUB5923 is used.
 - Only the RHUB5775/RHUB5961/RHUB5963e supports the LampSite trunk chain topology.
- Between the RHUB and pRRU: branch chain (single-link) and branch load sharing

4.1.2.2 Basic RFU Information

This section describes basic information about the RFUs, including names, logical types, supported RATs, TX/RX modes, and supported topologies.

 NOTE

- The logical type is indicated by the value of **Board Type** or **RRU type** that must be set on the local maintenance terminal (LMT), MAE-Deployment, or MAE-Access during RF module configuration.
- *ATBR* in the **TX/RX Mode** column indicates that the RF module uses *A* transmit channels and *B* receive channels.

Table 4-6 Basic RFU information

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
CRFUd	2100/AWS	MRFU	LTE FDD	2T2R	Star
CRFUe	2100	MRFU	LTE FDD	2T2R	Star
DRFU	900/1800	DRFU	GSM	2T2R	Star, chain
GRFU V1	1900	GRFU	GSM	1T2R	Star, chain
GRFU V2	900/1800	GRFU	GSM	1T2R	Star, chain
GRFU V2a	900/1800	GRFU	GSM	1T2R	Star, chain
LRFU	2600	LRFU	LTE FDD	2T2R	Star
LRFUe	700/800	LRFU	LTE FDD, LM	2T2R	Star
LRFUe	2600	LRFU	LTE FDD	2T2R	Star
MRFU V1	900	MRFU	GSM, UMTS, GU	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V1	1800	MRFU	GSM, LTE FDD, GL	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V1	1900	MRFU	GSM, UMTS	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V2	850	MRFU	GSM, UMTS, LTE FDD, GU, UL	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V2	900	MRFU	GSM, UMTS, LTE FDD, GU, GL, LTE NB-IoT, GM, UM, LM, GLM, ULM	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V2	1800	MRFU	GSM, LTE FDD, GL	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
MRFU V2a	900	MRFU	GSM, UMTS, LTE FDD, GU, GL, LTE NB-IoT, GM, LM, UM	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V2a	1800	MRFU	GSM, LTE FDD, GL	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V6	900	MRFU	GSM, UMTS, LTE FDD, GU, GL, LTE NB-IoT, GM, LM, UM	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFU V6	1800	MRFU	GSM, LTE FDD, GL	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFUc	2100	MRFU	UMTS, LTE FDD, UL	2T2R	Star, intra-board load sharing, inter-board load sharing, dual-star
MRFUd	1800	MRFU	GSM, LTE FDD, GL, LM, GLM	2T2R	Star, intra-board load sharing, inter-board load sharing, dual-star
MRFUd	900	MRFU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFUd V6	1800	MRFU	GSM, LTE FDD, GL, LM, GLM	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFUd V6	2100	MRFU	UMTS, LTE FDD, UL	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFUdw V6	1800	MRFU	GSM, LTE FDD, GL	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
MRFUe	900	MRFU	GSM, UMTS, LTE FDD, LTE NB-IoT, GU, GL, GM, UM, LM, GUM, ULM, GLM	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
MRFUe	1800	MRFU	GSM, LTE FDD, GL, LM, GLM	1T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
40 W WRFU	2100	WRFU	UMTS	1T2R	Star, chain
80 W WRFU	850/2100	WRFU	UMTS	1T2R	Star, chain
80 W WRFU Optimized	2100	WRFU	UMTS	1T2R	Star, chain
WRFUa	2100	WRFU	UMTS	1T2R	Star, chain
WRFUd	2100	WRFU	UMTS	2T2R	Star
WRFUe	2100	WRFU	UMTS	2T2R	Star

NOTE

The DRFU does not apply to the eGBTS.

4.1.2.3 Basic AAU Information

This section describes basic information about the AAUs, including names, logical types, supported RATs, TX/RX modes, and supported topologies.

NOTE

- The logical type is indicated by the value of **Board Type** or **RRU type** that must be set on the LMT, MAE-Deployment, or MAE-Access during RF module configuration.
- $ATBR$ in the **TX/RX Mode** column indicates that the RF module uses A transmit channels and B receive channels.

Table 4-7 Basic AAU information

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU3902	2100	AARU	UMTS	2T4R	Star, chain, intra-board load sharing, inter-board load sharing
AAU3902	1800	AARU	LTE FDD, GL	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
AAU3902	1800+2100	AARU	1800 MHz: LTE FDD, GL 2100 MHz: UMTS	4T4R (RU44) 2T4R (RU44)	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
AAU3910	1800	AIRU	LTE FDD, GL	4T4R (RU3971)	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU3910	2100	AIRU	UMTS	2T4R (RU3232)	Star, intra-board load sharing, inter-board load sharing
AAU3910	2600	AIRU	LTE FDD	2T4R (RU3260)	Star, intra-board load sharing, inter-board load sharing
AAU3910	AWS	AIRU	UMTS, LTE FDD, UL	2T4R (RU3832)	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU3910	2100+2600	AIRU	RU3832 (2100 MHz): UMTS RU3260 (2600 MHz): LTE FDD	2T4R (RU3832) 2T4R (RU3260)	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU3910	AWS+PCS	AIRU	AWS: UMTS, LTE FDD, LM, UL PCS: UMTS, LTE FDD, GU, GL, UL, GUL	4T4R (RU3971)	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU3911	1800	AIRU	RU3952m: GSM, LTE FDD, GL RU3971: LTE FDD, GL	2T4R (RU3952m) 4T4R (RU3971)	Star, chain, intra-board load sharing, inter-board load sharing, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU3911	2100	AIRU	RU3832: UMTS RU3952m: UMTS, LTE FDD, UL RU3971: UMTS, LTE FDD, UL	2T4R (RU3832) 2T4R (RU3952m) 4T4R (RU3971)	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, dual-star, intra-board load sharing, inter-board load sharing
AAU3911	2600	AIRU	LTE FDD	2T4R (RU3260) 4T4R (RU3261)	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
AAU3911	AWS	AIRU	UMTS, LTE FDD, UL	2T4R (RU3832)	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, dual-star, intra-board load sharing, inter-board load sharing
AAU3911	PCS	AIRU	UMTS, LTE FDD, GU, GL, UL, GUL	4T4R (RU3971)	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU3911	1800+2100	AIRU	1800 MHz+2100 MHz: GSM, LTE FDD, GL 2100 MHz+1800 MHz: UMTS, LTE FDD, UL	2T4R (RU3952m)	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU3911	2100+2600	AIRU	2100 MHz: UMTS, LTE FDD, UL 2600 MHz: LTE FDD	2T4R (RU3260) 2T4R (RU3952m)	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
AAU3911	2600+AWS	AIRU	2600 MHz: LTE FDD AWS: UMTS, LTE FDD, UL	2T4R (RU3260) 2T4R (RU3832)	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU3920*	1800+2100	AIRU	1800 MHz: GSM, LTE FDD, GL, LTE NB-IoT, LM, GLM 2100 MHz: UMTS, LTE FDD, UL	1800 MHz: 2T4R 2100 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
AAU3940	1800+2100	AIRU	1800 MHz: LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD), LN (SUL), ULN (SUL)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
AAU3940	AWS+PCS	AIRU	AWS: UMTS, LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD), LN (SUL), ULN (SUL) PCS: UMTS, LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD), LN (SUL), ULN (SUL)	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU3961	1800	AIRU	LTE FDD, GL	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
AAU3961	1800+2600	AIRU	RU3971 (1800 MHz): LTE FDD, GL RU3261 (2600 MHz): LTE FDD RU5301 (2600 MHz): LTE FDD	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
AAU5243	1800+2600	AIRU	1800 MHz: LTE FDD, LM, NR FDD 2600 MHz: LTE TDD, NR TDD, TN (TDD)	1800 MHz: 4T4R 2600 MHz: 8T8R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5246	1800+2600	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD), LM, LMN (FDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	1800 MHz: 4T4R 2600 MHz: 8T8R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU5271	2300	AIRU	LTE TDD	64T64R	Trunk chain
AAU5271	2600	AIRU	LTE TDD	64T64R	Trunk chain
AAU5281	3500	AIRU	LTE TDD	64T64R	Trunk chain
AAU5310	3700	AIRU	NR TDD	32T32R	Star
AAU5313	3500	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5313	3700	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5336	2300	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5336w	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5336w	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5339	3500	AIRU	LTE TDD, NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5339	3700	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5339w	3500/3700	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5349	3600	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5356	2300	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5356	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5356	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5356	3700	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5366	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5366	3600	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5368	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5368 md	2300+3500	AIRU	2300 MHz: LTE TDD, NR TDD, TN (TDD) 3500 MHz: NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5368 md	2300+3600	AIRU	2300 MHz: LTE TDD, NR TDD, TN (TDD) 3600 MHz: NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5368 md	2600+3600	AIRU	2600 MHz: LTE TDD, NR TDD, TN (TDD) 3600 MHz: NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5613	3500/3700	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5614	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5631	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5636	2300/2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5636	3400	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5636m	2600/3500/3600/3700	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5636w	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5636w	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5636w	3700	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5639	2300	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5639	3500/3700	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5639m	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5639w	2600/3500/3600/3700	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5646	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5649	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5656	2300	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5656	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5656	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5656	3700	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5665 md	2600+3500	AIRU	2600 MHz: LTE TDD, NR TDD, TN (TDD) 3500 MHz: NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5668	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5668 m	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5668 m	3800	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5685 md	2300+2600	AIRU	2300 MHz: LTE TDD, NR TDD, TN (TDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5711	1800	AIRU	LTE FDD	32T32R	Star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5711a	1800	AIRU	LTE FDD, NR FDD, LN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5711a	2600	AIRU	LTE FDD, NR FDD, LN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5726	1800+2100	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD) 2100 MHz: LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5726	AWS+PCS	AIRU	AWS: LTE FDD, NR FDD, LN (FDD) PCS: LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5726e	1800+2100	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD) 2100 MHz: LTE FDD, NR FDD, UL, UN (FDD), LN (FDD), ULN (FDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5726e	AWS+PCS	AIRU	AWS: LTE FDD, NR FDD, LN (FDD) PCS: LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5733	2600	AIRU	LTE FDD, NR FDD, LN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5736	2600	AIRU	LTE FDD, NR FDD, LN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5765	AWS+PCS	AIRU	AWS: LTE FDD, NR FDD, LN (FDD), LM PCS: LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD), LM, ULM When the AAU works in two frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5767t	1800+2100+2600	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD), LM 2100 MHz: LTE FDD, NR FDD, UL, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) When the AAU works in multiple frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5767t	AWS+PCS+2600	AIRU	AWS: LTE FDD, NR FDD, LN (FDD), LM PCS: LTE FDD, UL, NR FDD, LN (FDD), UN (FDD), ULN (FDD), LM, ULM 2600 MHz: LTE FDD, NR FDD, LN (FDD) When the AAU works in multiple frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band. However, in LTE NB-IoT, only LM and ULM are supported.	AWS: 32T32R PCS: 32T32R 2600 MHz: 32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5768	1800+2100	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD), LM 2100 MHz: LTE FDD, NR FDD, UL, UN (FDD), LN (FDD), ULN (FDD) When the AAU works in two frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band. However, in LTE NB-IoT, only LM and ULM are supported.	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5785t	700+800+900	AIRU	700 MHz: LTE FDD, NR FDD, LN (FDD), LM, LMN (FDD) 800 MHz: LTE FDD, NR FDD, LN (FDD), LM, LMN (FDD) 900 MHz: LTE FDD, NR FDD, GU, GL, GN (FDD), UL, UN (FDD), LN (FDD), ULN (FDD), GLN (FDD), LM, MN (FDD), GM, GLM, GMN (FDD), ULM, UMN (FDD), LMN (FDD) When the AAU works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	16T16R	Star, dual-star, intra-board load sharing, inter-board load sharing
AAU5811	3700	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5821g	2600	AIRU	LTE FDD, NR FDD, LN (FDD)	32T32R	Star, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5822a	1800+2100	AIRU	1800 MHz: LTE FDD, NR FDD, LN (FDD) 2100 MHz: LTE FDD, NR FDD, UL, UN (FDD), LN (FDD), ULN (FDD)	1800 MHz: 32T32R 2100 MHz: 32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5825a	2300/2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5825g	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5826g	2600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5831	3500	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5831	3700	AIRU	LTE TDD, NR TDD, TN (TDD)	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5832a	3500	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5832f	3700	AIRU	NR TDD	32T32R	Star, intra-board load sharing, inter-board load sharing
AAU5833f	3500/3700	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5835f	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5835f	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5835f	3400/3700	AIRU	NR TDD	64T64R	Star, intra-board load sharing, inter-board load sharing
AAU5835h	3600	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5836f	3500	AIRU	LTE TDD, NR TDD, TN (TDD)	64T64R	Star, intra-board load sharing, inter-board load sharing, dual-star
AAU5940	1800+2100	AIRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU5942	1800+2100	AIRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, LM, GLM, NR FDD, LN (FDD), GLN (FDD), LMN (FDD), GMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
AAU5942	AWS+PCS	AIRU	AWS: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: GSM, UMTS, LTE FDD, GL, UL, GU, GUL, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD)	AWS: 4T4R PCS: 4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
AAU5942	PCS+2600	AIRU	PCS: UMTS, LTE FDD, NR FDD, GU, GL, UL, GN (FDD), UN (FDD), LN (FDD), GUL, GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	PCS: 4T4R 2600 MHz: 4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing, dual-star
AAU5972	1800	AIRU	LTE FDD, GL, NR FDD, GN (FDD), LN (FDD), GLN (FDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing
HAAU5213	26000/28000	AIRU	NR TDD	4T4R	Star
HAAU5222	26000/28000	AIRU	NR TDD	400 MHz: 4T4R 800 MHz: 2T2R	Star
HAAU5233w	26000	AIRU	NR TDD	1200 MHz: 4T4R	Star or intra-board load sharing
HAAU5235w	28000	AIRU	NR TDD	4T4R	Star or intra-board load sharing
HAAU5323	26000/28000	AIRU	NR TDD	400 MHz: 8T8R 800 MHz: 4T4R	Star

 NOTE

*The AAU3920 RU provides four RF channels. The restrictions between RF channels and frequency bands are as follows:

- Channels A and B support only the 1800 MHz transmit frequency band.
- Channels C and D support only the 2100 MHz transmit frequency band.
- Channels A, B, C, and D support both the 1800 MHz and 2100 MHz receive frequency bands.

The HAAU5323 supports intra-board load sharing and works only in 4T4R when the 2:3 slot configuration (2_3_DSUUU) is used.

4.1.2.4 Basic RRU Information

This section describes basic information about the RRUs, including names, logical types, supported RATs, TX/RX modes, and supported topologies.

 NOTE

- The logical type is indicated by the value of **Board Type** or **RRU type** that must be set on the LMT, MAE-Deployment, or MAE-Access during RF module configuration.
- *ATBR* in the **TX/RX Mode** column indicates that the RF module uses *A* transmit channels and *B* receive channels.
- You can set the logical type for RF modules with **Logical Type** being **MRRU** and **Supported RAT** including **LTE FDD** in the tables to **MRRU** or **LRRU**, whereas **MRRU** is recommended.
- For an RRU that can work in multiple frequency bands at the same time, only the RATs supported by the RRU when it works in all frequency bands are displayed. For details about the RATs supported by an RRU working in a single frequency band or multiple frequency bands, see *RRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*.

Table 4-8 Basic RRU information

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3004	900/1800	DRRU	GSM	2T2R	Star, chain, intra-board cold backup ring
RRU3008 V1	850/900/1800/1900	GRRU	GSM	2T2R	Star, chain, intra-board cold backup ring
RRU3008 V2	900	GRRU	GSM	2T2R	Star, chain, intra-board cold backup ring
RRU3168	1900	MRRU	LTE TDD	8T8R	Star, chain, intra-board load sharing, inter-board load sharing
RRU3201	700/2600	MRRU	LTE FDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3203	700	MRRU	LTE FDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3220	800	MRRU	LTE FDD, LM	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3220E	1800	PRRU	LTE FDD	2T2R	Star, chain
RRU3221	2600	MRRU	LTE FDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3222	800	MRRU	LTE FDD, LM	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3229	2600	MRRU	LTE FDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3230E	1800	MRRU	LTE FDD, LTE NB-IoT, LM, NR FDD, LN (FDD), MN (FDD), LMN (FDD)	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3230E	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3232	1800/2300/2600/3500	MRRU	LTE TDD	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3236E	2300/3500	MRRU	LTE TDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing
RRU3240	2600	MRRU	LTE FDD	2T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3249	700	MRRU	LTE FDD, NR FDD, LN (FDD)	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3251	2300	MRRU	LTE TDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3252	2300/2600	MRRU	LTE TDD	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3253	2600	MRRU	LTE TDD	8T8R	Star, intra-board load sharing, inter-board load sharing
RRU3256	2300/2600/3500	MRRU	LTE TDD	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3259	2600	MRRU	LTE TDD	8T8R	Star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3260	2600	MRRU	LTE FDD	2T4R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3262	700	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	2T4R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3262	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	2T4R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3262	850	MRRU	LTE FDD	2T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3268	700	MRRU	LTE FDD, LM	2T2R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3268	800	MRRU	LTE FDD, LM	2T2R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3268	2600	MRRU	LTE FDD	2T2R	Star, chain, dual-star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3269	700	MRRU	LTE FDD, LM	2T2R	Star, chain, intra-board load sharing, inter-board load sharing
RRU3276	2300/2600	MRRU	LTE TDD	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3278	3500/3650/3700	MRRU	LTE TDD	8T8R	Star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3278u	3500	MRRU	LTE TDD	8T8R	Star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3279	2300/2500/2600	MRRU	LTE TDD	8T8R	Star, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3281	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3668	450	MRRU	LTE FDD	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU3801E	850/1900/2100	MRRU	UMTS	1T2R	Star, chain, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3804	850/1900 / 2100/AWS	MRRU	UMTS	1T2R	Star, chain, inter-board cold backup ring
RRU3805	850/1900	MRRU	UMTS	2T2R	Star, chain, inter-board cold backup ring
RRU3806	2100	MRRU	UMTS	1T2R	Star, chain, inter-board cold backup ring
RRU3808	2100/AWS	MRRU	UMTS	2T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing, dual-star
RRU3821E	1800	PRRU	LTE FDD	2T2R	Star, chain, inter-board cold backup ring
RRU3824	2100	MRRU	UMTS	1T2R	Star, chain, inter-board cold backup ring
RRU3826	2100	MRRU	UMTS	1T2R	Star, chain, inter-board cold backup ring
RRU3828	2100	MRRU	UMTS	2T2R	Star, chain, inter-board cold backup ring
RRU3829	2100	MRRU	UMTS	2T2R	Star, chain, inter-board cold backup ring
RRU3832	2100	MRRU	UMTS	2T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing, dual-star
RRU3832	AWS	MRRU	UMTS, LTE FDD, UL	2T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3832	AWS-3	MRRU	UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD)	2T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing, dual-star
RRU3838	2100	MRRU	UMTS	2T2R	Star, chain, inter-board cold backup ring
RRU3839	2100	MRRU	UMTS	2T2R	Star, chain, inter-board cold backup ring
RRU3841	AWS	MRRU	LTE FDD, LM	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU3853	2100	MRRU	UMTS	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3908 V1	850	MRRU	GSM, UMTS	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3908 V1	900	MRRU	GSM, UMTS, GU	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3908 V1	1800	MRRU	GSM, LTE FDD, GL	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3908 V1	1900	MRRU	GSM, UMTS	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3908 V2	850	MRRU	GSM, UMTS, GU	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3908 V2	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, LTE NB-IoT, GM, UM, LM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3922E	1800	MRRU	GSM	2T2R	Star, chain, intra-board cold backup ring
RRU3926	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, LTE NB-IoT, GM, LM, UM, GUM, GLM, ULM	1T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, dual-star, intra-board load sharing, inter-board load sharing
RRU3926	1800	MRRU	GSM, LTE FDD, GL	1T2R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, dual-star, intra-board load sharing, inter-board load sharing
RRU3928	1800	MRRU	GSM, LTE FDD, GL	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3928	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3929	1800	MRRU	GSM, LTE FDD, GL, LM, GLM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3929	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3930E	2100	MRRU	UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3936	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM	1T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3936	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM	1T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3936	1800	MRRU	GSM, LTE FDD, GL, LM, GLM	1T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3936	1900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL	1T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3938	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, NR FDD	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3938	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3939	1800	MRRU	GSM, LTE FDD, GL, LM, GLM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU3942	850	MRRU	GSM, UMTS, LTE FDD, GU, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3942	1900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, UM, LM, ULM, GULM	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU3952	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, ULM, GLM, GUM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), UMN (FDD), LMN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU3952	2100	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU3952m	1800+2100	MRRU	1800 MHz: GSM, LTE FDD, GL, LM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD) 2100 MHz: UMTS, LTE FDD, UL	1800 MHz: 2T4R (common mode) 1800 MHz +2100 MHz: 2T2R+0T2R (inter-frequency mutual aid mode)	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU395 2m	2100+1800	MRRU	2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL) 1800 MHz: GSM, LTE FDD, GL	2100 MHz: 2T4R (common mode) 2100 MHz +1800 MHz: 2T2R+0T2R (inter-frequency mutual aid mode)	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU395 3	2100	MRRU	UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU395 3	1900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, UM, LM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD)	2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU395 3	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU395 3	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), GUMN (FDD)	2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU395 3w	1800	MRRU	GSM, LTE FDD, GL, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU395 8	2100	MRRU	UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU395 9	2100	MRRU	UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU395 9	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU395 9	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU395 9a	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T2R	Star, chain, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU395 9a	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star
RRU395 9a	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star
RRU395 9w	1800	MRRU	GSM, LTE FDD, GL, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU396 1	800+900 EGSM	MRRU	GL, UL, GUL	800 MHz: 2T2R 900 EGSM: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU396 2	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LN (SUL), GLN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD)	1800 MHz: 2T4R 2100 MHz: 2T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU3965	800+900	MRRU	LTE FDD, GL, UL, GUL, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GUN (FDD), GULN (FDD), GUMN (FDD), GULMN (FDD)	800 MHz: 2T2R 900 MHz: 2T2R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU3971	1800	MRRU	LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU3971	1900	MRRU	UMTS, LTE FDD, GU, GL, UL, GUL, NR FDD, LN (FDD), GLN (FDD), ULN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU3971	AWS	MRRU	UMTS, LTE FDD, UL, LM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LMN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU3971	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU3971a	1800	MRRU	LTE FDD, GL, GLM, LM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU5251	2300	MRRU	LTE TDD	4T4R	Star, chain, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5254	2300+2600	MRRU	LTE TDD, NR TDD, TN (TDD)	2300 MHz: 4T4R 2600 MHz: 4T4R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5254	2600	MRRU	LTE TDD, NR TDD	8T8R	Star, dual-star, intra-board load sharing, inter-board load sharing
RRU5258	2300	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5258	2600	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5258	3500	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5258	3650	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5258	3700	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, dual-star, chain, intra-board load sharing, inter-board load sharing
RRU5263E	2100+3500	MRRU	2100 MHz: UMTS, LTE FDD, UL, NR FDD, LN (FDD), ULN (FDD) 3500 MHz: NR TDD	2100 MHz: 4T4R 3500 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5265E	3500	MRRU	LTE TDD, NR TDD, TN (TDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing
RRU5265E	3600	MRRU	LTE TDD, NR TDD, TN (TDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU530 1	800	LRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU530 1	1400	LRRU	LTE FDD, NR SDL, LN (SDL)	4T0R	Star, chain, intra-board load sharing, inter-board load sharing
RRU530 1	2600	LRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU530 4	2600	LRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU530 4w	700	LRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU530 4w	2600	LRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU530 5	1400	LRRU	LTE FDD, NR SDL, LN (SDL)	2T2R	Star, chain, intra-board load sharing, inter-board load sharing
RRU530 8	800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	2T2R	Star, chain, dual-star
RRU530 9	800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	2T2R	Star, chain, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5309	700	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	2T2R	Star, chain, dual-star
RRU5309w	800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star
RRU5310	700	MRRU	LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD)	4T4R	Star, chain, intra-board cold backup ring, inter-board cold backup ring, hot backup ring, intra-board load sharing, inter-board load sharing
RRU5336E	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU5501	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU550 2	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU550 2	AWS +PCS	MRRU	UMTS, LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	AWS: 4T4R PCS: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU550 2N	1800+2100	MRRU	LTE FDD, UL, LM, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LMN (FDD), ULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU550 2w	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU550 5	1800+2100	MRRU	LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 2T2R 2100 MHz: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU550 5N	1800+2100	MRRU	LTE FDD, UL, LM, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LMN (FDD), ULMN (FDD)	1800 MHz: 2T2R 2100 MHz: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5507	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), LN (FDD), UN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 2T4R 2100 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5508	700+900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 2T4R 900 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5508	700+850	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 2T4R 850 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5508	800+900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	800 MHz: 2T4R 900 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5508	1800+2100	MRRU	UMTS, LTE FDD, GL, UL, GUL, LN (SUL), ULN (SUL), GLN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD)	1800 MHz: 2T4R 2100 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5509t	700+800+900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 2T4R 800 MHz: 2T4R 900 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5512	700+850	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 850 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, LN (FDD), GN (FDD), UN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD) 700 MHz+850 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 4T4R 850 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5512	700+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD) 700 MHz+900 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 4T4R 900 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU551 2f	700+800, 700+900, 800+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) 700 MHz+800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 700 MHz+900 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD) 800 MHz+900 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL),	700: 4T4R 800: 4T4R 900: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequenc y Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
			ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)		
RRU551 2t	700+800 +900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 4T4R 800 MHz: 4T4R 900 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5513	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5513t	1400+1800+2100	MRRU	1400 MHz: LTE FDD, NR FDD, LN (FDD), NR SDL, LN (SDL) 1800 MHz: LTE FDD, GL, NR FDD, GN (FDD), LN (SUL), GLN (SUL), LN (FDD), GLN (FDD), LM (FDD), GLM (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	1400 MHz: 4T0R 1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU551 3t	1800+2100+2600	MRRU	1800 MHz: LTE FDD, GL, NR FDD, GN (FDD), LN (SUL), GLN (SUL), LN (FDD), GLN (FDD), LM (FDD), GLM (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU551 3w	AWS +PCS	MRRU	AWS: UMTS, LTE FDD, UL, LN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD) AWS+PCS: UMTS, LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	AWS: 4T4R PCS: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5513w	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5515	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800: 2T2R 2100: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU5515t	700+800+900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 2T2R 800 MHz: 2T2R 900 MHz: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5516	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5516	700+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) 700 MHz+900 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 4T4R 900 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5516t	700+800+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) When the RRU5516t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	700: 4T4R 800: 4T4R 900: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU551 6w	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU551 7t	1800+2100+2600	MRRU	1800 MHz: LTE FDD, GL, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LM (FDD), GLM (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU551 7t	AWS +PCS +2600	MRRU	AWS: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	AWS: 4T4R PCS: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU551 7et	1800+2100+2600	MRRU	1800 MHz: LTE FDD, GL, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LM (FDD), GLM (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	1800 MHz: 4T4R 2100 MHz: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5519et	700+800+900	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, LN (SUL), GLN (SUL), ULN (SUL), GULN (SUL), NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GUMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	700 MHz: 2T4R 800 MHz: 2T4R 900 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU5522	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5522 works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5522w	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5522w works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU552 2t	700+800 +900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) When the RRU552t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	700: 4T4R 800: 4T4R 900: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5525t	700+800+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: GSM, UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD) When the RRU5525t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	700 MHz: 2T2R 800 MHz: 2T2R 900 MHz: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequenc y Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU552 6	600+850	MRRU	600 MHz: LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD) 850 MHz: UMTS, LTE FDD, GU, GL, UL, GM, UM, LM, GUM, GLM, ULM, NR FDD, LN (FDD), GN (FDD), UN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD) When the RRU5526 works in the 650 MHz and 850 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	600 MHz: 4T4R 850 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526	700+850	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD) 850 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, LN (FDD), GN (FDD), UN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD) When the RRU5526 works in the 700 MHz and 850 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	700: 4T4R 850: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5526 works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526et	700+800+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) When the RRU5526et works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	700 MHz: 4T4R 800 MHz: 4T4R 900 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526t	600+700+900	MRRU	600 MHz: LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD) 700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) When the RRU5526t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	600: 4T4R 700: 4T4R 900: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526t	600+800+900	MRRU	600 MHz: LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 900 MHz: UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) When the RRU5526t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	600 MHz: 4T4R 800 MHz: 4T4R 900 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526t	1800+2100+2600	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) When the RRU5526t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526t	1400+1800+2100	MRRU	1400 MHz: LTE FDD, NR SDL, LN (SDL) 1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5526t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1400 MHz: 4T0R 1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526w	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5526w works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5526w	AWS+PCS	MRRU	AWS: UMTS, LTE FDD, UL, LM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD) When the RRU5526w works in the AWS and PCS frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	AWS: 4T4R PCS: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5527et	AWS +PCS +2600	MRRU	AWS: UMTS, LTE FDD, UL, LM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) When the RRU5527et works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	AWS: 4T4R PCS: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5527t	1800+2100+2600	MRRU	1800 MHz: LTE FDD, GL, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LM (FDD), GLM (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) When the RRU5527t works in two or three frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 4T4R 2100 MHz: 4T4R 2600 MHz: 4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5529et	700+800+900	MRRU	700 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD) 800 MHz: LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD) 900 MHz: GSM, UMTS, LTE FDD, GU, GL, UL, GUL, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD)	700 MHz: 2T4R 800 MHz: 2T4R 900 MHz: 2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU5810	3600	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU5815	3500	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU5818	2300	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing
RRU5818	2600 (Band 38/n38)	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing
RRU5818	2600 (Band 41/n41)	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, intra-board load sharing, inter-board load sharing, dual-star
RRU5818	3500	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5818w	3500	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5827	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5827 works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 8T8R 2100 MHz: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU5829	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) When the RRU5829 works in the 1800 MHz and 2100 MHz frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	1800 MHz: 8T8R 2100 MHz: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU583 6E	3500 (Band 42/n78)	MRRU	LTE TDD, NR TDD	4T4R	Star, chain
RRU583 6E	3500 (n78)	MRRU	NR TDD	4T4R	Star, chain
RRU583 6E	3700	MRRU	LTE TDD, NR TDD	4T4R	Star, chain
RRU586 2	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 8T8R 2100 MHz: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5862	AWS+PCS	MRRU	AWS: UMTS, LTE FDD, UL, LM, ULM, NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GUN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) AWS+PCS: UMTS, LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	AWS: 8T8R PCS: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5865	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: • GSM, 4T4R • LTE FDD/NR FDD, 8T8R 2100 MHz: • UMTS, 2T4R • LTE FDD/NR FDD, 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5866	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 8T8R 2100 MHz: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequenc y Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU586 6	AWS +PCS	MRRU	AWS: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, LTE FDD, GU, GL, UL, GUL, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), GUN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD) AWS+PCS: UMTS, LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	AWS: 8T8R PCS: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5867	1800+2100	MRRU	1800 MHz: LTE FDD, GL, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) 1800 MHz+2100 MHz: LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 8T8R 2100 MHz: 8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring, hot backup ring
RRU5872	2600	MRRU	LTE TDD, NR TDD, TN (TDD)	8T8R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5901	1800	MRRU	LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5901	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5901	AWS	MRRU	UMTS, LTE FDD, UL, LM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5901N	1800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5901w	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5903	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GLMN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5903	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), GUMN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5903	1900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, UM, LM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5903w	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GLM, ULM, GUM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), UMN (FDD), GLMN (FDD) ^a a: When the RRU5903w works in the RX frequency band of 814 MHz to 824 MHz or the TX frequency band of 859 MHz to 869 MHz, it cannot work in GSM, UMTS, or NR FDD mode.	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5904	2100	MRRU	UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LM	4T4R	Star, chain, dual-star
RRU5904	1800	MRRU	GSM, LTE FDD, GL, LTE NB-IoT, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU590 4N	1800	MRRU	LTE FDD, LTE NB-IoT, LM, LN (SUL), NR FDD, LN (FDD), MN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star
RRU590 4w	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GLM, ULM, GUM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), UMN (FDD), GLMN (FDD) ^a a: When the RRU5904w works in the RX frequency band of 814 MHz to 824 MHz or the TX frequency band of 859 MHz to 869 MHz, it cannot work in GSM, UMTS, or NR FDD mode.	4T4R	Star, chain, dual-star
RRU590 4w	1800	MRRU	GSM, LTE FDD, GL, LTE NB-IoT, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD)	4T4R	Star, chain, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU590 4w	1900	MRRU	UMTS, LTE FDD, GU, GL, UL, GUL, LM ^a , GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD) a: To support LTE NB-IoT on the RRU5904 (1900 MHz), the two LTE FDD cells must use contiguous spectrum resources.	4T4R	Star, chain, dual-star
RRU590 5	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU590 5	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU590 5	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), ULM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5905w	1800	MRRU	GSM, LTE FDD, GL, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5909	700	MRRU	LTE FDD, LM, NR SUL, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5909	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), GUN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5909	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD), MN (FDD), GMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5909	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), ULM UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), ULM, LM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU590 9c	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU590 9N	1800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU590 9s	1800	MRRU	LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU590 9s	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU591 0	800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD), MN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5910	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), LMN (FDD), UMN (FDD), GLMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5910	1800	MRRU	GSM, LTE FDD, GL, LTE NB-IoT, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5911	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5913	850	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GLMN (FDD)	2T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5915	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), ULN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5915	1800	MRRU	GSM, LTE FDD, GL, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5915	2100	MRRU	UMTS, LTE FDD, UL, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), ULM, LM	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5916	1800	MRRU	GSM, LTE FDD, GL, LTE NB-IoT, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLN (FDD), GMN (FDD), GLMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5916s	1400	MRRU	LTE FDD, NR SDL, LN (SDL)	4T0R	Star, chain, intra-board load sharing, inter-board load sharing
RRU5919	700	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5919	800	MRRU	LTE FDD, LM, LN (SUL), NR FDD, LN (FDD), LMN (FDD)	2T2R	Star, chain, dual-star
RRU5919	900	MRRU	GSM, UMTS, LTE FDD, GU, GL, UL, GUL, LTE NB-IoT, GM, UM, LM, GUM, GLM, ULM, GULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUN (FDD), GLN (FDD), MN (FDD), GMN (FDD), UMN (FDD), LMN (FDD), GUMN (FDD)	2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing, intra-board cold backup ring, inter-board cold backup ring
RRU5920	600	MRRU	LTE FDD, LM, NR FDD, LN (FDD), LMN (FDD)	4T4R	Star, chain, dual-star
RRU5920	2100	MRRU	UMTS, LTE FDD, UL, ULM, LN (SUL), ULN (SUL), NR FDD, UN (FDD), LN (FDD), ULN (FDD), LM	4T4R	Star, chain, dual-star
RRU5920w	1800	MRRU	GSM, LTE FDD, GL, LTE NB-IoT, GM, LM, GLM, LN (SUL), GLN (SUL), NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5920w	2600	MRRU	LTE FDD, NR FDD, LN (FDD)	4T4R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
RRU5935E	1800+2100	MRRU	LTE FDD, GL, UL, GUL, LM, GLM, ULM, GULM, NR FDD, LN (FDD), GLN (FDD), ULN (FDD), GULN (FDD), LMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	1800 MHz: 2T2R 2100 MHz: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing
RRU5935E	AWS+PCS	MRRU	AWS: UMTS, LTE FDD, UL, NR FDD, UN (FDD), LN (FDD), ULN (FDD) PCS: UMTS, GU, LTE FDD, GL, UL, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), GUL, GUN (FDD), GLN (FDD), ULN (FDD), LMN (FDD), GULM, GLMN (FDD), ULMN (FDD) When the RRU5935E works in the AWS and PCS frequency bands, the supported RATs are the combinations of the RATs supported in each single frequency band.	AWS: 2T2R PCS: 2T2R	Star, chain, dual-star, intra-board load sharing, inter-board load sharing

Table 4-9 Basic LampSite RRU information

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU3901	ALL	MPMU	GSM, UMTS, LTE FDD, LTE TDD, UL, GU	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE TDD: 2T2R	Star, branch load sharing
pRRU3902	ALL	MPMU	GSM, UMTS, LTE FDD, UL, GU, GL, LTE NB-IoT, LM, ULM, GLM	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R	Star
pRRU3907	1800+2100	MPMU	GSM, UMTS, LTE FDD, UL, GU, GL, LTE NB-IoT, LM, ULM, GLM	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R	Star, branch load sharing
pRRU3911	ALL	MPMU	CDMA, GSM, UMTS, LTE FDD, LTE TDD, UL, GU, GL, GUL, CU, CL, CUL, LTE NB-IoT, LM, ULM, GLM, GULM	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE TDD, LTE NB-IoT: 2T2R CDMA: 1T1R	Star, branch load sharing
pRRU3912	900+1800+2100+2600	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, UL, LM 1800 MHz: GSM, LTE FDD, LTE NB-IoT, LM 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, LM 2600 MHz: LTE FDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R CDMA: 1T1R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU3916	850+AWS+PCS+2600	MPMU	850 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, LM AWS: UMTS, LTE FDD, UL PCS: UMTS, LTE FDD, LTE NB-IoT, UL, LM 2600 MHz: LTE FDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R CDMA: 1T1R	Star, branch load sharing
pRRU3916	800+1800+2100+2600	MPMU	800 MHz: LTE FDD, LTE NB-IoT, LM 1800 MHz: GSM, LTE FDD, LTE NB-IoT, LM 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, LM 2600 MHz: LTE FDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R CDMA: 1T1R	Star, branch load sharing
pRRU3916	900+1800+2100+2600	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, UL, LM 1800 MHz: GSM, LTE FDD, LTE NB-IoT, LM 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, LM 2600 MHz: LTE FDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT: 2T2R CDMA: 1T1R	Star, branch load sharing
pRRU5551JR	26000	EPRRU	NR TDD	NR TDD: 2T2R	Star
pRRU5631CR	3500	MPMU	NR TDD	NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU563 3GR	1800+2100+3500	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU563 3GR	1800+2100+2600	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE TDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU563 3GR	1800+2100+3700	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3700 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU563 3GR	AWS+PCS+3500	MPMU	AWS: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) PCS: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3500 MHz: NR TDD	UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU565 3GR	1800+2100+2300	EPRRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2300 MHz: LTE TDD, NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT, LTE TDD, NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU565 3GR	1800+2100+2600	EPRRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GN (FDD), LM, LN (FDD), GLM, GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT, LTE TDD, NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU573 1GH	3500	MPMU	NR TDD	4T4R	Star

Model	Frequenc y Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU573 1GR	3500	MPMU	NR TDD	4T4R	Star
pRRU573 3GR	1800+210 0+2600	MPMU	1800 MHz: GSM, LTE FDD, LTE NB- IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB- IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	GSM: 1T1R UMTS: 1T1R/ 1T2R LTE FDD, LTE NB-IoT, NR FDD: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU573 3GR	1800+2100+3500	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD, LTE NB-IoT, NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU575 3GR	1800+2100+3500	EPRRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), GLM, GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UN (FDD), LM, LN (FDD), ULM, ULN (FDD), LMN (FDD), ULMN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU575 4GR	AWS+PCS +2600 MHz +3500 MHz	EPRRU	AWS: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UN (FDD), LM, LN (FDD), ULM, ULN (FDD), LMN (FDD), ULMN (FDD) PCS: GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GN (FDD), UL, UN (FDD), LM, LN (FDD), GUL, GUN (FDD), GLM, GLN (FDD), ULM, ULN (FDD), LMN (FDD), GULM, GULN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU575 5GR	900+1800 +2100+2600+3500	EPRRU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GN (FDD), UL, UN (FDD), LM, LN (FDD), GUL, GUN (FDD), GLM, GLN (FDD), ULM, ULN (FDD), LMN (FDD), GULM, GULN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GN (FDD), LM, LN (FDD), GLM, GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UN (FDD), LM, LN (FDD), ULM, ULN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 900 MHz: 1T1R/2T2R 1800 MHz/2100 MHz/2600 MHz: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU575 5GR	1800+2100+2300+2600+3500	EPRRU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GN (FDD), LM, LN (FDD), GLM, GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UN (FDD), LM, LN (FDD), ULM, ULN (FDD), LMN (FDD), ULMN (FDD) 2300 MHz: LTE TDD, NR TDD, TN (TDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R LTE TDD: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU575 5GT	900+1800 +2100+2600+3500	EPRRU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GN (FDD), UL, UN (FDD), LM, LN (FDD), GUL, GUN (FDD), GLM, GLN (FDD), ULM, ULN (FDD), LMN (FDD), GULM, GULN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GN (FDD), LM, LN (FDD), GLM, GLN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UN (FDD), LM, LN (FDD), ULM, ULN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 900 MHz: 1T1R/2T2R 1800 MHz/2100 MHz/2600 MHz: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5913	900+1800+2100+2600	MPMU	GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GUL, GUM, GUN (FDD), GLM, GLN (FDD), GMN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), GULM, GULN (FDD), GUMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, NR FDD: 900 MHz: 1T1R/2T2R 1800 MHz/2100 MHz/2600 MHz: 2T2R LTE NB-IoT: 2T2R	Star
pRRU5918	900+1800+2100+2600	MPMU	GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GUL, GUM, GUN (FDD), GLM, GLN (FDD), GMN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), GULM, GULN (FDD), GUMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, NR FDD: 900 MHz: 1T1R/2T2R 1800 MHz/2100 MHz/2600 MHz: 2T2R LTE NB-IoT: 2T2R	Star
pRRU5920	1800+2100	MPMU	GSM, UMTS, LTE FDD, UL, LTE NB-IoT, GM, LM, ULM, GL, GLM	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, LTE NB-IoT: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5920	1800+2300	MPMU	GSM, LTE FDD, LTE TDD, LTE NB-IoT, GM, LM, GL, GLM	GSM: 1T1R LTE FDD, LTE NB-IoT: 2T2R	Star
pRRU5921	900+1800+2100+2300	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, GU, GL, GM, UL, UM, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), MN (FDD), UMN (FDD), GMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), MN (FDD), GMN (FDD), LMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), MN (FDD), UMN (FDD) 2300 MHz: LTE TDD, NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE TDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R	Star, chain, load sharing

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5923	900+1800+2100+2600	MPMU	GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GUL, GUM, GUN (FDD), GLM, GLN (FDD), GMN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), GULM, GULN (FDD), GUMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, NR FDD: 900 MHz: 1T1R/2T2R 1800 MHz/2100 MHz/2600 MHz: 2T2R LTE NB-IoT: 2T2R	Star
pRRU5923	700+850+1800+2300	MPMU	GSM, UMTS, LTE TDD, NR TDD, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GUL, GUM, GUN (FDD), GLM, GLN (FDD), GMN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), GULM, GULN (FDD), GUMN (FDD), GLMN (FDD), ULMN (FDD), GULMN (FDD)	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, NR FDD: 850 MHz: 1T1R/2T2R 700 MHz/1800 MHz: 2T2R LTE NB-IoT, LTE TDD: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5923	850+PCS+AWS+2600	MPMU	UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD)	UMTS: 1T2R/1T1R LTE FDD, NR FDD: 850 MHz: 2T2R 2600 MHz: 2T2R LTE NB-IoT: 2T2R	Star
pRRU5923	850+AWS+PCS+2600	MPMU	850 MHz/AWS/PCS: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN FDD	UMTS: 1T1R/1T2R LTE FDD: 850 MHz: 1T1R/2T2R - AWS/PCS/2600 MHz: 2T2R LTE NB-IoT: 2T2R NR FDD: 850 MHz: 2T2R 2600 MHz: 2T2R	Star
pRRU5928	850+AWS+PCS+2600	MPMU	850 MHz/AWS/PCS: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE FDD, NR FDD, LN (FDD)	UMTS: 1T1R/1T2R LTE FDD: 850 MHz: 1T1R/2T2R - AWS/PCS/2600 MHz: 2T2R LTE NB-IoT: 2T2R NR FDD: 850 MHz: 1T1R/1T2R - AWS/PCS/2600 MHz: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5935	3500/3700	MPMU	NR TDD	4T4R	3500 MHz: Star, load sharing 3700 MHz: Star
pRRU5935E	3500	MPMU	NR TDD	4T4R	Star, load sharing
pRRU5935H	3500	MPMU	NR TDD	4T4R	Star
pRRU5936	1800+2100+3500	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), ULN (FDD), MN (FDD), UMN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5961	1800+2100+3500	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), ULN (FDD), MN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3500 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star
pRRU5961	1800+2100+3700	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), ULN (FDD), MN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3700 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5961	900+1800+2100+2300	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, GU, GL, GM, UL, UM, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), MN (FDD), UMN (FDD), LMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), ULN (FDD), MN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2300 MHz: LTE TDD, NR TDD	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE TDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5961	900+1800+2100+2600	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, GU, GL, GM, UL, UM, LM, GLM, ULM, NR FDD, GN (FDD), UN (FDD), LN (FDD), MN (FDD), UMN (FDD), LMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, GL, GM, LM, GLM, NR FDD, GN (FDD), LN (FDD), GLN (FDD), MN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, UL, UM, LM, ULM, NR FDD, UN (FDD), ULN (FDD), MN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	GSM: 1T1R UMTS: 1T1R/1T2R LTE FDD: 2T2R LTE TDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 2T2R	Star

Model	Frequenc y Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU596 1E	1800+210 0+3700	MPMU	1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 3700 MHz: NR TDD	GSM: 1T1R UMTS: 1T1R/ 1T2R LTE FDD: 2T2R LTE NB-IoT: 2T2R NR FDD: 2T2R NR TDD: 4T4R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5961E	900+1800+2100+2300	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GLM, GMN (FDD), ULM, UMN (FDD), LMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2300 MHz: LTE TDD, LN (TDD), NR TDD	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, LTE TDD, LTE NB-IoT, NR FDD, NR TDD: 2T2R	Star

Model	Frequency Band (MHz)	Logical Type	Supported RAT	TX/RX Mode	Supported Topology
pRRU5961E	900+1800+2100+2600	MPMU	900 MHz: GSM, UMTS, LTE FDD, LTE NB-IoT, NR FDD, GU, GL, GM, GN (FDD), UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), GLM, GMN (FDD), ULM, UMN (FDD), LMN (FDD) 1800 MHz: GSM, LTE FDD, LTE NB-IoT, NR FDD, GL, GM, GN (FDD), LM, LN (FDD), MN (FDD), GLM, GLN (FDD), GMN (FDD), LMN (FDD), GLMN (FDD) 2100 MHz: UMTS, LTE FDD, LTE NB-IoT, NR FDD, UL, UM, UN (FDD), LM, LN (FDD), MN (FDD), ULM, ULN (FDD), UMN (FDD), LMN (FDD), ULMN (FDD) 2600 MHz: LTE TDD, NR TDD, TN (TDD)	GSM: 1T1R UMTS: 1T2R/1T1R LTE FDD, LTE TDD, LTE NB-IoT, NR FDD, NR TDD: 2T2R	Star
pRRU5963	3500	MPMU	NR TDD	4T4R	Star
pRRU5973	3500	MPMU	NR TDD	4T4R	Star
pRRU5973H	3500	MPMU	NR TDD	4T4R	Star

4.2 Delay Requirements

The relationship between the maximum delay and the maximum distance between a BBU and an RF module is as follows: Maximum delay = Maximum distance between the BBU and the RF module (km) x 5 μ s/km.

For example, if the maximum distance between the BBU and the RF module is 20 km, the maximum transmission delay between them is 100 μ s.

NOTE

- For details about the maximum distance between a BBU and an RF module, see the technical specification document of the corresponding RF module in *3900 & 5900 Series Base Station Product Documentation*.
- When the BBU and RF module are connected through transmission equipment, the supported maximum distance between the BBU and the RF module will be shorter than the theoretical maximum distance as extra delay will be introduced by the transmission equipment. If the actual distance exceeds the supported maximum distance, the cell servicing capabilities, such as uplink and downlink throughputs and access success rate, will deteriorate.

4.3 Network Analysis

4.3.1 Benefits

None

4.3.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

This function has no impact on the network.

4.4 Requirements

4.4.1 Licenses

None

4.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.2.1 Prerequisite Functions

None

4.4.2.2 Mutually Exclusive Functions

None

4.4.2.3 Function Impacts

None

4.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

The BTS5929R does not support Multi-mode BS RRU/RFU star-connection with separate CPRI Interface(eNodeB).

Boards

A limited number of CPRI ports on the BBU are supported by different main control boards.

- When the GTMUc/UMDU/UMPTb is used, a maximum of 36 CPRI ports on the BBU are supported.
- When the UMPTe/UMPTg/UMPTga/UMPTHa is used, a maximum of 216 CPRI ports on the BBU are supported.

NOTICE

If the hot backup ring or trunk chain topology is used, CPRI optical fibers of the same length are required.

RF Modules

RF modules are installed and connected to the BBU in a specific topology. The chain/ring configurations of the RF modules are consistent with the actual cable connections. There are no redundant optical or electrical modules and non-connected cables on the optical or electrical ports on the lowest level RF module. For details about how to install the RF modules, see the installation guide for the corresponding base station and RF module. Use the optical or electrical modules that are delivered with the RF modules and confirm that the optical or electrical module rate conforms to the network planning.

If the topology structure needs to be adjusted on the live network (that is, the CPRI port used by the BBU or RF module is changed), perform network reconstruction after the RF modules are powered off. Do not hot swap the RF modules. If the topology does not need to be adjusted, for example, the optical module or optical fiber on the same CPRI port needs to be replaced, the optical module or optical fiber can be hot-swapped.

Adjusting the RF module topology by hot swapping the RF modules will lead to the following problems: RF module service interruption, incorrect configuration relationship between the cell and the RF module, inter-RF module communication conflict, inconsistency between the CPRI rate and the planned value, and RF module failure in startup.

 NOTE

- In hot swap mode, the optical fiber, electrical cable, optical module, or electrical module is connected to or removed from the CPRI port on the RF module or BBU when the RF module is running.
- If there are redundant optical or electrical modules and non-connected cables on the optical or electrical ports on the RF modules, the CPRI line rate on the chain/ring may be inconsistent with the planned value.

4.4.4 Others

None

4.5 Operation and Maintenance (GBTS)

4.5.1 Precautions

N/A

4.5.2 Data Configuration

Comply with the following rules when single-mode or multimode RF modules work in single-mode scenarios:

- The slot numbers of RFUs that work in different modes in the same RFU subrack must be different.
- Before adding an RXU, the RXU chain/ring must be added.

4.5.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/RFU/AAU model: Run the **DSP BTSELABEL** command to query the electronic label of the RRU/RFU/AAU.
 - RRU/RFU/AAU topology: Use it to determine how to configure RRU/RFU/AAU CPRI links.
1. The following table describes key parameters in a **BTSRXUCHAIN** MO to configure an RXU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Chain No.	BTSRXUCHAIN.R CN	This parameter uniquely identifies a chain/ring in a base station.
Topo Type	BTSRXUCHAIN.T T	• In the star or chain topology, set this parameter to CHAIN(CHAIN). In this case, the TCN, TSRN, TSN, and TPN parameters are invalid. In the star topology, only one RRU/RFU/AAU can be configured on a chain. • In the ring topology, set this parameter to RING(RING) and specify two CPRI ports.
Head Cabinet No.	BTSRXUCHAIN.H CN	Recommended configurations are as follows: • Cabinet number: 0 to 7. • Subrack number: 0 or 1. • When the GTMU is used to connect an RRU/RFU/AAU, the slot number must be set to 6. • When the UBRI is used to connect an RRU/RFU/AAU, the slot number must be set to 2. • When the UBBP is used to connect an RRU/RFU/AAU, the slot number is preferentially set to 1. • Optical port number: 0 to 5. The values of Head Port No. and Tail Port No. must be different.
Head Subrack No.	BTSRXUCHAIN.H SRN	
Head Slot No.	BTSRXUCHAIN.H SN	
Head Port No.	BTSRXUCHAIN.H PN	
Tail Cabinet No.	BTSRXUCHAIN.T CN	
Tail Subrack No.	BTSRXUCHAIN.T SRN	
Tail Slot No.	BTSRXUCHAIN.T SN	
Tail Port No.	BTSRXUCHAIN.T PN	

2. The following table describes key parameters in a **BTSRXUBRD** MO to add an RXU board.

Parameter Name	Parameter ID	Setting Notes
Board Type	BTSRXUBRD.BT	Set this parameter based on the type of the in-use RF module. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules .
RXU Specification	BTSRXUBRD.RXU SPEC	You are advised to set this parameter to BYNAME(By Name) . To configure a specific specification, set this parameter according to the value of RXU Specification Enumerated Description in the LST BTSINTRXUSPEC command output.
RXU Spec Name	BTSRXUBRD.RXU SPECNAME	Specify this parameter when RXU Specification is set to BYNAME(By Name) . Set this parameter according to the value of RXU Specification Enumerated Name in the LST BTSINTRXUSPEC command output.
RXU Spec Value	BTSRXUBRD.RXU SPECVALUE	Specify this parameter when RXU Specification is set to BYVALUE(By Value) . Set this parameter according to the value of RXU Specification Enumerated Value in the LST BTSINTRXUSPEC command output.
Cabinet No.	BTSRXUBRD.CN	- This parameter must be set to 0 for an AAU. - This parameter can be set to 0 for an RRU. - This parameter can be set to a value ranging from 0 to 62 for an RFU.
Subrack No.	BTSRXUBRD.SRN	- This parameter can be set to a value ranging from 60 to 254 for an RRU/AAU. - This parameter can be set to 4 or 5 for an RFU in a BTS3900L/BTS5900L cabinet or a virtual cabinet, and must be set to 4 for an RFU in any other cabinet.
Slot No.	BTSRXUBRD.SN	- This parameter must be set to 0 for an RRU/AAU. - This parameter can be set to a value ranging from 0 to 8 for an RFU in a BTS3900AL cabinet or a virtual cabinet, and can be set to a value ranging from 0 to 5 for an RFU in any other cabinet.
RXU Chain No.	BTSRXUBRD.RXU CHAINNO	Set this parameter based on the actual physical connection.
RXU Board Position	BTSRXUBRD.RXU POS	This parameter specifies the position of an RRU/RFU/AAU on a chain/ring. The value of this parameter is set sequentially from 1. The value of this parameter for an RRU/RFU/AAU must differ from that for any other RRU/RFU/AAU configured on the same chain/ring. In the chain topology, the end close to the BBU is the chain head, and the sequence number starts from 1.

Parameter Name	Parameter ID	Setting Notes
Power Class	BTSRXUBRD.PW RMODE	Set this parameter based on the specified power mode.
Receive Frequency Band Mutual Aid Switch	BTSRXUBP.RxFre qBandMutualSw	This parameter is valid only for RF modules whose hardware capabilities support inter-frequency mutual aid, such as the RRU3952m. Dynamically modifying the value of this parameter for an inter-frequency mutual aid RF module will reset the RF module.

3. The following table describes key parameters in a **BTSRXUBP** MO to configure RXU board parameters.

Parameter Name	Parameter ID	Setting Notes
R XU Chain No.	BTSRXUBP.RXUC HAINNO	Set this parameter based on the actual physical connection.
R XU Name	BTSRXUBP.RXUN AME	This parameter is optional and can be customized to contain 1 to 64 characters.
R XU Board Type	BTSRXUBP.RXUT YPE	Set this parameter based on the type of the in-use RF module. The setting must be consistent with that of the BT parameter. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules .
Working Standard	BTSRXUBP.WOR KINGSTANDARD	When an inter-frequency mutual aid module works in inter-frequency mutual aid mode, and the two RX frequency bands are in different RATs, set this parameter to a value including multiple RATs. For example, if the RRU3952m (1800 MHz) is configured to work in 2T2R in LTE 1800 MHz and 2R in UMTS 2100 MHz, set this parameter to UMTS_AND_LTE .

Scenario-specific Data Preparation

Scenario 1: star topology

- The following table describes key parameters in a **BTSRXUCHAIN** MO to configure an RXU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	BTSRXUCHAIN.T T	Set this parameter to CHAIN(CHAIN) .

In the dual-star topology, set the **TT** parameter to **CHAIN(CHAIN)** for each RAT of the separate-MPT base station.

- The following table describes key parameters in a **BTSRXUBRD** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RXU Chain No.	BTSRXUBRD.RXU CHAINNO	Set this parameter to the actual chain/ring number.
RXU Board Position	BTSRXUBRD.RXU POS	Set this parameter to 1 .

Scenario 2: chain topology

- The following table describes key parameters in a **BTSRXUCHAIN** MO to configure an RXU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	BTSRXUCHAIN.T T	Set this parameter to CHAIN(CHAIN) .

- The following table describes key parameters in a **BTSRXUBRD** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RXU Chain No.	BTSRXUBRD.RXU CHAINNO	Set this parameter to the actual chain/ring number.
RXU Board Position	BTSRXUBRD.RXU POS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 1 for the first RRU/RFU/AAU and 2 for the second RRU/RFU/AAU.

Scenario 3: ring topology

- The following table describes key parameters in a **BTSRXUCHAIN** MO to configure an RXU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	BTSRXUCHAIN.T T	Set this parameter to RING(RING) .

- The following table describes key parameters in a **BTSRXUBRD** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RXU Chain No.	BTSRXUBRD.RXU CHAINNO	Set this parameter to the actual chain/ring number.

Parameter Name	Parameter ID	Setting Notes
RXU Board Position	BTSRXUBRD.RXU POS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 1 for the first RRU/RFU/AAU and 2 for the second RRU/RFU/AAU.

4.5.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

```
//Adding an RXU chain
ADD BTSRXUCHAIN: IDTYPE=BYID, BTSID=9, RCN=0, TT=CHAIN, HCN=0, HSRN=0, HSN=6, HPN=0;

//Adding an RXU ring
ADD BTSRXUCHAIN: IDTYPE=BYID, BTSID=9, RCN=0, TT=RING, HCN=0, HSRN=0, HSN=6, HPN=1, TCN=0,
TSRN=0, TSN=6, TPN=2;

//(Optional) Configuring the breakpoints
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=1, BP1=0, BP2=0;

//Adding an RXU board
ADD BTSRXUBRD: IDTYPE=BYID, BTSID=9, BT=MRRU, RXUSPEC=BYNAME, RXUSPECNAME= "RRU3929",
CN=0, SRN=60, SN=0, RXUNAME="RRU_1", RXUCHAINNO=0, RXUPOS=1;

//Setting RXU board parameters
SET BTSRXUBP: IDTYPE=BYID, BTSID=9, RXUIDTYPE=SRNSN, CN=0, SRN=60, SN=0, RXUTYPE=MRRU,
SndRcvMode2=DOUBLE_ANTENNA;

//(Optional) Removing the breakpoints
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=1, BP1=NULL, BP2=NULL;
```

4.5.2.3 Using the MAE-Deployment

- For detailed operations of configuring a single NE, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide: GBTS Initial Configuration Guide (MAE-Deployment-based) > Creating a Single GBTS > Configuring GBTS Device Data > Configuring RF Units*.
- For detailed operations of configuring NEs in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide: GBTS Initial Configuration Guide (MAE-Deployment-based) > Creating GBTSs in Batches > Preparing GBTS Data > Device Data for a GBTS > RF Unit or AAU Data*.

4.5.3 Activation Verification

After configuration data is activated, ensure that no RRU/RFU/AAU related alarm is displayed on the **Alarm/Event** tab page on the BSC LMT.

On the BSC LMT or MAE-Access, run the **DSP BTSBRD** command to check the status of the RRU/RFU/AAU.

The status of the RRU/RFU/AAU is normal if the value of **Board state** is **Normal**.

Run the **DSP BTSCPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

4.5.4 Network Monitoring

N/A

4.5.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps.

MML-based Adjustment

After the adjustment, run the **DSP BTSCPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

Adjusting an RXU Topology

An RXU topology is adjusted by modifying the connection mode of RXUs during capacity expansion. In GSM mode, topology adjustment includes splitting one RXU ring into one or more RXU chains, combining two RXU chains into one RXU ring, and changing the RXU chain to the RXU ring.

- To split one RXU ring into one or more RXU chains, perform the following operations:

Step 1 Run the **SET BTSBREAKPOINT** command to configure breakpoints on an RXU ring.

- If you want to split one RXU ring into one RXU chain, set **Position of Break Point 1** and **Position of Break Point 2** to the ring tail.
- If you want to split one RXU ring into two RXU chains, set **Position of Break Point 1** and **Position of Break Point 2** to the split position on the ring.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=1, BP1=2, BP2=2;
```

Step 2 Run the **SPT BTRSXRUCHAIN** command to split the RXU ring into one or more RXU chains.

If you want to split one RXU ring into one RXU chain, set **Separate Position** to the ring tail. If you want to split one RXU ring into two RXU chains, set **Separate Position** to the split position on the ring.

```
SPT BTRSXRUCHAIN: IDTYPE=BYID, BTSID=9, RCN=1, SEPARATEPOS=2;
```

Step 3 Run the **SET BTSBREAKPOINT** command to remove the breakpoints configured in step 1. In this step, set **Position of Break Point 1** and **Position of Break Point 2** to **NULL(NULL)**.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=1, BP1=NULL, BP2=NULL;
```

----End

- To combine two RXU chains into one RXU ring, perform the following operations:

- Step 1** Run the **DSP BTSCPRILBR** command to check whether values of **Current CPRI Line Bit Rate** of the two RXU chains are the same. If they are not the same, go to step 3 after the combination. If they are the same, skip step 3.

```
DSP BTSCPRILBR: IDTYPE=BYID, BTSID=9, RCN=0;
DSP BTSCPRILBR: IDTYPE=BYID, BTSID=9, RCN=2;
```

- Step 2** Run the **CBN BTSRXUCHAIN** command to combine two RXU chains into one RXU ring. After the two RXU chains are combined successfully, the value of **Chain No.** of the newly created ring is the same as that of **First RXU Chain No.**

```
CBN BTSRXUCHAIN: IDTYPE=BYID, BTSID=9, CHAINTORING=COMBINECHAINS, RCN1=0, RCN2=2;
```

- Step 3** (Optional) Run the **STR BTSCPRILBRNEG** command to perform CPRI line rate negotiation.

```
STR BTSCPRILBRNEG: IDTYPE=BYID, BTSID=9, RCN=0;
```

----End

- To change the RXU chain to the RXU ring, perform the following operations:

- Step 1** Run the **CBN BTSRXUCHAIN** command to change the RXU chain to the RXU ring.

```
CBN BTSRXUCHAIN: IDTYPE=BYID, BTSID=9, CHAINTORING=CHAINTORING, RCN1=0, TCN=0, TSN=0, TSN=6, TPN=4;
```

----End

Adding an RXU to a Chain/Ring

An RXU can be added to any position on an RXU chain/ring without changing the original topology.

- Step 1** (Optional) Run the **SET BTSBREAKPOINT** command to configure the breakpoint(s) for adding an RXU to the head or middle of a chain/ring. The breakpoint position is the same as the position of the added RXU on the chain/ring.

- For the RXU chain, configure one breakpoint with **Position of Break Point 1** specified.
- For the RXU ring, configure two breakpoints with **Position of Break Point 1** and **Position of Break Point 2** set to the same value.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=0, BP1=1;
```

NOTE

If a breakpoint is configured on the RXU chain/ring, the RXU boards following the breakpoint will not provide services.

- Step 2** Run the **ADD BTSRXUBRD** command to add an RXU.

```
ADD BTSRXUBRD: IDTYPE=BYID, BTSID=9, BT= MRRU, RXUSPEC=BYNAME,
RXUSPECNAME="RRU3929",CN=0, SRN=61, SN=0, RXUENAME=" RRU_2", RXUCHAINNO=0, RXUPOS=2;
```

- Step 3** Run the **SET BTSRXUBP** command to configure RXU board parameters.

```
SET BTSRXUBP: IDTYPE=BYID, BTSID=9, RXUIDTYPE=SRNSN, CN=0, SRN=61, SN=0, RXUTYPE=MRRU,
SndRcvMode2=DOUBLE_ANTENNA;
```

- Step 4** (Optional) Run the **SET BTSBREAKPOINT** command to remove the breakpoint(s) configured in step 1. In this step, set **Position of Break Point 1** and **Position of Break Point 2** to **NULL(NULL)**.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=0, BP1=NULL, BP2=NULL;
```

----End

Removing an RXU from a Chain/Ring

Step 1 (Optional) Run the **SET BTSBREAKPOINT** command to configure the breakpoint(s) for removing an RXU from the head or middle of a chain/ring. The breakpoint position is the same as the position of the removed RXU on the chain/ring.

- For the RXU chain, configure one breakpoint with **Position of Break Point 1** specified.
- For the RXU ring, configure two breakpoints with **Position of Break Point 1** and **Position of Break Point 2** set to the same value.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=0, BP1=1;
```

NOTE

If a breakpoint is configured on the RXU chain/ring, the RXU boards following the breakpoint will not provide services.

Step 2 Run the **RMV BTRSXBURD** command to remove the specified RXU.

```
RMV BTRSXBURD: IDTYPE=BYID, BTSID=9, RXUIDTYPE=RXUPOS, RXUCHAINNO=0, RXUPOS=2;
```

Step 3 (Optional) Run the **SET BTSBREAKPOINT** command to remove the breakpoint(s) configured in step 1. In this step, set **Position of Break Point 1** and **Position of Break Point 2** to **NULL(NULL)**.

```
SET BTSBREAKPOINT: IDTYPE=BYID, BTSID=9, RCN=0, BP1=NULL, BP2=NULL;
```

----End

Modifying the RXU Type

Step 1 Run the **SET BTRSXBUP** command to modify the value of **RXU Board Type**.

```
SET BTRSXBUP: IDTYPE=BYID, BTSID=9, RXUIDTYPE=SRNSN, CN=0, SRN=60, SN=0, RXUTYPE=MRRU;
```

NOTE

- The value ranges of **Cabinet No.**, **Subrack No.**, and **Slot No.** for the RRU differ from those for the RFU. Therefore, the value of **RXU Board Type** for an RRU cannot be changed to the value of **RXU Board Type** for an RFU and the value of **RXU Board Type** for an RFU cannot be changed to the value of **RXU Board Type** for an RRU.
- The value of **RXU Board Type** for the pRRU cannot be modified.

----End

Modifying the RXU Working Standard

Before the modification, ensure that the RXU hardware supports the target working standard. For details about the RXU hardware capability, see the **Supported RAT** columns of the tables in [4.1.2 RF Modules](#). In addition, ensure that the license for the target working standard has been purchased.

Step 1 Run the **SET BTRSXBUP** command with **Working Standard** set to the target working standard.

```
SET BTRSXBUP: IDTYPE=BYID, BTSID=9, RXUIDTYPE=SRNSN, CN=0, SRN=60, SN=0, RXUTYPE=MRRU,  
WORKINGSTANDARD=GSM;
```

 NOTE

Working standard of the pRRU cannot be modified.

----End

MAE-Deployment-based Adjustment

- **Adjusting an RXU topology**

For details about the MAE-Deployment-based adjustment, see **Reconfiguring a GBTS > Changing CPRI Networking of a 3900 Series Base Station > Changing RXU Chains to an RXU Ring and Changing an RXU Ring to an RXU Chain** in *GBSS Reconfiguration Guide*.

- **Adding an RXU to a chain/ring**

For details about the MAE-Deployment-based adjustment, see **Reconfiguring a GBTS > Adding an RF Module** in *GBSS Reconfiguration Guide*.

- **Removing an RXU from a chain/ring**

For details about the MAE-Deployment-based adjustment, see **Reconfiguring a GBTS > Removing an RF Module** in *GBSS Reconfiguration Guide*.

4.6 Operation and Maintenance (eGBTS)

4.6.1 Precautions

None

4.6.2 Data Configuration

Comply with the following rules when single-mode or multimode RF modules work in single-mode scenarios:

- The slot numbers of RFUs that work in different modes in the same RFU subrack must be different.
- Before adding an RRU/RFU/AAU, the RRU chain/ring must be added.

4.6.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/RFU/AAU model: Run the **DSP ELABEL** command to query the electronic label of the RRU/RFU/AAU.
 - RRU/RFU/AAU topology: Use it to determine how to configure RRU/RFU/AAU links.
1. The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Chain No.	RRUCHAIN.RCN	This parameter uniquely identifies a chain/ring in a base station.
Topo Type	RRUCHAIN.TT	• In the star or chain topology, set this parameter to CHAIN(CHAIN). In this case, the TCN, TSRN, TSN, and TPN parameters are invalid. In the star topology, only one RRU/RFU/AAU can be configured on a chain. • In the ring topology, set this parameter to RING(RING) and specify two CPRI ports and the backup mode.
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	When this parameter is set to LOCALPORT(local port) , the BBP is connected to the RRU/RFU/AAU. When this parameter is set to PEERPORT(peer port) , the local BBP (of one RAT) is connected to the peer BBP (of another RAT) through the virtual CPRI subchannel on the backplane, and the peer BBP is then connected to the RRU/RFU/AAU.
Protocol Type	RRUCHAIN.PROTOCOL	This parameter specifies the communication protocol used by the chain/ring.
Head Cabinet No.	RRUCHAIN.HCN	These parameters specify the cabinet number, subrack number, slot number of the board providing the CPRI port serving as the chain/ring head or ring tail, and CPRI port number. Set these parameters based on hardware installation information.
Head Subrack No.	RRUCHAIN.HSRN	
Head Slot No.	RRUCHAIN.HSN	
Head Port No.	RRUCHAIN.HPN	
Tail Cabinet No.	RRUCHAIN.TCN	
Tail Subrack No.	RRUCHAIN.TSRN	
Tail Slot No.	RRUCHAIN.TSN	
Tail Port No.	RRUCHAIN.TPN	

Parameter Name	Parameter ID	Setting Notes
BreakPoint Position1	RRUCHAIN.BRKP OS1	These parameters specify the sequence number (counting from the chain/ring head) of an RRU/RFU/AAU after which a chain/ring breaks. - For a chain, only one breakpoint can be configured. - For a ring, two breakpoints must be configured. The value of BreakPoint Position1 must not be greater than the value of BreakPoint Position2. If there is only one breakpoint, the values of BreakPoint Position1 and BreakPoint Position2 must be the same, indicating that the two breakpoints are the same. - To remove a breakpoint, set BreakPoint Position1 and BreakPoint Position2 to 255.
BreakPoint Position2	RRUCHAIN.BRKP OS2	
CPRI Line Rate	RRUCHAIN.CR	This parameter is set by the user or negotiated automatically. ¹
User-Defined Rate Negotiation Switch	RRUCHAIN.USER DEFRATENEGOS W	If this switch is turned on, users can specify a bit rate set and a rate change period for automatic rate negotiation.
Bit Rate Set	RRUCHAIN.BITR ATESET	This parameter specifies the bit rate set for rate negotiation. This parameter takes effect only when the user-defined rate negotiation switch is turned on.
Rate Change Period	RRUCHAIN.RATE CHGPERIOD	This parameter specifies the duration of a port working in each negotiated rate. This parameter takes effect only when the user-defined rate negotiation switch is turned on.

 NOTE

If intermediate devices (such as OTN or RTN) exist on the chain/ring between the BBU and RF modules, set this parameter to a specific value based on the CPRI port specifications of the BBU, RF modules, and intermediate devices.

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RRU.CV	- This parameter can be set to a value ranging from 0 to 15 for an AAU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 15 for an RRU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 62 for an RFU.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	RRU.SRN	- This parameter can be set to a value ranging from 60 to 254 for an RRU/AAU. - This parameter can be set to 4 or 5 for an RFU in a BTS3900L/BTS5900L cabinet or a virtual cabinet, and must be set to 4 for an RFU in any other cabinet.
Slot No.	RRU.SN	- This parameter must be set to 0 for an RRU/AAU installed outside a cabinet. - This parameter can be set to a value ranging from 0 to 8 for an RFU in a BTS3900AL cabinet or a virtual cabinet, and can be set to a value ranging from 0 to 5 for an RFU in any other cabinet.
Topo Position	RRU.TP	Set this parameter to TRUNK(TRUNK) .
RRU Chain No.	RRU.RCN	Set this parameter based on the actual physical connection.
RRU Position	RRU.PS	This parameter specifies the position of an RRU/RFU/AAU on a chain/ring. The value of this parameter is set sequentially from 0. The value of this parameter for an RRU/RFU/AAU must differ from that for any other RRU/RFU/AAU configured on the same chain/ring. In the chain topology, the end close to the BBU is the chain head, and the sequence number starts from 0.
RRU type	RRU.RT	Set this parameter based on the type of the in-use RF module. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules . ¹
RF Unit Working Mode	RRU.RS	When an inter-frequency mutual aid module works in inter-frequency mutual aid mode, and the two RX frequency bands are in different RATs, set this parameter to a value including multiple RATs. For example, if the RRU3952m (1800 MHz) is configured to work in 2T2R in LTE 1800 MHz and 2R in UMTS 2100 MHz, set this parameter to UL .
RRU Name	RRU.RN	This parameter is optional and can be customized to contain 0 to 99 characters. This parameter takes effect only when an RRU is configured.
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of an RRU/RFU/AAU.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of an RRU/RFU/AAU.

Parameter Name	Parameter ID	Setting Notes
RRU Slave CPRI Line Rate	RRU.SCR	- If this parameter is set to a fixed value, this configuration takes effect only when the CPRI Line Rate of the corresponding chain/ring is set to MANUAL(Manual). - In other scenarios, the CPRI line rate is automatically negotiated.
Remote Flag	RRU.REMOTEFLAG	If you do not need to define this information, set this parameter to UNDEFINED(Undefined) .
User Label	RRU.USERLABEL	This parameter specifies the user-defined RRU/AAU descriptions using 0 to 255 characters.
Receive Frequency Band Mutual Aid Switch	RRU.RXFREQBANDMUTUALSW	This parameter is valid only for RF modules whose hardware capabilities support inter-frequency mutual aid, such as the RRU3952m. Dynamically modifying the value of this parameter for an inter-frequency mutual aid RF module will reset the RF module.

 NOTE

The eGBTS does not support the DRFU and RRU3004.

- The following table describes key parameters in an **AAS** MO to configure an active antenna system.

Parameter Name	Parameter ID	Setting Notes
Subrack No.	AAS.SRN	This parameter can be set to a value ranging from 60 to 254 for an AAS.
AAS Name	AAS.AN	This parameter is optional and can be customized to contain 0 to 31 characters.
AAU Specification	AAS.AAUSPEC	Set this parameter based on the actual hardware type of the module.

- The following table describes the parameters related to **AARU**. Set the key parameters according to the descriptions in the following table.

Parameter Name	Parameter ID	Setting Notes
Intermediate Frequency Offset	AARU.IFOFFSET	This parameter indicates the intermediate frequency offset of the AARU. The default value 0 is recommended.

Scenario-specific Data Preparation

Scenario 1: star topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

In the dual-star topology, set the **TT** parameter to **CHAIN(CHAIN)** for each RAT of the separate-MPT base station.

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain number.
RRU Position	RRU.PS	Set this parameter to 0 .

Scenario 2: chain topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain number.
RRU Position	RRU.PS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU.

Scenario 3: ring topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to RING(RING) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual ring number.
RRU Position	RRU.PS	Set this parameter based on the number of RRU/RFUs/AAUs. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU.

4.6.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

Step 2 (Optional) Configure breakpoints before adding an RRU/RFU/AAU if **Topo Type** is set to **RING(RING)** for the RRU chain/ring added in step 1.

Run the **MOD RRUCHAIN** command to configure breakpoints on the RRU chain/ring. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **0**.

Step 3 Run the corresponding command to add an RF module and configure related parameters based on the following table.

If the RRU Type Is...	Then...
MRRU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
MRFU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
AIRU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
GRFU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .

If the RRU Type Is...	Then...
GRRU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
AARU	1. Run the ADD AAS command to add an AAS. In this step, set Topo Position to TRUNK(TRUNK). 2. Run the ADD AARU command to add an AARU.

Step 4 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoints configured in step 2. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

----End

MML Command Examples

- For an RF module whose **RRU type** value is **LRRU/PRRU/MRRU/MRFU/AIRU**
 //Adding an RRU chain
 ADD RRUCHAIN: RCN=0, **TT=CHAIN**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

 //Adding an RRU ring
 ADD RRUCHAIN: RCN=1, **TT=RING**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0, TSN=3, TPN=2;

 //Adding a load sharing topology
 ADD RRUCHAIN: RCN=2, **TT=LOADBALANCE**, HCN=0, HSRN=0, HSN=3, HPN=3, TCN=0, TSRN=0, TSN=3, TPN=4;

 //(Optional) Configuring the breakpoints
 MOD RRUCHAIN: RCN=0, **BRKPOS1=0, BRKPOS2=0**;

 //Adding an RRU/RFU/AAU
 ADD RRU: CN=0, SRN=60, SN=0, **TP=TRUNK**, RCN=0, PS=0, RT=MRRU, RS=GO, RN= "MRRU_1", RXNUM=2, TXNUM=2;

 //(Optional) Removing the breakpoints
 MOD RRUCHAIN: RCN=1, **BRKPOS1=255, BRKPOS2=255**;
- For an RF module whose **RRU type** value is **AARU**
 //Adding an RRU chain
 ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

 //Adding an RRU ring
 ADD RRUCHAIN: RCN=0, TT=RING, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0, TSN=3, TPN=2;

 //Adding an AAS
 ADD AAS: CN=0, SRN=70, **TP=TRUNK**, RCN=0, PS=1, AN="AAS_1", AAUSPEC="AAU3902";

 //Adding an AARU
 ADD AARU: CN=0, SRN=70, SN=1, RS=GO, AN="AARU1", RXNUM=2, TXNUM=2, IFOFFSET=0, VRETNO=70;

4.6.2.3 Using the MAE-Deployment

- For detailed operations of configuring a single NE, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide: Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based) > Creating a Single Co-MPT Base Station > Configuring Device Data for a Co-MPT Base Station > Configuring RF Units*.

- For detailed operations of configuring NEs in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide: Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based)* > *Creating Co-MPT Base Stations in Batches (MOC Export)/Creating Co-MPT Base Stations in Batches (Customization Tool)*.

4.6.3 Activation Verification

After configuration data is activated, ensure that no RRU/RFU/AAU related alarm is displayed on the **Alarm/Event** tab page on the LMT.

On the LMT or MAE-Access, run the **DSP BRD** command to check the status of the RRU/RFU/AAU.

The status of the RRU/RFU/AAU is normal if the value of **Availability Status** is **Normal**.

Run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

4.6.4 Network Monitoring

None

4.6.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps.

MML-based Adjustment

After the adjustment, run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

Combining RRU Chains into an RRU Ring

A cold backup ring topology can be set up by combining two RRU chains. The precautions for combining RRU chains into an RRU ring are as follows:

- The **Chain No.** value of the two RRU chains must differ from each other, and the value of **Topo Type** must be **CHAIN(CHAIN)**.
- The total number of RRU cascading levels on the two RRU chains cannot exceed 8.

Step 1 Run the **DSP CPRILBR** command to check whether values of **Current CPRI Line Bit Rate** of the two RRU chains are the same. If they are not the same, go to step 3 after the combination. If they are the same, skip step 3.

```
DSP CPRILBR: RCN=1;
```

Step 2 Run the **CMB RRUCHAIN** command to combine two RRU chains and set up a cold backup ring topology. After the two RRU chains are combined successfully, the value of **Chain No.** of the newly created ring is the same as that of **First RRU Chain No.**

CMB RRUCHAIN: RCN1=0, RCN2=2;

Step 3 (Optional) Run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

STR CPRILBRNEG: RCN=0;

----End

Adding an RRU/RFU/AAU to a Chain/Ring

An RRU/RFU/AAU can be added to any position on an RRU chain/ring without changing the original topology.

Step 1 (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for adding an RRU/RFU/AAU to the head or middle of a chain/ring.

- For the RRU chain, configure one breakpoint with **BreakPoint Position1** set to 0.
- For the RRU ring, configure two breakpoints with **BreakPoint Position1** set to 0 and **BreakPoint Position2** set to the number of RRU cascading levels.

MOD RRUCHAIN: RCN=0, BRKPOS1=0;

Step 2 Run the **ADD RRU** command to add an RRU/RFU/AAU and set related information.

ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=GO, RN="MRRU_2", RXNUM=2, TXNUM=2;

Step 3 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;

----End

Removing an RRU/RFU/AAU from a Chain/Ring

Step 1 (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for removing an RRU/RFU/AAU from the head or middle of a chain/ring.

- For the chain, configure one breakpoint with **BreakPoint Position1** set to 0.
- For the ring, configure two breakpoints with **BreakPoint Position1** set to 0 and **BreakPoint Position2** set to the number of RRU cascading levels.

MOD RRUCHAIN: RCN=0, BRKPOS1=1;

Step 2 Run the **RMV RRU** command to remove the RRU/RFU/AAU.

RMV RRU: CN=0, SRN=61, SN=0;

Step 3 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;

----End

Modifying the RRU/RFU Logical Type

- Step 1** Run the **MOD RRU** command to reconfigure the **RRU type** parameter for an RRU/RFU.

```
MOD RRU: CN=0, SRN=60, SN=0, RT=MRRU;
```

NOTE

- The value ranges of **Cabinet No.**, **Subrack No.**, and **Slot No.** for the RRU differ from those for the RFU. Therefore, the value of **RRU type** for an RRU cannot be changed to the value of **RRU type** for an RFU, and the value of **RRU type** for an RFU cannot be changed to the value of **RRU type** for an RRU.
- The value of **RRU type** for the pRRU and AAU cannot be modified.

----End

Modifying the RRU/RFU/AAU Working Standard

Before the modification, ensure that the RF module hardware supports the target working standard. For details about the RF module hardware capability, see the **Supported RAT** columns of the tables in [4.1.2 RF Modules](#). In addition, ensure that the license for the target working standard has been purchased.

- Step 1** Run the **MOD RRU** command with **RF Unit Working Mode** set to the target working standard.

```
MOD RRU: CN=0, SRN=60, SN=0, RS=GO;
```

NOTE

Working standard of the pRRU cannot be modified.

----End

MAE-Deployment-based Adjustment

- **Combining RRU chains into an RRU ring**
For details about the MAE-Deployment-based adjustment, see **Reconfiguring an eGBTS > Changing the CPRI-based topologies of the Base Stations > Changing RRU Chains to an RRU Ring** in *GBSS Reconfiguration Guide*.
- **Adding an RRU/RFU to a chain/ring**
For details about the MAE-Deployment-based adjustment, see **Reconfiguring an eGBTS > Adding an RF Module** in *GBSS Reconfiguration Guide*.
- **Removing an RRU/RFU from a chain/ring**
For details about the MAE-Deployment-based adjustment, see **Reconfiguring an eGBTS > Removing an RF Module** in *GBSS Reconfiguration Guide*.

4.7 Operation and Maintenance (NodeB)

4.7.1 Precautions

None

4.7.2 Data Configuration

Comply with the following rules when single-mode or multimode RF modules work in single-mode scenarios:

- The slot numbers of RFUs that work in different modes in the same RFU subrack must be different.
- Before adding an RRU/RFU/AAU, the RRU chain/ring must be added.

4.7.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/RFU/AAU model: Run the **DSP ELABEL** command to query the electronic label of the RRU/RFU/AAU.
 - RRU/RFU/AAU topology: Use it to determine how to configure RRU/RFU/AAU links.
1. The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Chain No.	RRUCHAIN.RCN	This parameter uniquely identifies a chain/ring in a base station.
Topo Type	RRUCHAIN.TT	• In the star or chain topology, set this parameter to CHAIN(CHAIN). In this case, the TCN, TSRN, TSN, and TPN parameters are invalid. In the star topology, only one RRU/RFU/AAU can be configured on a chain. • In the ring topology, set this parameter to RING(RING) and specify two CPRI ports and the backup mode. • In the branch chain (single-link) topology, set this parameter to CHAIN(CHAIN). • In the trunk chain topology, set this parameter to TRUNK_CHAIN(TRUNK_CHAIN).
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD .

Parameter Name	Parameter ID	Setting Notes
Access Type	RRUCHAIN.AT	When this parameter is set to LOCALPORT(local port) , the BBP is connected to the RRU/RFU/AAU. When this parameter is set to PEERPORT(peer port) , the local BBP (of one RAT) is connected to the peer BBP (of another RAT) through the virtual CPRI subchannel on the backplane, and the peer BBP is then connected to the RRU/RFU/AAU.
Protocol Type	RRUCHAIN.PROTOCOL	This parameter specifies the communication protocol used by the chain/ring.
Head Cabinet No.	RRUCHAIN.HCN	These parameters specify the cabinet number, subrack number, slot number of the board providing the CPRI port serving as the chain/ring head or ring tail, and CPRI port number. Set these parameters based on hardware installation information.
Head Subrack No.	RRUCHAIN.HSRN	
Head Slot No.	RRUCHAIN.HSN	
Head Port No.	RRUCHAIN.HPN	
Tail Cabinet No.	RRUCHAIN.TCN	
Tail Subrack No.	RRUCHAIN.TSRN	
Tail Slot No.	RRUCHAIN.TSN	
Tail Port No.	RRUCHAIN.TPN	
BreakPoint Position1	RRUCHAIN.BRKP OS1	These parameters specify the sequence number (counting from the chain/ring head) of an RRU/RFU/AAU after which a chain/ring breaks. - For a chain, only one breakpoint can be configured. - For a ring, two breakpoints must be configured. The value of BreakPoint Position1 must not be greater than the value of BreakPoint Position2. If there is only one breakpoint, the values of BreakPoint Position1 and BreakPoint Position2 must be the same, indicating that the two breakpoints are the same. - To remove a breakpoint, set BreakPoint Position1 and BreakPoint Position2 to 255.
BreakPoint Position2	RRUCHAIN.BRKP OS2	
CPRI Line Rate	RRUCHAIN.CR	This parameter is set by the user or negotiated automatically. ¹
User-Defined Rate Negotiation Switch	RRUCHAIN.USER DEFRATENEGOS W	If this switch is turned on, users can specify a bit rate set and a rate change period for automatic rate negotiation.
Bit Rate Set	RRUCHAIN.BITR ATESET	This parameter specifies the bit rate set for rate negotiation. This parameter takes effect only when the user-defined rate negotiation switch is turned on.

Parameter Name	Parameter ID	Setting Notes
Rate Change Period	RRUCHAIN.RATE CHGPERIOD	This parameter specifies the duration of a port working in each negotiated rate. This parameter takes effect only when the user-defined rate negotiation switch is turned on.
Head SubPort No.	RRUCHAIN.HSP N	- When a DSFP optical module is used, subchannel 0 indicates the optical fiber on the left, and subchannel 1 indicates the optical fiber on the right. - In common star, chain, ring, and load sharing topologies, either subchannel 0 or subchannel 1 can be used. - In CPRI MUX topology, only subchannel 0 can be configured. - When a non-DSFP optical module is used, only subchannel 0 can be used.

 NOTE

1. If intermediate devices (such as OTN or RTN) exist on the chain/ring between the BBU and RF modules, set this parameter to a specific value based on the CPRI port specifications of the BBU, RF modules, and intermediate devices.
2. The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RRU.CN	- This parameter can be set to a value ranging from 0 to 15 for an AAU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 15 for an RRU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 62 for an RFU.
Subrack No.	RRU.SRN	- This parameter can be set to a value ranging from 60 to 254 for an RRU/AAU. - This parameter can be set to 4 or 5 for an RFU in a BTS3900L/BTS5900L cabinet or a virtual cabinet, and must be set to 4 for an RFU in any other cabinet.
Slot No.	RRU.SN	- This parameter must be set to 0 for an RRU/AAU installed outside a cabinet. - This parameter must be set to a value ranging from 1 to 3 for a pRRU. - This parameter can be set to a value ranging from 0 to 8 for an RFU in a BTS3900AL cabinet or a virtual cabinet, and can be set to a value ranging from 0 to 5 for an RFU in any other cabinet.

Parameter Name	Parameter ID	Setting Notes
Topo Position	RRU.TP	- Set this parameter to TRUNK(TRUNK) when this parameter indicates the position where the RRU/RFU/AAU is installed in the topology. - Set this parameter to BRANCH(BRANCH) for the pRRU.
RRU Chain No.	RRU.RCN	Set this parameter based on the actual physical connection.
RRU Position	RRU.PS	This parameter specifies the position of an RRU/RFU/AAU on a chain/ring. The value of this parameter is set sequentially from 0. The value of this parameter for an RRU/RFU/AAU must differ from that for any other RRU/RFU/AAU configured on the same chain/ring. In the chain topology, the end close to the BBU is the chain head, and the sequence number starts from 0.
RRU type	RRU.RT	Set this parameter based on the type of the in-use RF module. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules .
RF Unit Working Mode	RRU.RS	When an inter-frequency mutual aid module works in inter-frequency mutual aid mode, and the two RX frequency bands are in different RATs, set this parameter to a value including multiple RATs. For example, if the RRU3952m (1800 MHz) is configured to work in 2T2R in LTE 1800 MHz and 2R in UMTS 2100 MHz, set this parameter to UL .
RRU Name	RRU.RN	This parameter is optional and can be customized to contain 0 to 99 characters. This parameter takes effect only when an RRU is configured.
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of an RRU/RFU/AAU.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of an RRU/RFU/AAU.
RRU Slave CPRI Line Rate	RRU.SCR	- If this parameter is set to a fixed value, this configuration takes effect only when the CPRI line rate mode of the corresponding chain/ring is set to MANUAL(Manual). - In other scenarios, the CPRI line rate is automatically negotiated.
Remote Flag	RRU.REMOTEFL AG	If you do not need to define this information, set this parameter to UNDEFINED(Undefined) .

Parameter Name	Parameter ID	Setting Notes
User Label	RRU.USERLABEL	This parameter specifies the user-defined RRU/AAU descriptions using 0 to 255 characters.
Receive Frequency Band Mutual Aid Switch	RRU.RXFREQBANDMUTUALSW	This parameter is valid only for RF modules whose hardware capabilities support inter-frequency mutual aid, such as the RRU3952m. Dynamically modifying the value of this parameter for an inter-frequency mutual aid RF module will reset the RF module.

3. (Optional) The following table describes key parameters in an **RHUB** MO required for DBS3900 LampSite base stations.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RHUB.CN	This parameter must be set to 0 for an RHUB.
Subrack No.	RHUB.SRN	This parameter can be set to a value ranging from 60 to 254 for an RHUB. The subrack No. of an RHUB must be different from that of another RHUB or an RRU.
Slot No.	RHUB.SN	This parameter must be set to 0 for an RHUB.
RRU Chain No.	RHUB.RCN	This parameter uniquely identifies an RHUB chain/ring in a base station.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.
RHUB Name	RHUB.RN	This parameter is optional and can be customized to contain 0 to 31 characters.

4. The following table describes key parameters in an **AAS** MO to configure an active antenna system.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	AAS.CN	This parameter must be set to 0 for an AAS.
Subrack No.	AAS.SRN	This parameter can be set to a value ranging from 60 to 254 for an AAS.
AAS Name	AAS.AN	This parameter is optional and can be customized to contain 0 to 31 characters.
AAU Specification	AAS.AAUSPEC	Set this parameter based on the actual hardware type of the module.

5. The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
Topo Position	AAMU.TP	Set this parameter to TRUNK(TRUNK) .
AAS Chain No.	AAMU.RCN	Set this parameter based on the actual physical connection.
AAS Position	AAMU.PS	This parameter specifies the position of an AAS on an RRU chain/ring. It is numbered from the start port in sequence.

6. The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	AARU.CN	This parameter must be set to 0 for an AARU.
Subrack No.	AARU.SRN	This parameter can be set to a value ranging from 60 to 254 for an AARU. The subrack number of the AARU must be the same as that of the AAS.
Slot No.	AARU.SN	This parameter can be set to a value ranging from 1 to 3 for an AARU.
AARU Name	AARU.AN	This parameter is optional and can be customized to contain 0 to 31 characters.
AARU Work Standard	AARU.RS	This parameter specifies the working standard of an AARU. Set this parameter to UO(UMTS_ONLY) .
Number of RX channels	AARU.RXNUM	Set this parameter to the actual number of RX channels of an AARU.
Number of TX channels	AARU.TXNUM	Set this parameter to the actual number of TX channels of an AARU.
VRET No.	AARU.VRETNO	This parameter specifies the number of a virtual antenna corresponding to the AARU.
Intermediate Frequency Offset	AARU.IFOFFSET	This parameter indicates the intermediate frequency offset of the AARU. The default value 0 is recommended.

Scenario-specific Data Preparation

Scenario 1: star topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

In the dual-star topology, set the **TT** parameter to **CHAIN(CHAIN)** for each RAT of the separate-MPT base station.

For an RF module whose logical type is MRRU/PRRU/MRFU/WRFU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.
RRU Position	RRU.PS	Set this parameter to 0 .

For an RF module whose logical type is AARU:

- The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
AAS Chain No.	AAMU.RCN	Set this parameter to the actual chain/ring number.
AAS Position	AAMU.PS	Set this parameter to 0 .

- The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
AARU Work Standard	AARU.RS	Set this parameter to UO(UMTS_ONLY) .

Scenario 2: chain topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

For an RF module whose logical type is MRRU/PRRU/MRFU/WRFU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.
RRU Position	RRU.PS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU.
Cabinet No.	RHUB.CN	Set this parameter to 0 .
Subrack No.	RHUB.SRN	Set this parameter according to the network plan. The subrack No. of an RHUB must be different from that of another RHUB or an RRU.
Slot No.	RHUB.SN	Set this parameter to 0 .
RHUB Name	RHUB.RN	Set this parameter based on the network plan.

For an RF module whose logical type is AARU:

- The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
AAS Chain No.	AAMU.RCN	Set this parameter to the actual chain/ring number.
AAS Position	AAMU.PS	Set this parameter based on the number of AAUs. For example, set this parameter to 0 for the first AAU and 1 for the second AAU.

- The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
AARU Work Standard	AARU.RS	Set this parameter to UO(UMTS_ONLY) .

Scenario 3: ring topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to RING(RING) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.
RRU Position	RRU.PS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU.

Scenario 4: branch chain (single-link) topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RHUB chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RHUB** MO to configure an RHUB.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RHUB.RCN	Set this parameter to the actual chain/ring number.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.

- The following table describes key parameters in an **RRUCHAIN** MO to configure a pRRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RRU** MO to configure a pRRU.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.
RRU Position	RRU.PS	Set this parameter to 0 .
RRU type	RRU.RT	Set this parameter to MPMU .

Parameter Name	Parameter ID	Setting Notes
Topo Position	RRU.TP	Set this parameter to BRANCH(BRANCH) .
RF Unit Working Mode	RRU.RS	Set this parameter to UO(UMTS_ONLY) .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of a pRRU. - Set this parameter to 0 for the pRRU3901. - Set this parameter to 4 for the pRRU3902 and pRRU3907. - Set this parameter to 6 for the pRRU3911.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of a pRRU. The number of TX channels of a pRRU is the same as the number of RX channels of this pRRU.
a: For pRRU565x <i>AB</i> modules (<i>x</i> is a digit; <i>A</i> and <i>B</i> stand for letters), set RRU type to EPRRU . For other pRRU models, set RRU type to MPMU .		

4.7.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

Step 2 (Optional) Configure breakpoints before adding an RRU/RFU if **Topo Type** is set to **RING(RING)** for the RRU chain/ring added in step 1.

Run the **MOD RRUCHAIN** command to configure breakpoints on the RRU chain/ring. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **0**.

Step 3 Run the corresponding command to add an RF module and configure related parameters based on the following table.

If the Logical Type Is...	Then...
MRRU/PRRU/MRFU/WRFU/AIRU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
AARU	- Run the ADD AAS command to add an AAS. In this step, set Topo Position to TRUNK(TRUNK). - Run the ADD AARU command to add an AARU.

If the Logical Type Is...	Then...
MPMU	1. Run the ADD RHUB command to add an RHUB to the RRU chain added in step 1. 2. Run the ADD RRUCHAIN command to add an RRU chain between the RHUB and pRRU. 3. Run the ADD RRU command to add a pRRU. Set Topo Position to BRANCH(BRANCH) and RRU type to MPMU.

Step 4 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoints configured in step 2. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

----End

MML Command Examples

- For an RF module whose **RRU type** value is **LRRU/PRRU/MRRU/MRFU/AIRU**
 //Adding an RRU chain
 ADD RRUCHAIN: RCN=0, **TT=CHAIN**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

 //Adding an RRU ring
 ADD RRUCHAIN: RCN=1, **TT=RING**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0, TSN=3, TPN=2;

 //Adding a load sharing topology
 ADD RRUCHAIN: RCN=2, **TT=LOADBALANCE**, HCN=0, HSRN=0, HSN=3, HPN=3, TCN=0, TSRN=0, TSN=3, TPN=4;

 //(Optional) Configuring the breakpoints
 MOD RRUCHAIN: RCN=0, **BRKPOS1=0, BRKPOS2=0**;

 //Adding an RRU/RFU/AAU
 ADD RRU: CN=0, SRN=60, SN=0, **TP=TRUNK**, RCN=0, PS=0, RT=MRRU, RS=UO, RN= "MRRU_1", RXNUM=2, TXNUM=2;

 //(Optional) Removing the breakpoints
 MOD RRUCHAIN: RCN=1, **BRKPOS1=255, BRKPOS2=255**;
- For an RF module whose **RRU type** value is **AARU**
 //Adding an RRU chain
 ADD RRUCHAIN: RCN=0, **TT=CHAIN**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

 //Adding an RRU ring
 ADD RRUCHAIN: RCN=0, **TT=RING**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0, TSN=3, TPN=2;

 //Adding an AAS
 ADD AAS: CN=0, SRN=70, **TP=TRUNK**, RCN=0, PS=1, AN="AAS_1", AAUSPEC="AAU3902";

 //Adding an AARU
 ADD AARU: CN=0, SRN=70, SN=1, RS=GO, AN="AARU1", RXNUM=2, TXNUM=2, IFOFFSET=0, VRETNO=70;
- For an RF module whose **RRU type** value is **MPMU**
 //Adding a main chain/ring
 ADD RRUCHAIN: RCN=1, **TT=CHAIN**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1;

 //Adding an RHUB
 ADD RHUB: CN=0, SRN=100, SN=0, RCN=0, PS=0, RN="RHUB_1";

 //Adding a branch chain/ring

```
//If a branch chain (single-link) topology is used
ADD RRUCHAIN: RCN=2, TT=CHAIN, BM=COLD, HCN=0, HSRN=100, HSN=0, HPN=0;
//If a branch load sharing topology is used
ADD RRUCHAIN: RCN=2, TT=LOADBALANCE, HCN=0, HSRN=100, HSN=0, HPN=1, TCN=0,
TSRN=100, TSN=0, TPN=2;

//Adding a pRRU
ADD RRU: CN=0, SRN=110, SN=0, TP=BRANCH, RCN=2, PS=0, RT=MPMU, RS=UO, RN="pRRU_1",
RXNUM=0, TXNUM=0;
```

4.7.2.3 Using the MAE-Deployment

- For detailed operations of configuring a single NE, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: NodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single NodeB > Configuring NodeB Device Data > Configuring RF Units*.
- For detailed operations of configuring NEs in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: NodeB Initial Configuration Guide (MAE-Deployment-based) > Creating NodeBs in Batches (MOC Export)/Creating NodeBs in Batches (Customization Tool)*.

4.7.3 Activation Verification

After configuration data is activated, ensure that no RRU/RFU/AAU related alarm is displayed on the **Alarm/Event** tab page on the LMT.

On the LMT or MAE-Access, run the **DSP BRD** command to check the status of the RRU/RFU/AAU.

The status of the RRU/RFU/AAU is normal if the value of **Availability Status** is **Normal**.

Run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

NOTE

For pRRU565x*AB* modules (*x* is a digit; *A* and *B* stand for different letters), if the pRRUs have been configured and are running properly, CPRI line rate negotiation will be performed automatically for their branch chains. Therefore, you do not need to run the **STR CPRILBRNEG** command. You can run the **DSP CPRIPORT** command to query the CPRI rates supported by the branch chains where the pRRUs reside.

4.7.4 Network Monitoring

None

4.7.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps.

MML-based Adjustment

After the adjustment, run the **DSP CPILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPILBRNEG** command to perform CPRI line rate negotiation.

Combining RRU Chains into an RRU Ring

A cold backup ring topology can be set up by combining two RRU chains. The precautions for combining RRU chains into an RRU ring are as follows:

- The **Chain No.** value of the two RRU chains must differ from each other, and the value of **Topo Type** must be **CHAIN(CHAIN)**.
- The total number of RRU cascading levels on the two RRU chains cannot exceed 8.

Step 1 Run the **DSP CPILBR** command to check whether values of **Current CPRI Line Bit Rate** of the two RRU chains are the same. If they are not the same, go to step 3 after the combination. If they are the same, skip step 3.

```
DSP CPILBR: RCN=1;
```

Step 2 Run the **CMB RRUCHAIN** command to combine two RRU chains and set up a cold backup ring topology. After the two RRU chains are combined successfully, the value of **Chain No.** of the newly created ring is the same as that of **First RRU Chain No.**

```
CMB RRUCHAIN: RCN1=0, RCN2=2;
```

Step 3 (Optional) Run the **STR CPILBRNEG** command to perform CPRI line rate negotiation.

```
STR CPILBRNEG: RCN=0;
```

----End

Adding an RRU/RFU/AAU to a Chain/Ring

An RRU/RFU/AAU can be added to any position on an RRU chain/ring without changing the original topology.

Step 1 (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for adding an RRU/RFU/AAU to the head or middle of a chain/ring.

- For the RRU chain, configure one breakpoint with **BreakPoint Position1** set to 0.
- For the RRU ring, configure two breakpoints with **BreakPoint Position1** set to 0 and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

Step 2 Run the **ADD RRU** command to add an RRU/RFU/AAU and set related information.

```
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=GO, RN= "MRRU_2", RXNUM=2, TXNUM=2;
```

Step 3 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Removing an RRU/RFU/AAU from a Chain/Ring

Step 1 (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for removing an RRU/RFU/AAU from the head or middle of a chain/ring.

- For the chain, configure one breakpoint with **BreakPoint Position1** set to **0**.
- For the ring, configure two breakpoints with **BreakPoint Position1** set to **0** and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=1;
```

Step 2 Run the **RMV RRU** command to remove the RRU/RFU/AAU.

```
RMV RRU: CN=0, SRN=61, SN=0;
```

Step 3 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Modifying the RRU/RFU Logical Type

Step 1 Run the **MOD RRU** command to reconfigure the **RRU type** parameter for an RRU/RFU.

```
MOD RRU: CN=0, SRN=60, SN=0, RT=MRRU;
```

NOTE

- The value ranges of **Cabinet No.**, **Subrack No.**, and **Slot No.** for the RRU differ from those for the RFU. Therefore, the value of **RRU type** for an RRU cannot be changed to the value of **RRU type** for an RFU, and the value of **RRU type** for an RFU cannot be changed to the value of **RRU type** for an RRU.
- The value of **RRU type** for the pRRU and AAU cannot be modified.

----End

Modifying the RRU/RFU/AAU Working Standard

Before the modification, ensure that the RF module hardware supports the target working standard. For details about the RF module hardware capability, see the **Supported RAT** columns of the tables in [4.1.2 RF Modules](#). In addition, ensure that the license for the target working standard has been purchased.

Step 1 Run the **MOD RRU** command with **RF Unit Working Mode** set to the target working standard.

```
MOD RRU: CN=0, SRN=60, SN=0, RS=UO;
```

NOTE

Working standard of the pRRU cannot be modified.

----End

Adding an RHUB to a Chain

An RHUB can be added to any position on an RHUB chain.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure a breakpoint for adding an RHUB to the head of a chain.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=0;
```

- Step 2** Run the **ADD RHUB** command to add an RHUB.

```
ADD RHUB: CN=0, SRN=101, SN=0, RCN=1, PS=0, RN="RHUB_2";
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Adding a pRRU to an RHUB

Before adding a pRRU, a branch chain/ring must be added between the pRRU and RHUB.

- Step 1** Run the **ADD RRUCHAIN** command to add a branch chain/ring between the RHUB and pRRU.

```
ADD RRUCHAIN: RCN=2, TT=CHAIN, BM=COLD, HCN=0, HSRN=100, HSN=0, HPN=1;
```

- Step 2** Run the **ADD RRU** command to add a pRRU.

```
ADD RRU: CN=0, SRN=111, SN=0, TP=BRANCH, RCN=3, PS=0, RT=MPMU, RS=UO, RXNUM=0, TXNUM=0;
```

----End

Removing an RHUB from a Chain

Before removing an RHUB, remove the pRRUs and pRRU branch chains/rings connected to this RHUB.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure a breakpoint for removing an RHUB from the head or middle of a chain.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=0;
```

- Step 2** Run the **RMV RHUB** command to remove an RHUB.

```
RMV RHUB: CN=0, SRN=100, SN=0;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Removing a pRRU from an RHUB

- Step 1** Run the **RMV RRU** command to remove a pRRU.

```
RMV RRU: CN=0, SRN=111, SN=0;
```

----End

MAE-Deployment-based Adjustment

- Adding an RRU/RFU to a chain/ring

For details about the MAE-Deployment-based adjustment, see **Reconfiguring a NodeB > Device Data Reconfiguration > Adding an RF Unit** in *RAN Reconfiguration Guide*.

4.8 Operation and Maintenance (eNodeB)

4.8.1 Precautions

N/A

4.8.2 Data Configuration

Comply with the following rules when single-mode or multimode RF modules work in single-mode scenarios:

- The slot numbers of RFUs that work in different modes in the same RFU subrack must be different.
- For a hot backup ring topology, only one RRU/RFU/AAU can be configured, and RRU/RFU/AAU cascading is not supported.
- Add RRU chains/rings before adding RRUs/RFUs/AAUs.

4.8.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/RFU/AAU model: Run the **DSP ELABEL** command to query the electronic label of the RRU/RFU/AAU.
 - RRU/RFU/AAU topology: Use it to determine how to configure RRU/RFU/AAU links.
1. The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Chain No.	RRUCHAIN.RCN	This parameter uniquely identifies a chain/ring in a base station.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.77	- In the star or chain topology, set this parameter to CHAIN(CHAIN). In this case, the TCN, TSRN, TSN, and TPN parameters are invalid. In the star topology, only one RRU/RFU/AAU can be configured on a chain. - In the ring topology, set this parameter to RING(RING) and specify two CPRI ports and the backup mode. - In the load sharing topology, set this parameter to LOADBALANCE(LOADBALANCE). In this case, two CPRI ports need to be specified. - In the branch chain (single-link) topology, set this parameter to CHAIN(CHAIN). - In the branch load sharing topology, set this parameter to LOADBALANCE(LOADBALANCE). In this case, two CPRI ports need to be specified. - In the trunk chain topology, set this parameter to TRUNK_CHAIN(TRUNK_CHAIN).
Backup Mode	RRUCHAIN.BM	- If the Topo Type parameter is set to CHAIN(CHAIN), set this parameter to COLD(COLD). - If the Topo Type parameter is set to RING(RING), - Set this parameter to COLD(COLD) in the cold backup ring topology. - Set this parameter to HOT(HOT) in the hot backup ring topology. In this case, the ring head and tail must belong to different BRIs, and only one RRU/RFU/AAU can be added. - Before changing the backup mode of a chain/ring, ensure that there is no RRU/RFU/AAU on the chain/ring. - If the non-load-sharing topology is changed to the load sharing topology, RRUs/RFUs/AAUs can be configured on the chain/ring. When the load sharing topology is changed to the non-load-sharing topology, ensure that no RRU/RFU/AAU is configured on the chain/ring.
Access Type	RRUCHAIN.A7	When this parameter is set to LOCALPORT(local port) , the BBP is connected to the RRU/RFU/AAU. When this parameter is set to PEERPORT(peer port) , the local BBP (of one RAT) is connected to the peer BBP (of another RAT) through the virtual CPRI subchannel on the backplane, and the peer BBP is then connected to the RRU/RFU/AAU.

Parameter Name	Parameter ID	Setting Notes
Protocol Type	RRUCHAIN.PROTOCOL	This parameter specifies the communication protocol used by the chain/ring. - For RF modules that support the CPRI protocol, it is recommended that this parameter be set to CPRI. In this case, the protocol type that takes effect on the chain/ring is CPRI. In this case, the value of Protocol Type in the SET EQUIPMENT command is invalid. - For RF modules that support the eCPRI protocol, it is recommended that this parameter be set to eCPRI. In this case, the protocol type that takes effect on the chain/ring is eCPRI. In this case, the value of Protocol Type in the SET EQUIPMENT command is invalid. - When this parameter is set to DEFAULT, the protocol type that takes effect on the chain/ring is determined by the value of Protocol Type in the SET EQUIPMENT command. - For a port that is not configured with a chain/ring: - If the interface board supports only the CPRI protocol, the CPRI protocol is used by default. - If the interface board supports only the eCPRI protocol, the eCPRI protocol is used by default. - If the interface board supports both the CPRI and eCPRI protocols, the CPRI and eCPRI protocols are tried in polling mode.
Head Cabinet No.	RRUCHAIN.HCN	These parameters specify the cabinet number, subrack number, slot number of the board providing the CPRI port serving as the chain/ring head or ring tail, and CPRI port number. Set these parameters based on hardware installation information.
Head Subrack No.	RRUCHAIN.HSRN	
Head Slot No.	RRUCHAIN.HSN	
Head Port No.	RRUCHAIN.HPN	
Tail Cabinet No.	RRUCHAIN.TCN	
Tail Subrack No.	RRUCHAIN.TSRN	
Tail Slot No.	RRUCHAIN.TSN	
Tail Port No.	RRUCHAIN.TPN	

Parameter Name	Parameter ID	Setting Notes
BreakPoint Position1	RRUCHAIN.BRKP OS1	These parameters specify the sequence number (counting from the chain/ring head) of an RRU/RFU/AAU after which a chain/ring breaks. - For the star and chain topologies, only one breakpoint can be configured. For the ring and load sharing topology, two breakpoints must be configured. - For the ring and load sharing topologies, the value of BreakPoint Position1 must be less than or equal to the value of BreakPoint Position2. If there is only one breakpoint, the values of BreakPoint Position1 and BreakPoint Position2 must be the same, indicating that the two breakpoints are the same. - When adjusting a topology, perform the following operations: - For the star and chain topologies, configure a breakpoint at the chain head, that is, set BreakPoint Position1 to 0. - For the ring and load sharing topologies, configure two breakpoints, one at the ring head and the other at the ring tail. That is, set BreakPoint Position1 to 0 and BreakPoint Position2 to a value equal to the number of RRU/RFU/AAU cascading levels. - To remove a breakpoint, set BreakPoint Position1 and BreakPoint Position2 to 255.
BreakPoint Position2	RRUCHAIN.BRKP OS2	
CPRI Line Rate	RRUCHAIN.CR	This parameter is set by the user or negotiated automatically. ¹
User-Defined Rate Negotiation Switch	RRUCHAIN.USER DEFRATENEGOS W	This parameter specifies the user-defined rate negotiation switch. If this switch is turned on, users can specify a bit rate set and a rate change period for automatic rate negotiation.
Bit Rate Set	RRUCHAIN.BITR ATESET	This parameter specifies the bit rate set for rate negotiation. This parameter takes effect only when the user-defined rate negotiation switch is turned on.
Rate Change Period	RRUCHAIN.RATE CHGPERIOD	This parameter specifies the duration of a port working in each negotiated rate. This parameter takes effect only when the user-defined rate negotiation switch is turned on.

Parameter Name	Parameter ID	Setting Notes
Head SubPort No.	RRUCHAIN.HSP N	- When a DSFP optical module is used, subchannel 0 indicates the optical fiber on the left, and subchannel 1 indicates the optical fiber on the right. - In common star, chain, ring, and load sharing topologies, either subchannel 0 or subchannel 1 can be used. - In CPRI MUX topology, only subchannel 0 can be configured. - When a non-DSFP optical module is used, only subchannel 0 can be used.

 NOTE

- If intermediate devices (such as OTN or RTN) exist on the chain/ring between the BBU and RF modules, set this parameter to a specific value based on the CPRI port specifications of the BBU, RF modules, and intermediate devices.
- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RRU.CV	- This parameter can be set to a value ranging from 0 to 15 for an AAU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 15 for an RRU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 62 for an RFU.
Subrack No.	RRU.SRV	- This parameter can be set to a value ranging from 60 to 254 for an RRU/AAU. - This parameter can be set to 4 or 5 for an RFU in a BTS3900L/BTS5900L cabinet or a virtual cabinet, and must be set to 4 for an RFU in any other cabinet.
Slot No.	RRU.SV	- This parameter must be set to 0 for an RRU/AAU installed outside a cabinet. - This parameter must be set to a value ranging from 1 to 3 for a pRRU. - This parameter can be set to a value ranging from 0 to 8 for an RFU in a BTS3900AL cabinet or a virtual cabinet, and can be set to a value ranging from 0 to 5 for an RFU in any other cabinet.
RRU Name	RRU.RV	This parameter is optional and can be customized to contain 0 to 99 characters. This parameter takes effect only when an RRU is configured.

Parameter Name	Parameter ID	Setting Notes
RRU type	RRU.RT	Set this parameter based on the type of the in-use RF module. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules .
RRU Position	RRU.PS	This parameter specifies the position of an RRU/RFU/AAU on a chain/ring. The value of this parameter is set sequentially from 0 . The value of this parameter for an RRU/RFU/AAU must differ from that for any other RRU/RFU/AAU configured on the same chain/ring. In the chain topology, the end close to the BBU is the chain head, and the sequence number starts from 0.
RRU Chain No.	RRU.RCN	Set this parameter based on the actual physical connection.
Topo Position	RRU.TP	- Set this parameter to TRUNK(TRUNK) when this parameter indicates the position where the RRU/RFU/AAU is installed in the topology. - Set this parameter to BRANCH(BRANCH) for the pRRU.
RF Unit Working Mode	RRU.RS	When an inter-frequency mutual aid module works in inter-frequency mutual aid mode, and the two RX frequency bands are in different RATs, set this parameter to a value including multiple RATs. For example, if the RRU3952m (1800 MHz) is configured to work in 2T2R in LTE 1800 MHz and 2R in UMTS 2100 MHz, set this parameter to UL .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of an RRU/RFU/AAU.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of an RRU/RFU/AAU.
RRU Slave CPRI Line Rate	RRU.SCR	- If this parameter is set to a fixed value, this configuration takes effect only when the CPRI line rate mode of the corresponding chain/ring is set to MANUAL(Manual). - In other scenarios, the CPRI line rate is automatically negotiated.
Remote Flag	RRU.REMOTEFL AG	If you do not need to define this information, set this parameter to UNDEFINED(Undefined) .
User Label	RRU.USERLABEL	This parameter specifies the user-defined RRU/AAU descriptions using 0 to 255 characters.

Parameter Name	Parameter ID	Setting Notes
Receive Frequency Band Mutual Aid Switch	RRU.RXFREQBANDMUTUALSW	This parameter specifies whether to enable the receive frequency band mutual aid function on an RF module. This parameter is valid only for RF modules whose hardware capabilities support inter-frequency mutual aid, such as the RRU3952m. Dynamically modifying the value of this parameter for an inter-frequency mutual aid RF module will reset the RF module.

3. (Optional) The following table describes key parameters in an **RHUB** MO required for DBS3900 LampSite base stations.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RHUB.CN	This parameter must be set to 0 for an RHUB.
Subrack No.	RHUB.SRN	This parameter can be set to a value ranging from 60 to 254 for an RHUB. The subrack No. of an RHUB must be different from that of another RHUB or an RRU/RFU/AAU.
Slot No.	RHUB.SN	This parameter must be set to 0 for an RHUB.
RRU Chain No.	RHUB.RCN	This parameter uniquely identifies an RHUB chain/ring in a base station.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.
RHUB Name	RHUB.RN	This parameter is optional and can be customized to contain 0 to 31 characters.

4. The following table describes key parameters in an **AAS** MO to configure an active antenna system.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	AAS.CN	This parameter must be set to 0 for an AAS.
Subrack No.	AAS.SRN	This parameter can be set to a value ranging from 60 to 254 for an AAS.
AAS Name	AAS.AN	This parameter is optional and can be customized to contain 0 to 31 characters.
AAU Specification	AAS.AAUSPEC	This parameter specifies the type of an AAS. Set this parameter based on the actual hardware type.

5. The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
Topo Position	AAMU.TP	This parameter specifies the position where an AAS is installed in the topology. Set this parameter to TRUNK(TRUNK) .
AAS Chain No.	AAMU.RCN	Set this parameter based on the actual physical connection.
AAS Position	AAMU.PS	This parameter specifies the position of an AAS on an RRU chain/ring. It is numbered from the start port in sequence.

6. The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	AARU.CN	This parameter must be set to 0 for an AARU.
Subrack No.	AARU.SRN	This parameter can be set to a value ranging from 60 to 254 for an AARU. The subrack number of the AARU must be the same as that of the AAS.
Slot No.	AARU.SN	This parameter can be set to a value ranging from 1 to 3 for an AARU.
AARU Name	AARU.AN	This parameter is optional and can be customized to contain 0 to 31 characters.
AARU Work Standard	AARU.RS	This parameter specifies the working standard of an AARU. Set this parameter based on site configuration.
Number of RX channels	AARU.RXNUM	Set this parameter to the actual number of RX channels of an AARU.
Number of TX channels	AARU.TXNUM	Set this parameter to the actual number of TX channels of an AARU.
VRET No.	AARU.VRETNO	This parameter specifies the number of a virtual antenna corresponding to the AARU.
Intermediate Frequency Offset	AARU.IFOFFSET	This parameter indicates the intermediate frequency offset of the AARU. The default value 0 is recommended.

7. (Optional) The following table describes the key parameter in an **EQUIPMENT** MO to configure the automatic CPRI resource defragmentation switch.

Parameter Name	Parameter ID	Setting Notes
Auto CPRI Resource Defragment Switch	EQUIPMENT.AUT OCPRIRESDEFRA GSW	1. If multiple cells with different bandwidths, compression modes, TX/RX modes, or cell types exist on one link, CPRI fragments may be generated when users add or remove a cell, modify the bandwidth or the compression mode, or activate or deactivate a cell. This will lead to a cell activation failure. In this case, this switch can be turned on. 2. When this switch is turned on, if an LTE cell activation fails because of CPRI fragments and the total bandwidth of CPRI fragments exceeds the CPRI bandwidth required by the to-be-activated cell, automatic CPRI resource defragmentation will be triggered. All services in the cells on the same CPRI link as the to-be-activated cell will be interrupted and recovered after the defragmentation. 3. Chain, load sharing, and cold backup ring topologies are supported.

8. (Optional) The following table describes key parameters in the **RFPHYSITE** and **DEVSITEINFO** MOs to configure a physical RF site.

Parameter Name	Parameter ID	Setting Notes
Physical RF Site ID	RFPHYSITE.RFP HYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023 .
Physical RF Site Name	RFPHYSITE.RFP HYSITENAME	This parameter specifies the name of a physical RF site. The name should be detailed and straightforward enough to locate a physical RF site. The name of physical RF sites must be unique within an NE.
User Label	RFPHYSITE.USE RLABEL	This parameter specifies the user-defined information of a physical RF site.
Cabinet No.	DEVSITEINFO.C N	This parameter specifies the cabinet number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 15 .
Subrack No.	DEVSITEINFO.S RN	This parameter specifies the subrack number of an RF module, CXU, or RHUB. It can be set to a value ranging from 60 to 254 .
Slot No.	DEVSITEINFO.S N	This parameter specifies the slot number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 5 .

Parameter Name	Parameter ID	Setting Notes
Physical RF Site ID	DEVSITEINFO.RFPHYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023. After a physical RF site is configured, you need to set this parameter to associate an RF module, CXU, or RHUB with the physical RF site. The physical RF site ID set for the RF module, CXU, or RHUB must be consistent with the actual device deployment. Upon configuration of physical RF sites using the RFPHYSITE MO, the RF modules, CXUs, or RHUBs at the same site must be configured with the same physical RF site ID.

 NOTE

- A physical RF site can refer to the RF modules and their supporting devices deployed on a pole, tower, or rooftop site. It needs to be configured based on the actual situation. For details about how to configure a physical RF site, see *3900 & 5900 Series Base Station Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*.
- It is recommended that physical RF sites be configured in the scenario where multiple sites are configured in one BBU subrack.

Scenario-specific Data Preparation

Scenario 1: star topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

In the dual-star topology, set the **TT** parameter to **CHAIN(CHAIN)** for each RAT of the separate-MPT base station.

For an RF module whose logical type is LRRU/MRRU/PRRU/MRFU/LRFU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain number.

Parameter Name	Parameter ID	Setting Notes
RRU Position	RRU.PS	Set this parameter to 0 .

For an RF module whose logical type is AARU:

- The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
AAS Chain No.	AAMU.RCN	Set this parameter to the actual chain number.
AAS Position	AAMU.PS	Set this parameter to 0 .

- The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
AARU Work Standard	AARU.RS	Set this parameter to LO(LTE_FDD) .

Scenario 2: chain topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

For an RF module whose logical type is LRRU/MRRU/PRRU/MRFU/LRFU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Position	RRU.PS	Set this parameter based on the number of RRUs/RFUs/AAUs. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU.

For an RF module whose logical type is AARU:

- The following table describes key parameters in an **AAMU** MO to configure an active antenna management unit.

Parameter Name	Parameter ID	Setting Notes
AAS Chain No.	AAMU.RCN	Set this parameter to the actual chain number.
AAS Position	AAMU.PS	Set this parameter based on the number of AAUs. For example, set this parameter to 0 for the first AAU and 1 for the second AAU.

- The following table describes key parameters in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
AARU Work Standard	AARU.RS	Set this parameter to LO(LTE_FDD) .

Scenario 3: ring topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to RING(RING) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) or HOT(HOT) based on the actual configuration.

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual ring number.
RRU Position	RRU.PS	- Set this parameter based on the number of RRUs/ RFUs/AAUs when the cold backup ring topology is used. For example, set this parameter to 0 for the first RRU/RFU/AAU and 1 for the second RRU/RFU/AAU. - Set this parameter to 0 when the hot backup ring topology is used.

Scenario 4: load sharing topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to LOADBALANCE(LOADBALANCE) .

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.
RRU Position	RRU.PS	Set this parameter to 0 . The load sharing topology supports only one-level RRU/RFU/AAU cascading.

Scenario 5: branch chain (single-link) topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RHUB chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RHUB** MO to configure an RHUB.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RHUB.RCN	Set this parameter to the actual chain number.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.

- The following table describes key parameters in an **RRUCHAIN** MO to configure a pRRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Head Cabinet No.	RRUCHAIN.HCN	This parameter specifies the cabinet number of the RHUB that is connected to the pRRU.
Head Subrack No.	RRUCHAIN.HSR N	This parameter specifies the subrack number of the RHUB that is connected to the pRRU.
Head Slot No.	RRUCHAIN.HSN	This parameter specifies the slot number of the RHUB that is connected to the pRRU.

Parameter Name	Parameter ID	Setting Notes
Head Port No.	RRUCHAIN.HPN	This parameter specifies the number of the head CPRI port on the RHUB that is connected to the pRRU.

- The following table describes key parameters in an **RRU** MO to configure a pRRU.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the chain number of the RHUB that is connected to the pRRU.
RRU Position	RRU.PS	Set this parameter to 0 .
RRU type	RRU.RT	Set this parameter to MPMU .
Topo Position	RRU.TP	Set this parameter to BRANCH(BRANCH) .
RF Unit Working Mode	RRU.RS	Set this parameter to LO(LTE_FDD) or TDL(LTE_TDD) .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of a pRRU. - Set this parameter to 0 for the pRRU3901. - Set this parameter to 4 for the pRRU3902 and pRRU3907. - Set this parameter to 6 for the pRRU3911.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of a pRRU. The number of TX channels of a pRRU is the same as the number of RX channels of this pRRU.
a: For pRRU565x4B modules (x is a digit; A and B stand for letters), set RRU type to EPRRU . For other pRRU models, set RRU type to MPMU .		

Scenario 6: branch load sharing topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RHUB chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RHUB** MO to configure an RHUB.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RHUB.RCN	Set this parameter to the actual chain number.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.

- The following table describes key parameters in an **RRUCHAIN** MO to configure a pRRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to LOADBALANCE(LOADBALANCE) .
Head Cabinet No.	RRUCHAIN.HCN	This parameter specifies the cabinet number of the RHUB that is connected to the pRRU.
Head Subrack No.	RRUCHAIN.HSRN	This parameter specifies the subrack number of the RHUB that is connected to the pRRU.
Head Slot No.	RRUCHAIN.HSN	This parameter specifies the slot number of the RHUB that is connected to the pRRU.
Head Port No.	RRUCHAIN.HPN	This parameter specifies the number of the head CPRI port on the RHUB that is connected to the pRRU.

- The following table describes key parameters in an **RRU** MO to configure a pRRU.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the chain number of the RHUB that is connected to the pRRU.
RRU Position	RRU.PS	Set this parameter to 0 . The load sharing topology supports only one-level pRRU cascading.
RRU type	RRU.RT	Set this parameter to MPMU .
Topo Position	RRU.TP	Set this parameter to BRANCH(BRANCH) .
RF Unit Working Mode	RRU.RS	Set this parameter to LO(LTE_FDD) or TDL(LTE_TDD) .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of a pRRU. - Set this parameter to 0 for the pRRU3901. - Set this parameter to 4 for the pRRU3902 and pRRU3907. - Set this parameter to 6 for the pRRU3911.

Parameter Name	Parameter ID	Setting Notes
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of a pRRU. The number of TX channels of a pRRU is the same as the number of RX channels of this pRRU.

4.8.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

Step 2 (Optional) Configure breakpoints before adding an RRU/RFU/AAU if **Topo Type** is set to **RING(RING)** or **LOADBALANCE(LOADBALANCE)** for the RRU chain/ring added in step 1.

Run the **MOD RRUCHAIN** command to configure breakpoints on the RRU chain/ring. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **0**.

Step 3 Run the corresponding command to add an RF module and configure related parameters based on the following table.

If the Logical Type Is...	Then...
LRRU/PRRU/ MRRU/MRFU/ LRFU/AIRU	Run the ADD RRU command to add an RRU/RFU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
AARU	1. Run the ADD AAS command to add an AAS. In this step, set Topo Position to TRUNK(TRUNK) . 2. Run the ADD AARU command to add an AARU.
MPMU	1. Run the ADD RHUB command to add an RHUB to the chain added in step 1. 2. Run the ADD RRUCHAIN command to add an RRU chain between the RHUB and pRRU. 3. Run the ADD RRU command to add a pRRU with RRU type set to MPMU .

Step 4 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoints configured in step 2. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

Step 5 (Optional) Run the **ADD RFPHYSITE** command to configure a physical RF site.

Step 6 (Optional) After configuring a physical RF site, run the **ADD DEVSITEINFO** command to associate an RRU, CXU, or RHUB with the physical RF site through the **DEVSITEINFO** subobject of the RRU, CXU, and RHUB.

----End

MML Command Examples

- For an RF module whose **RRU type** value is **LRRU/PRRU/MRRU/MRFU/AIRU**

```
//Adding an RRU chain
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

//Adding an RRU ring
ADD RRUCHAIN: RCN=1, TT=RING, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0,
TSN=3, TPN=2;

//Adding a load sharing topology
ADD RRUCHAIN: RCN=2, TT=LOADBALANCE, HCN=0, HSRN=0, HSN=3, HPN=3, TCN=0, TSRN=0,
TSN=3, TPN=4;

//(Optional) Configuring the breakpoints
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;

//Adding an RRU/RFU/AAU
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=LO, RN= "MRRU_1",
RXNUM=2, TXNUM=2;

//(Optional) Removing the breakpoints
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

- For an RF module whose **RRU type** value is **AARU**

```
//Adding an RRU chain
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

//Adding an RRU ring
ADD RRUCHAIN: RCN=0, TT=RING, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0,
TSN=3, TPN=2;

//Adding an AAS
ADD AAS: CN=0, SRN=70, TP=TRUNK, RCN=0, PS=1, AN="AAS_1", AAUSPEC="AAU3902";

//Adding an AARU
ADD AARU: CN=0, SRN=70, SN=1, RS=LO, AN="AARU1", RXNUM=2, TXNUM=2, IFOFFSET=0,
VRETNO=70;
```

- For an RF module whose **RRU type** value is **MPMU**

```
//Adding a main chain/ring
ADD RRUCHAIN: RCN=1, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1;

//Adding an RHUB
ADD RHUB: CN=0, SRN=100, SN=0, RCN=0, PS=0, RN="RHUB_1";

//Adding a branch chain/ring
//If a branch chain (single-link) topology is used
ADD RRUCHAIN: RCN=2, TT=CHAIN, BM=COLD, HCN=0, HSRN=100, HSN=0, HPN=0;
//If a branch load sharing topology is used
ADD RRUCHAIN: RCN=2, TT=LOADBALANCE, HCN=0, HSRN=100, HSN=0, HPN=1, TCN=0,
TSRN=100, TSN=0, TPN=2;

//Adding a pRRU
ADD RRU: CN=0, SRN=110, SN=0, TP=BRANCH, RCN=2, PS=0, RT=MPMU, RS=LO, RN="pRRU_1",
RXNUM=0, TXNUM=0;
```

- (Optional) Configure a physical RF site named RFPHYSICALSITE1.

```
//Configuring a physical RF site
ADD RFPHYSITE: RFPHYSITEID=1, RFPHYSITENAME="RFPHYSICALSITE1";

//Associating RXUs with the physical RF site
//Associating three 2100 MHz RRUs with RFPHYSICALSITE1
```

```
ADD DEVSITEINFO: CN=0, SRN=150, SN=0, RFPHYSITEID=1;  
ADD DEVSITEINFO: CN=0, SRN=151, SN=0, RFPHYSITEID=1;  
ADD DEVSITEINFO: CN=0, SRN=152, SN=0, RFPHYSITEID=1;  
//Associating three 3500 MHz AAUs with RFPHYSICALSITE1  
ADD DEVSITEINFO: CN=0, SRN=180, SN=0, RFPHYSITEID=1;  
ADD DEVSITEINFO: CN=0, SRN=181, SN=0, RFPHYSITEID=1;  
ADD DEVSITEINFO: CN=0, SRN=182, SN=0, RFPHYSITEID=1;
```

```
//Checking the configurations consistency of physical RF sites  
DSP MBTSRFPHYSITECFG;
```

4.8.2.3 Using the MAE-Deployment

- For detailed operations of configuring a single NE, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: eNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single eNodeB > Configuring eNodeB Device Data > Configuring RF Units*.
- For detailed operations of configuring NEs in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: eNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating eNodeBs in Batches > Preparing eNodeB Data > Device Data for an eNodeB > RF Unit Data*.

4.8.3 Activation Verification

After configuration data is activated, ensure that no RRU/RFU/AAU related alarm is displayed on the **Alarm/Event** tab page on the LMT.

On the LMT or MAE-Access, run the **DSP BRD** command to check the status of the RRU/RFU/AAU.

The status of the RRU/RFU/AAU is normal if the value of **Availability Status** is **Normal**.

Run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

NOTE

For pRRU565x4B modules (*x* is a digit; *A* and *B* stand for different letters), if the pRRUs have been configured and are running properly, CPRI line rate negotiation will be performed automatically for their branch chains. Therefore, you do not need to run the **STR CPRILBRNEG** command. You can run the **DSP CPRIPORT** command to query the CPRI rates supported by the branch chains where the pRRUs reside.

4.8.4 Network Monitoring

The performance of eCPRI can be monitored by checking the following counters.

Counter Name	Counter Description
VS.ECPRIPORT.TxBytes	Number of Bytes Transmitted from an eCPRI Port
VS.ECPRIPORT.RxBytes	Number of Bytes Received by an eCPRI Port

Counter Name	Counter Description
VS.ECPRIPORT.TxPackets	Number of Packets Transmitted from an eCPRI Port
VS.ECPRIPORT.RxPackets	Number of Packets Received by an eCPRI Port
VS.ECPRIPORT.TxMaxSpeed	Maximum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.TxMinSpeed	Minimum Transmit Rate of an eCPRI Port
VS.ECPRIPORT.TxMeanSpeed	Average Transmit Rate of an eCPRI Port
VS.ECPRIPORT.RxMaxSpeed	Maximum Receive Rate of an eCPRI Port
VS.ECPRIPORT.RxMinSpeed	Minimum Receive Rate of an eCPRI Port
VS.ECPRIPORT.RxMeanSpeed	Average Receive Rate of an eCPRI Port
VS.ECPRIPORT.TxTotalBW	Total Transmit Bandwidth of an eCPRI Port
VS.ECPRIPORT.RxTotalBW	Total Receive Bandwidth of an eCPRI Port
VS.ECPRIPORT.RxErrPackets	Number of Error Packets Received by an eCPRI Port

4.8.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps. After the adjustment, observe the adjusted items. For details, see [4.8.3 Activation Verification](#).

MML-based Adjustment

After the adjustment, run the **DSP CPRI LBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRI LBR NEG** command to perform CPRI line rate negotiation.

Combining RRU Chains into an RRU Ring

A cold backup ring topology can be set up by combining two RRU chains. The precautions for combining RRU chains into an RRU ring are as follows:

- The **Chain No.** value of the two RRU chains must differ from each other, and the value of **Topo Type** must be **CHAIN(CHAIN)**.
- The total number of RRU cascading levels on the two RRU chains cannot exceed 4.
- If the **Access Type** parameter is set to **PEERPORT(peer port)** for the RRU chains to be combined, these RRU chains cannot be combined.
- The chain, cold backup ring, and hot backup ring topologies cannot be concurrently configured on one LBBPc but can be concurrently configured on one LBBPd or UBBP.

- Step 1** Run the **DSP CPRILBR** command to check whether values of **Current CPRI Line Bit Rate** of the two RRU chains are the same. If they are not the same, go to step 3 after the combination. If they are the same, skip step 3.

```
DSP CPRILBR: RCN=1;
```

- Step 2** Run the **CMB RRUCHAIN** command to combine two RRU chains and set up a cold backup ring topology. After the two RRU chains are combined successfully, the value of **Chain No.** of the newly created ring is the same as that of **First RRU Chain No.**

```
CMB RRUCHAIN: RCN1=0, RCN2=2;
```

- Step 3** (Optional) Run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

```
STR CPRILBRNEG: RCN=0;
```

----End

Restructuring a Star Topology to a Load Sharing Topology

When the bandwidth of a single CPRI link for a star topology is insufficient, the star topology needs to be restructured to an intra-board or inter-board load sharing topology. The precautions during the restructuring are as follows:

- Ensure that the RRU in the star topology supports load sharing topology.
- Ensure that the configuration of the BBP to be used is complete, and CPRI ports for load sharing topology are available and idle.

- Step 1** Run the **MOD RRUCHAIN** command to configure a breakpoint.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

- Step 2** Run the **MOD RRUCHAIN** command with **Topo Type** set to **LOADBALANCE(LOADBALANCE)**.

```
MOD RRUCHAIN: RCN=0, TT=LOADBALANCE;
```

- Step 3** Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Adding an RRU/RFU/AAU to a Chain/Ring

An RRU/RFU/AAU can be added to any position on an RRU chain/ring without changing the original topology.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for adding an RRU/RFU/AAU to the head or middle of a chain/ring.

- For the RRU chain, configure one breakpoint with **BreakPoint Position1** set to **0**.
- For the RRU ring, configure two breakpoints with **BreakPoint Position1** set to **0** and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

- Step 2** Run the **ADD RRU** command to add an RRU/RFU/AAU and set related information.

```
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=GO, RN= "MRRU_2", RXNUM=2, TXNUM=2;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Removing an RRU/RFU/AAU from a Chain/Ring

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for removing an RRU/RFU/AAU from the head or middle of a chain/ring.

- For the chain, configure one breakpoint with **BreakPoint Position1** set to 0.
- For the ring, configure two breakpoints with **BreakPoint Position1** set to 0 and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=1;
```

- Step 2** Run the **RMV RRU** command to remove the RRU/RFU/AAU.

```
RMV RRU: CN=0, SRN=61, SN=0;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Modifying the RRU/RFU Logical Type

- Step 1** Run the **MOD RRU** command to reconfigure the **RRU type** parameter for an RRU/RFU.

```
MOD RRU: CN=0, SRN=60, SN=0, RT=MRRU;
```

NOTE

- The value ranges of **Cabinet No.**, **Subrack No.**, and **Slot No.** for the RRU differ from those for the RFU. Therefore, the value of **RRU type** for an RRU cannot be changed to the value of **RRU type** for an RFU, and the value of **RRU type** for an RFU cannot be changed to the value of **RRU type** for an RRU.
- The value of **RRU type** for the pRRU and AAU cannot be modified.

----End

Modifying the RRU/RFU/AAU Working Standard

Before the modification, ensure that the RF module hardware supports the target working standard. For details about the RF module hardware capability, see the **Supported RAT** columns of the tables in [4.1.2 RF Modules](#). In addition, ensure that the license for the target working standard has been purchased.

- Step 1** Run the **MOD RRU** command with **RF Unit Working Mode** set to the target working standard.

```
MOD RRU: CN=0, SRN=60, SN=0, RS=LO;
```

 NOTE

Working standard of the pRRU cannot be modified.

----End

Adding an RHUB to a Chain

An RHUB can be added to any position on an RHUB chain.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure a breakpoint for adding an RHUB to the head of a chain.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=0;
```

- Step 2** Run the **ADD RHUB** command to add an RHUB.

```
ADD RHUB: CN=0, SRN=101, SN=0, RCN=1, PS=0, RN="RHUB_2";
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Adding a pRRU to an RHUB

Before adding a pRRU, a branch chain/ring must be added between the pRRU and RHUB.

- Step 1** Run the **ADD RRUCHAIN** command to add a branch chain/ring between the RHUB and pRRU.

```
ADD RRUCHAIN: RCN=2, TT=CHAIN, BM=COLD, HCN=0, HSRN=100, HSN=0, HPN=1;
```

- Step 2** Run the **ADD RRU** command to add a pRRU.

```
ADD RRU: CN=0, SRN=111, SN=0, TP=BRANCH, RCN=3, PS=0, RT=MPPMU, RS=UO, RXNUM=0, TXNUM=0;
```

----End

Removing an RHUB from a Chain

Before removing an RHUB, remove the pRRUs and pRRU branch chains/rings connected to this RHUB.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure a breakpoint for removing an RHUB from the head or middle of a chain.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=0;
```

- Step 2** Run the **RMV RHUB** command to remove an RHUB.

```
RMV RHUB: CN=0, SRN=100, SN=0;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Removing a pRRU from an RHUB

Step 1 Run the **RMV RRU** command to remove a pRRU.

```
RMV RRU: CN=0, SRN=111, SN=0;
```

----End

MAE-Deployment-based Adjustment

- **Adding an RRU/RFU to a chain/ring**

For details about the MAE-Deployment-based adjustment, see the following section in *eRAN Reconfiguration Guide of 3900 & 5900 Series Base Station Product Documentation*: **Device Data Reconfigurations > Adding or Deleting an RF Module**.

- **Removing an RRU/RFU from a chain/ring**

For details about the MAE-Deployment-based adjustment, see the following section in *eRAN Reconfiguration Guide of 3900 & 5900 Series Base Station Product Documentation*: **Device Data Reconfigurations > Adding or Deleting an RF Module**.

4.9 Operation and Maintenance (gNodeB)

4.9.1 Precautions

N/A

4.9.2 Data Configuration

Comply with the following rules when single-mode or multimode RF modules work in single-mode scenarios:

- For a hot backup ring topology, only one RRU/AAU can be configured, and RRU/AAU cascading is not supported.
- Add RRU chains/rings before adding RRUs/AAUs.

4.9.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/AAU model: Run the **DSP ELABEL** command to query the electronic label of the RRU/AAU.
 - RRU/AAU topology: Use it to determine how to configure RRU/AAU links.
1. The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Chain No.	RRUCHAIN.RCN	This parameter uniquely identifies a chain/ring in a base station.
Topo Type	RRUCHAIN.TT	• In the star or chain topology, set this parameter to CHAIN(CHAIN). In this case, the TCN, TSRN, TSN, and TPN parameters are invalid. In the star topology, only one RRU/AAU can be configured on a chain. • In the ring topology, set this parameter to RING(RING) and specify two CPRI ports and the backup mode. • In the load sharing topology, set this parameter to LOADBALANCE(LOADBALANCE). In this case, two CPRI ports need to be specified. • In the branch chain (single-link) topology, set this parameter to CHAIN(CHAIN). • In the branch load sharing topology, set this parameter to LOADBALANCE(LOADBALANCE). In this case, two CPRI ports need to be specified. • In the trunk chain topology, set this parameter to TRUNK_CHAIN(TRUNK_CHAIN).
Backup Mode	RRUCHAIN.BM	• If the Topo Type parameter is set to CHAIN(CHAIN), set this parameter to COLD(COLD). • If the Topo Type parameter is set to RING(RING), – Set this parameter to COLD(COLD) in the cold backup ring topology. – Set this parameter to HOT(HOT) in the hot backup ring topology. In this case, the ring head and tail must belong to different BRIs, and only one RRU/AAU can be added. • Before changing the backup mode of a chain/ring, ensure that there is no RRU/AAU on the chain/ring. • If the non-load-sharing topology is changed to the load sharing topology, RRUs/AAUs can be configured on the chain/ring. When the load sharing topology is changed to the non-load-sharing topology, ensure that no RRU/AAU is configured on the chain/ring.

Parameter Name	Parameter ID	Setting Notes
Access Type	RRUCHAIN.A7	When this parameter is set to LOCALPORT(local port) , the BBP is connected to the RRU/AAU. When this parameter is set to PEERPORT(peer port) , the local BBP (of one RAT) is connected to the peer BBP (of another RAT) through the virtual CPRI subchannel on the backplane, and the peer BBP is then connected to the RRU/AAU.
Protocol Type	RRUCHAIN.PROTOCOL	This parameter specifies the communication protocol used by the chain/ring. - For RF modules that support the CPRI protocol, it is recommended that this parameter be set to CPRI. In this case, the protocol type that takes effect on the chain/ring is CPRI. In this case, the value of Protocol Type in the SET EQUIPMENT command is invalid. - For RF modules that support the eCPRI protocol, it is recommended that this parameter be set to eCPRI. In this case, the protocol type that takes effect on the chain/ring is eCPRI. In this case, the value of Protocol Type in the SET EQUIPMENT command is invalid. - When this parameter is set to DEFAULT, the protocol type that takes effect on the chain/ring is determined by the value of Protocol Type in the SET EQUIPMENT command. - For a port that is not configured with a chain/ring: - If the interface board supports only the CPRI protocol, the CPRI protocol is used by default. - If the interface board supports only the eCPRI protocol, the eCPRI protocol is used by default. - If the interface board supports both the CPRI and eCPRI protocols, the CPRI and eCPRI protocols are tried in polling mode.
Head Cabinet No.	RRUCHAIN.HCN	These parameters specify the cabinet number, subrack number, slot number of the board providing the CPRI port serving as the chain/ring head or ring tail, and CPRI port number. Set these parameters based on hardware installation information.
Head Subrack No.	RRUCHAIN.HSRN	
Head Slot No.	RRUCHAIN.HSN	
Head Port No.	RRUCHAIN.HPN	
Tail Cabinet No.	RRUCHAIN.TCN	
Tail Subrack No.	RRUCHAIN.TSRN	
Tail Slot No.	RRUCHAIN.TSN	
Tail Port No.	RRUCHAIN.TPN	

Parameter Name	Parameter ID	Setting Notes
BreakPoint Position1	RRUCHAIN.BRKP OS1	These parameters specify the sequence number (counting from the chain/ring head) of an RRU/AAU after which a chain/ring breaks. - For the star and chain topologies, only one breakpoint can be configured. For the ring and load sharing topology, two breakpoints must be configured. - For the ring and load sharing topologies, the value of BreakPoint Position1 must be less than or equal to the value of BreakPoint Position2. If there is only one breakpoint, the values of BreakPoint Position1 and BreakPoint Position2 must be the same, indicating that the two breakpoints are the same. - When adjusting a topology, perform the following operations: - For the star and chain topologies, configure a breakpoint at the chain head, that is, set BreakPoint Position1 to 0. - For the ring and load sharing topologies, configure two breakpoints, one at the ring head and the other one at the ring tail. That is, set BreakPoint Position1 to 0 and BreakPoint Position2 to a value equal to the number of RRU/AAU cascading levels. - To remove a breakpoint, set BreakPoint Position1 and BreakPoint Position2 to 255.
BreakPoint Position2	RRUCHAIN.BRKP OS2	
CPRI Line Rate	RRUCHAIN.CR	This parameter is set by the user or negotiated automatically. ¹
Head SubPort No.	RRUCHAIN.HSP N	- When a DSFP optical module is used, subchannel 0 indicates the optical fiber on the left, and subchannel 1 indicates the optical fiber on the right. - In common star, chain, ring, and load sharing topologies, either subchannel 0 or subchannel 1 can be used. - In CPRI MUX topology, only subchannel 0 can be configured. - When a non-DSFP optical module is used, only subchannel 0 can be used.

 NOTE

1. If intermediate devices (such as OTN or RTN) exist on the chain/ring between the BBU and RF modules, set this parameter to a specific value based on the CPRI port specifications of the BBU, RF modules, and intermediate devices.

2. The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RRU.CN	- This parameter can be set to a value ranging from 0 to 15 for an AAU. Generally, it is set to 0. - This parameter can be set to a value ranging from 0 to 15 for an RRU. Generally, it is set to 0.
Subrack No.	RRU.SRN	This parameter can be set to a value ranging from 60 to 254 for an RRU/AAU.
Slot No.	RRU.SN	This parameter must be set to 0 for an RRU/AAU.
RRU Name	RRU.RN	This parameter is optional and can be customized to contain 0 to 99 characters. This parameter takes effect only when an RRU is configured.
RRU type	RRU.RT	Set this parameter based on the type of the in-use RF module. For details, see the Logical Type columns of the tables in 4.1.2 RF Modules .
RRU Position	RRU.PS	This parameter specifies the position of an RRU/AAU on an RRU chain/ring. The value of this parameter is set sequentially from 0. The value of this parameter for an RRU/AAU must differ from that for any other RRU/AAU configured on the same chain/ring. In the chain topology, the end close to the BBU is the chain head, and the sequence number starts from 0.
RRU Chain No.	RRU.RCN	Set this parameter based on the actual physical connection.
Topo Position	RRU.TP	Set this parameter to TRUNK(TRUNK) when this parameter indicates the position where the RRU/AAU is installed in the topology.
RF Unit Working Mode	RRU.RS	When an inter-frequency mutual aid module works in inter-frequency mutual aid mode, and the two RX frequency bands are in different RATs, set this parameter to a value including multiple RATs. For example, if the RRU3952m (1800 MHz) is configured to work in 2T2R in LTE 1800 MHz and 2R in UMTS 2100 MHz, set this parameter to UL .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of an RRU/AAU.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of an RRU/AAU.

Parameter Name	Parameter ID	Setting Notes
RRU Slave CPRI Line Rate	RRU.SCR	- If this parameter is set to a fixed value, this configuration takes effect only when the CPRI line rate mode of the corresponding chain/ring is set to MANUAL(Manual). - In other scenarios, the CPRI line rate is automatically negotiated.
Remote Flag	RRU.REMOTEFLAG	If you do not need to define this information, set this parameter to UNDEFINED(Undefined) .
User Label	RRU.USERLABEL	This parameter specifies the user-defined RRU/AAU descriptions using 0 to 255 characters.
Receive Frequency Band Mutual Aid Switch	RRU.RXFREQBANDMUTUALSW	This parameter is valid only for RF modules whose hardware capabilities support inter-frequency mutual aid, such as the RRU3952m. Dynamically modifying the value of this parameter for an inter-frequency mutual aid RF module will reset the RF module.

3. (Optional) The following table describes the key parameter in an **EQUIPMENT** MO to configure the automatic CPRI resource defragmentation switch.

Parameter Name	Parameter ID	Setting Notes
Auto CPRI Resource Defragment Switch	EQUIPMENT.AUTOCPRIRESDEFRA GSW	1. If multiple cells with different bandwidths, compression modes, TX/RX modes, or cell types exist on one link, CPRI fragments may be generated when users add or remove a cell, modify the bandwidth or the compression mode, or activate or deactivate a cell. This will lead to a cell activation failure. In this case, this switch can be turned on. 2. When this switch is turned on, if an LTE cell activation fails because of CPRI fragments and the total bandwidth of CPRI fragments exceeds the CPRI bandwidth required by the to-be-activated cell, automatic CPRI resource defragmentation will be triggered. All services in the cells on the same CPRI link as the to-be-activated cell will be interrupted and recovered after the defragmentation. 3. Chain, load sharing, and cold backup ring topologies are supported.

4. (Optional) The following table describes key parameters in the **RFPHYSITE** and **DEVSITEINFO** MOs to configure a physical RF site.

Parameter Name	Parameter ID	Setting Notes
Physical RF Site ID	RFPHYSITE.RFP HYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023 .
Physical RF Site Name	RFPHYSITE.RFP HYSITENAME	This parameter specifies the name of a physical RF site. The name should be detailed and straightforward enough to locate a physical RF site. The name of physical RF sites must be unique within an NE.
User Label	RFPHYSITE.USE RLABEL	This parameter specifies the user-defined information of a physical RF site.
Cabinet No.	DEVSITEINFO.C N	This parameter specifies the cabinet number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 15 .
Subrack No.	DEVSITEINFO.S RN	This parameter specifies the subrack number of an RF module, CXU, or RHUB. It can be set to a value ranging from 60 to 254 .
Slot No.	DEVSITEINFO.S N	This parameter specifies the slot number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 5 .
Physical RF Site ID	DEVSITEINFO.R FPHYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023. After a physical RF site is configured, you need to set this parameter to associate an RF module, CXU, or RHUB with the physical RF site. The physical RF site ID set for the RF module, CXU, or RHUB must be consistent with the actual device deployment. Upon configuration of physical RF sites using the RFPHYSITE MO, the RF modules, CXUs, or RHUBs at the same site must be configured with the same physical RF site ID.

NOTE

- A physical RF site can refer to the RF modules and their supporting devices deployed on a pole, tower, or rooftop site. It needs to be configured based on the actual situation. For details about how to configure a physical RF site, see *3900 & 5900 Series Base Station Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*.
- It is recommended that physical RF sites be configured in the scenario where multiple sites are configured in one BBU subrack.

Scenario-specific Data Preparation

Scenario 1: star topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

In the dual-star topology, set the **TT** parameter to **CHAIN(CHAIN)** for each RAT of the separate-MPT base station.

For an RF module whose logical type is MRRU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain number.
RRU Position	RRU.PS	Set this parameter to 0 .
Cabinet No.	RHUB.CN	Set this parameter to 0 .
Subrack No.	RHUB.SRN	Set this parameter according to the network plan. The subrack No. of an RHUB must be different from that of another RHUB or an RRU/RFU/AAU.
Slot No.	RHUB.SN	Set this parameter to 0 .
RHUB Name	RHUB.RN	Set this parameter based on the network plan.

Scenario 2: chain topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Access Type	RRUCHAIN.AT	Set this parameter to LOCALPORT(local port) .

For an RF module whose logical type is MRRU/AIRU:

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain number.
RRU Position	RRU.PS	Set this parameter based on the number of RRUs/AAUs. For example, set this parameter to 0 for the first RRU/AAU and 1 for the second RRU/AAU.

Scenario 3: ring topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to RING(RING) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) or HOT(HOT) based on the actual configuration.

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual ring number.
RRU Position	RRU.PS	- Set this parameter based on the number of RRUs/AAUs when the cold backup ring topology is used. For example, set this parameter to 0 for the first RRU/AAU and 1 for the second RRU/AAU. - Set this parameter to 0 when the hot backup ring topology is used.

Scenario 4: load sharing topology

- The following table describes the key parameter in an **RRUCHAIN** MO to configure an RRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to LOADBALANCE(LOADBALANCE) .

- The following table describes key parameters in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the actual chain/ring number.

Parameter Name	Parameter ID	Setting Notes
RRU Position	RRU.PS	Set this parameter to 0 . The load sharing topology supports only one-level RRU/AAU cascading.

Scenario 5: branch chain (single-link) topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RHUB chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RHUB** MO to configure an RHUB.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RHUB.RCN	Set this parameter to the actual chain number.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.

- The following table describes key parameters in an **RRUCHAIN** MO to configure a pRRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .
Head Cabinet No.	RRUCHAIN.HCN	This parameter specifies the cabinet number of the RHUB that is connected to the pRRU.
Head Subrack No.	RRUCHAIN.HSR N	This parameter specifies the subrack number of the RHUB that is connected to the pRRU.
Head Slot No.	RRUCHAIN.HSN	This parameter specifies the slot number of the RHUB that is connected to the pRRU.
Head Port No.	RRUCHAIN.HPN	This parameter specifies the number of the head CPRI port on the RHUB that is connected to the pRRU.

- The following table describes key parameters in an **RRU** MO to configure a pRRU.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the chain number of the RHUB that is connected to the pRRU.
RRU Position	RRU.PS	Set this parameter to 0 .
RRU type	RRU.RT	Set this parameter to MPMU .
Topo Position	RRU.TP	Set this parameter to BRANCH(BRANCH) .
RF Unit Working Mode	RRU.RS	Set this parameter to NO(NR_ONLY) .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of a pRRU. - Set this parameter to 0 for the pRRU3901. - Set this parameter to 4 for the pRRU3902 and pRRU3907. - Set this parameter to 6 for the pRRU3911.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of a pRRU. The number of TX channels of a pRRU is the same as the number of RX channels of this pRRU.
a: For pRRU565xAB modules (x is a digit; A and B stand for letters), set RRU type to EPRRU . For other pRRU models, set RRU type to MPMU .		

Scenario 6: branch load sharing topology

- The following table describes key parameters in an **RRUCHAIN** MO to configure an RHUB chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to CHAIN(CHAIN) .
Backup Mode	RRUCHAIN.BM	Set this parameter to COLD(COLD) .

- The following table describes key parameters in an **RHUB** MO to configure an RHUB.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RHUB.RCN	Set this parameter to the actual chain number.
RHUB Position	RHUB.PS	This parameter specifies the position of an RHUB on a chain/ring. It is numbered from the start port in sequence. For example, set this parameter to 0 for the first-level RHUB and 1 for the second-level RHUB, and so on.

- The following table describes key parameters in an **RRUCHAIN** MO to configure a pRRU chain/ring.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT	Set this parameter to LOADBALANCE(LOADBALANCE) .
Head Cabinet No.	RRUCHAIN.HCN	This parameter specifies the cabinet number of the RHUB that is connected to the pRRU.
Head Subrack No.	RRUCHAIN.HSRN	This parameter specifies the subrack number of the RHUB that is connected to the pRRU.
Head Slot No.	RRUCHAIN.HSN	This parameter specifies the slot number of the RHUB that is connected to the pRRU.
Head Port No.	RRUCHAIN.HPN	This parameter specifies the number of the head CPRI port on the RHUB that is connected to the pRRU.

- The following table describes key parameters in an **RRU** MO to configure a pRRU.

Parameter Name	Parameter ID	Setting Notes
RRU Chain No.	RRU.RCN	Set this parameter to the chain number of the RHUB that is connected to the pRRU.
RRU Position	RRU.PS	Set this parameter to 0 . The load sharing topology supports only one-level pRRU cascading.
RRU type	RRU.RT	Set this parameter to MPMU .
Topo Position	RRU.TP	Set this parameter to BRANCH(BRANCH) .
RF Unit Working Mode	RRU.RS	Set this parameter to NO(NR_ONLY) .
Number of RX channels	RRU.RXNUM	Set this parameter to the actual number of RX channels of a pRRU. - Set this parameter to 0 for the pRRU3901. - Set this parameter to 4 for the pRRU3902 and pRRU3907. - Set this parameter to 6 for the pRRU3911.
Number of TX channels	RRU.TXNUM	Set this parameter to the actual number of TX channels of a pRRU. The number of TX channels of a pRRU is the same as the number of RX channels of this pRRU.

4.9.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact,

dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

Step 2 (Optional) Configure breakpoints before adding an RRU/AAU if **Topo Type** is set to **RING(RING)** or **LOADBALANCE(LOADBALANCE)** for the RRU chain/ring added in step 1.

Run the **MOD RRUCHAIN** command to configure breakpoints on the RRU chain/ring. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **0**.

Step 3 Run the corresponding command to add an RF module and configure related parameters based on the following table.

If the Logical Type Is...	Then...
MRRU/AIRU	Run the ADD RRU command to add an RRU/AAU. In this step, set Topo Position to TRUNK(TRUNK) .
MPMU	1. Run the ADD RHUB command to add an RHUB to the RRU chain added in step 1. 2. Run the ADD RRUCHAIN command to add an RRU chain between the RHUB and pRRU. 3. Run the ADD RRU command to add a pRRU. Set Topo Position to BRANCH(BRANCH) and RRU type to MPMU.

Step 4 (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoints configured in step 2. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

Step 5 (Optional) Run the **ADD RFPHYSITE** command to configure a physical RF site.

Step 6 (Optional) After configuring a physical RF site, run the **ADD DEVSITEINFO** command to associate an RRU, CXU, or RHUB with the physical RF site through the **DEVSITEINFO** subobject of the RRU, CXU, and RHUB.

Step 7 (Optional) After configuring a physical RF site and associating it with an RRU, CXU, or RHUB, run the **DSP MBTSRFPHYSITECFG** command to check whether the physical RF site configuration of an RF module shared by multiple RATs in an MBTS is consistent among those RATs.

----End

MML Command Examples

- For an RF module whose **RRU type** value is **MRRU/AIRU**

//Adding an RRU chain

ADD RRUCHAIN: RCN=0, **TT=CHAIN**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=0;

//Adding an RRU ring

ADD RRUCHAIN: RCN=1, **TT=RING**, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1, TCN=0, TSRN=0,

```
TSN=3, TPN=2;

//Adding a load sharing topology
ADD RRUCHAIN: RCN=2, TT=LOADBALANCE, HCN=0, HSRN=0, HSN=3, HPN=3, TCN=0, TSRN=0,
TSN=3, TPN=4;

//(Optional) Configuring the breakpoints
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;

//Adding an RRU/AAU
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=NO, RN= "MRRU_1",
RXNUM=2, TXNUM=2;

//(Optional) Removing the breakpoints
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

- For an RF module whose **RRU type** value is **MPMU**

```
//Adding a main chain/ring
ADD RRUCHAIN: RCN=1, TT=CHAIN, BM=COLD, HCN=0, HSRN=0, HSN=3, HPN=1;

//Adding an RHUB
ADD RHUB: CN=0, SRN=100, SN=0, RCN=0, PS=0, RN="RHUB_1";

//Adding a branch chain/ring
//If a branch chain (single-link) topology is used
ADD RRUCHAIN: RCN=2, TT=CHAIN, BM=COLD, HCN=0, HSRN=100, HSN=0, HPN=0;
//If a branch load sharing topology is used
ADD RRUCHAIN: RCN=2, TT= LOADBALANCE, HCN=0, HSRN=100, HSN=0, HPN=1, TCN=0,
TSRN=100, TSN=0, TPN=2;

//Adding a pRRU
ADD RRU: CN=0, SRN=110, SN=0, TP=BRANCH, RCN=2, PS=0, RT=MPMU, RS=NO, RN="pRRU_1",
RXNUM=0, TXNUM=0;
```

- (Optional) Configure a physical RF site named RFPHYSICALSITE1.

```
//Configuring a physical RF site
ADD RFPHYSITE: RFPHYSITEID=1, RFPHYSITENAME="RFPHYSICALSITE1";

//Associating RXUs with the physical RF site
//Associating three 2100 MHz RRUs with RFPHYSICALSITE1
ADD DEVSITEINFO: CN=0, SRN=150, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=151, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=152, SN=0, RFPHYSITEID=1;
//Associating three 3500 MHz AAUs with RFPHYSICALSITE1
ADD DEVSITEINFO: CN=0, SRN=180, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=181, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=182, SN=0, RFPHYSITEID=1;

//Checking the configurations consistency of physical RF sites
DSP MBTSRFPHYSITECFG;
```

4.9.2.3 Using the MAE-Deployment

- For detailed operations of configuring a single NE, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation*: **gNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating a Single gNodeB > Configuring gNodeB Device Data > Configuring RF Units**.
- For detailed operations of configuring NEs in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation*: **gNodeB Initial Configuration Guide (MAE-Deployment-based) > Creating gNodeBs in Batches > Preparing gNodeB Data > Device Data for a gNodeB > RF Unit Data**.

4.9.3 Activation Verification

After configuration data is activated, ensure that no RRU/AAU related alarm is displayed on the **Alarm/Event** tab page on the LMT.

On the LMT or MAE-Access, run the **DSP BRD** command to check the status of the RRU/AAU.

The status of the RRU/AAU is normal if the value of **Availability Status** is **Normal**.

Run the **DSP CPRI LBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRI LBR NEG** command to perform CPRI line rate negotiation.

NOTE

For pRRU565x4B modules (x is a digit; A and B stand for different letters), if the pRRUs have been configured and are running properly, CPRI line rate negotiation will be performed automatically for their branch chains. Therefore, you do not need to run the **STR CPRI LBR NEG** command. You can run the **DSP CPRI PORT** command to query the CPRI rates supported by the branch chains where the pRRUs reside.

4.9.4 Network Monitoring

The performance of eCPRI can be monitored by checking the following counters.

Counter Name	Counter Description
VS.ECPRI PORT.TxBytes	Number of Bytes Transmitted from an eCPRI Port
VS.ECPRI PORT.RxBytes	Number of Bytes Received by an eCPRI Port
VS.ECPRI PORT.TxPackets	Number of Packets Transmitted from an eCPRI Port
VS.ECPRI PORT.RxPackets	Number of Packets Received by an eCPRI Port
VS.ECPRI PORT.TxMaxSpeed	Maximum Transmit Rate of an eCPRI Port
VS.ECPRI PORT.TxMinSpeed	Minimum Transmit Rate of an eCPRI Port
VS.ECPRI PORT.TxMeanSpeed	Average Transmit Rate of an eCPRI Port
VS.ECPRI PORT.RxMaxSpeed	Maximum Receive Rate of an eCPRI Port
VS.ECPRI PORT.RxMinSpeed	Minimum Receive Rate of an eCPRI Port
VS.ECPRI PORT.RxMeanSpeed	Average Receive Rate of an eCPRI Port
VS.ECPRI PORT.TxTotalBW	Total Transmit Bandwidth of an eCPRI Port
VS.ECPRI PORT.RxTotalBW	Total Receive Bandwidth of an eCPRI Port
VS.ECPRI PORT.RxErrPackets	Number of Error Packets Received by an eCPRI Port

4.9.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps. After the adjustment, observe the adjusted items. For details, see [4.9.3 Activation Verification](#).

MML-based Adjustment

After the adjustment, run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

Combining RRU Chains into an RRU Ring

A cold backup ring topology can be set up by combining two RRU chains. The precautions for combining RRU chains into an RRU ring are as follows:

- The **Chain No.** value of the two RRU chains must differ from each other, and the value of **Topo Type** must be **CHAIN(CHAIN)**.
- The total number of RRU cascading levels on the two RRU chains cannot exceed 4.
- If the **Access Type** parameter is set to **PEERPORT(peer port)** for the RRU chains to be combined, these RRU chains cannot be combined.

Step 1 Run the **DSP CPRILBR** command to check whether values of **Current CPRI Line Bit Rate** of the two RRU chains are the same. If they are not the same, go to step 3 after the combination. If they are the same, skip step 3.

```
DSP CPRILBR: RCN=1;
```

Step 2 Run the **CMB RRUCHAIN** command to combine two RRU chains and set up a cold backup ring topology. After the two RRU chains are combined successfully, the value of **Chain No.** of the newly created ring is the same as that of **First RRU Chain No.**

```
CMB RRUCHAIN: RCN1=0, RCN2=2;
```

Step 3 (Optional) Run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

```
STR CPRILBRNEG: RCN=0;
```

----End

Restructuring a Star Topology to a Load Sharing Topology

When the bandwidth of a single CPRI link for a star topology is insufficient, the star topology needs to be restructured to an intra-board or inter-board load sharing topology. The precautions during the restructuring are as follows:

- Ensure that the RRU in the star topology supports load sharing topology.
- Ensure that the configuration of the BBP to be used is complete, and CPRI ports for load sharing topology are available and idle.

Step 1 Run the **MOD RRUCHAIN** command to configure a breakpoint.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

- Step 2** Run the **MOD RRUCHAIN** command with **Topo Type** set to **LOADBALANCE(LOADBALANCE)**.

```
MOD RRUCHAIN: RCN=0, TT=LOADBALANCE;
```

- Step 3** Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Adding an RRU/AAU to a Chain/Ring

An RRU/AAU can be added to any position on an RRU chain/ring without changing the original topology.

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for adding an RRU/AAU to the head or middle of a chain/ring.

- For the RRU chain, configure one breakpoint with **BreakPoint Position1** set to **0**.
- For the RRU ring, configure two breakpoints with **BreakPoint Position1** set to **0** and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

- Step 2** Run the **ADD RRU** command to add an RRU/AAU and set related information.

```
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=GO, RN= "MRRU_2", RXNUM=2, TXNUM=2;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Removing an RRU/AAU from a Chain/Ring

- Step 1** (Optional) Run the **MOD RRUCHAIN** command to configure the breakpoint(s) for removing an RRU/AAU from the head or middle of a chain/ring.

- For the chain, configure one breakpoint with **BreakPoint Position1** set to **0**.
- For the ring, configure two breakpoints with **BreakPoint Position1** set to **0** and **BreakPoint Position2** set to the number of RRU cascading levels.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=1;
```

- Step 2** Run the **RMV RRU** command to remove the RRU/AAU.

```
RMV RRU: CN=0, SRN=61, SN=0;
```

- Step 3** (Optional) Run the **MOD RRUCHAIN** command to remove the breakpoint(s) configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

----End

Modifying the RRU Logical Type

Step 1 Run the **MOD RRU** command to reconfigure the **RRU type** parameter for an RRU.

```
MOD RRU: CN=0, SRN=60, SN=0, RT=MRRU;
```

----End

Modifying the RRU/AAU Working Standard

Before the modification, ensure that the RF module hardware supports the target working standard. For details about the RF module hardware capability, see the **Supported RAT** columns of the tables in [4.1.2 RF Modules](#). In addition, ensure that the license for the target working standard has been purchased.

Step 1 Run the **MOD RRU** command with **RF Unit Working Mode** set to the target working standard.

```
MOD RRU: CN=0, SRN=60, SN=0, RS=LO;
```

----End

MAE-Deployment-based Adjustment

- **Adding an RRU to a chain/ring**

For details about the MAE-Deployment-based adjustment, see the following section in *5G RAN Reconfiguration Guide of 3900 & 5900 Series Base Station Product Documentation*: **Device Data Reconfigurations > Adding or Deleting an RF Module**.

- **Removing an RRU from a chain/ring**

For details about the MAE-Deployment-based adjustment, see the following section in *5G RAN Reconfiguration Guide of 3900 & 5900 Series Base Station Product Documentation*: **Device Data Reconfigurations > Adding or Deleting an RF Module**.

4.10 Operation and Maintenance (Multimode Base Station)

4.10.1 Precautions

None

4.10.2 Data Configuration

4.10.2.1 Data Preparation

This section contains the following two parts:

- "Common Data Preparation": It describes the required configurations for feature deployment in all scenarios.
- "Scenario-specific Data Preparation": It describes the scenario-specific configurations in typical scenarios.

In practice, certain items in "Common Data Preparation" and "Scenario-specific Data Preparation" are configured based on actual scenario requirements.

"Common Data Preparation" describes the required configurations for feature deployment in separate-MPT and co-MPT multimode base stations.

- Separate-MPT multimode base station
Basic information about RF modules must be configured on each RAT of the multimode base station. For details, see the method of configuring basic information about each RAT.
When an RF module works in multiple RATs, basic information about each RAT must comply with the configuration principles.
- Co-MPT multimode base station
Information about all RATs is configured on the same main control board. Therefore, data needs to be configured only once.

Common Data Preparation

Confirm the following information before common data preparation:

- RRU/RFU/AAU model: Run the **DSP BTSELABEL** (GBTs) or **DSP ELABEL** (eGBTs/NodeB/eNodeB) command to query the electronic label of the RRU/RFU/AAU.
 - RRU/RFU/AAU topology: Use it to determine how to configure RRU/RFU/AAU links.
1. When an RF module works in a single RAT, refer to one of the following sections for common data preparation.
 - For GBTs, see [4.5.2.1 Data Preparation](#).
 - For eGBTs, see [4.6.2.1 Data Preparation](#).
 - For NodeBs, see [4.7.2.1 Data Preparation](#).
 - For eNodeBs, see [4.8.2.1 Data Preparation](#).
 - For gNodeBs, see [4.9.2.1 Data Preparation](#).
 2. When an RF module in a separate-MPT base station works in multiple RATs, configure the RF module in compliance with the following rules:
 - The RF module is configured for and monitored by each RAT separately.
 - Basic information specific to each RAT is included in common data preparation specific to each RAT.
 - The common parameters of the RF module must be the same for every RAT, including **Cabinet No.**, **Subrack No.**, **Slot No.**, and other parameters described in the following tables. If the configurations are inconsistent, ALM-26274 Inter-System Board Object Configuration Conflict will be generated.

Table 4-10 GBTs parameters of the multimode base station

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Topo Type	BTSRXUCHAIN.77	ADD BTSRXUCHAIN	CHAIN(CHAIN)

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Board Type	BTSRXUCHAIN.BT	ADD BTSRXUBRD	Set this parameter based on the type of the in-use RF module. For details, see 4.1.2 RF Modules .
RXU Board Position	BTSRXUCHAIN.RXUP OS	ADD/MOD BTSRXUBRD	0
Working Standard	BTSRXUCHAIN.WORKINGSTANDARD	SET BTSRXUBP	Set this parameter based on the multimode services to be supported. If a module needs to carry multimode services, the corresponding RATs must be configured.
MRRU/GRRU Sending Receiving Mode	BTSRXUCHAIN.SndRcvMode2	SET BTSRXUBP	Set this parameter based on site configuration.
MRFU/GRFU Sending Receiving Mode	BTSRXUCHAIN.SndRcvMode3	SET BTSRXUBP	Set this parameter based on site configuration.
Associated Board CN	BTSRXUCHAIN.RELAT EDCN	SET BTSRXUBP	Set this parameter based on site configuration.
Associated Board SRN	BTSRXUCHAIN.RELAT EDSRN	SET BTSRXUBP	Set this parameter based on site configuration.

Table 4-11 GBTS parameters of the multimode base station

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Associated Board SN	BTSRXUBP.RELATEDSN	SET BTSRXUBP	Set this parameter based on site configuration.
IF Offset	BTSRXUBP.IFOFFSET	SET BTSRXUBP	0
Lvl1 Vswr	BTSRXUBP.Lvl1Vswr	SET BTSRXUBP	20
Serious VSWR Alarm Post-processing Mode	BTSRXUBP.VSWRALM DISPOSE	SET BTSRXUBP	ON(ON)

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Lvl2 Vswr	BTSRXUBP.Lvl2Vswr	SET BTSRXUBP	20
Antenna Tributary 1 Factor	BTSRXUBP.MRRUATTENFACTOR1	SET BTSRXUBP	0
Antenna Tributary 2 Factor	BTSRXUBP.MRRUATTENFACTOR2	SET BTSRXUBP	0

Table 4-12 eGBTS/NodeB/eNodeB/gNodeB parameters of the multimode base station (configured with the RF module whose logical type is MRRU/PRRU/MRFU/AIRU)

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Topo Type	RRUCHAIN.TT (5G gNodeB, LTE eNodeB)	ADD RRUCHAIN	CHAIN(CHAIN)
Backup Mode	RRUCHAIN.BM (5G gNodeB, LTE eNodeB)	ADD RRUCHAIN	COLD(COLD)
RRU type	RRU.RT (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	Set this parameter based on the type of the in-use RF module. For details, see 4.1.2 RF Modules .
Topo Position	RRU.TP (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	TRUNK(TRUNK)
RRU Position	RRU.PS (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	0
RF Unit Working Mode	RRU.RS (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	Set this parameter based on the multimode services to be supported. If a module needs to carry multimode services, the corresponding RATs must be configured.
Number of RX channels	RRU.RXNUM (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	Set this parameter based on site configuration.
Number of TX channels	RRU.TXNUM (5G gNodeB, LTE eNodeB)	ADD/MOD RRU	Set this parameter based on site configuration.

Table 4-13 eGBTS/NodeB/eNodeB/gNodeB parameters of the multimode base station (configured with the RF module whose logical type is MRRU/PRRU/MRFU/AIRU)

Parameter Name	Parameter ID	Configuration Command	Recommended Value
VSWR alarm threshold	RRU.ALMTHRHL <i>(5G gNodeB, LTE eNodeB)</i>	ADD/MOD RRU	20
VSWR alarm clearance threshold	RRU.ALMCLRTHRHL	ADD/MOD RRU	17 This parameter applies only to 5000 series RRUs.
VSWR Alarm Threshold Extend Switch	RRU.VSWRALMVALEXTSW	ADD/MOD RRU	OFF(Off) This parameter applies only to MRRU RF modules.
VSWR alarm post-processing switch	RRU.ALMPROCSW <i>(5G gNodeB, LTE eNodeB)</i>	ADD/MOD RRU	ON(ON)
VSWR alarm post-processing threshold	RRU.ALMPROCTHRHL <i>(5G gNodeB, LTE eNodeB)</i>	ADD/MOD RRU	20
Attenuation	RXBRANCH.ATTEN <i>(5G gNodeB, LTE eNodeB)</i>	MOD RXBRANCH	0

Table 4-14 eGBTS/NodeB/eNodeB parameters of the multimode base station (configured with the RF module whose logical type is AARU)

Parameter Name	Parameter ID	Configuration Command	Recommended Value
Topo Type	RRUCHAIN.TT	ADD RRUCHAIN	CHAIN(CHAIN)
Backup Mode	RRUCHAIN.BM	ADD RRUCHAIN	COLD(COLD)
Topo Position	AAS.TP	ADD AAS	TRUNK(TRUNK)
AAS Position	AAS.PS	ADD AAS	0

Parameter Name	Parameter ID	Configuration Command	Recommended Value
AARU Work Standard	AARU.RS	ADD/MOD AARU	Set this parameter based on the multimode services to be supported. If a module needs to carry multimode services, the corresponding RATs must be configured.
Number of RX channels	AARU.RXNUM	ADD/MOD AARU	Set this parameter based on site configuration.
Number of TX channels	AARU.TXNUM	ADD/MOD AARU	Set this parameter based on site configuration.
VRET No.	AARU.VRETNO	ADD/MOD AARU	Set this parameter based on site configuration.
Intermediate Frequency Offset	AARU.IFOFFSET	ADD/MOD AARU	The default value 0 is recommended.

3. When an RF module in a co-MPT base station works in multiple RATs, common data, except the working standard, is the same as that for an eNodeB.

- The following table describes the key parameter in an **RRU** MO to configure an RF module.

Parameter Name	Parameter ID	Setting Notes
RF Unit Working Mode	RRU.RS (5G gNodeB, LTE eNodeB)	This parameter specifies the working standard of an RRU/RFU/AAU. Set this parameter to a value including multiple RATs.

- The following table describes the key parameter in an **AARU** MO to configure an active antenna radio unit.

Parameter Name	Parameter ID	Setting Notes
AARU Work Standard	AARU.RS	This parameter specifies the working standard of an AARU. Set this parameter to a value including multiple RATs.

4. (Optional) The following table describes the key parameter in an **EQUIPMENT** MO to configure the automatic CPRI resource defragmentation switch.

Parameter Name	Parameter ID	Setting Notes
Auto CPRI Resource Defragment Switch	EQUIPMENT.AUTOCPRI RESDEFRAGSW	1. If multiple cells with different bandwidths, compression modes, TX/RX modes, or cell types exist on one link, CPRI fragments may be generated when users add or remove a cell, modify the bandwidth or the compression mode, or activate or deactivate a cell. This will lead to a cell activation failure. In this case, this switch can be turned on. 2. When this switch is turned on, if an LTE cell activation fails because of CPRI fragments and the total bandwidth of CPRI fragments exceeds the CPRI bandwidth required by the to-be-activated cell, automatic CPRI resource defragmentation will be triggered. All services in the cells on the same CPRI link as the to-be-activated cell will be interrupted and recovered after the defragmentation. 3. Chain, load sharing, and cold backup ring topologies are supported.

5. (Optional) The following table describes key parameters in the **RFPHYSITE** and **DEVSITEINFO** MOs to configure a physical RF site.

Parameter Name	Parameter ID	Setting Notes
Physical RF Site ID	RFPHYSITE.RFP HYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023 .
Physical RF Site Name	RFPHYSITE.RFP HYSITENAME	This parameter specifies the name of a physical RF site. The name should be detailed and straightforward enough to locate a physical RF site. The name of physical RF sites must be unique within an NE.
User Label	RFPHYSITE.USE RLABEL	This parameter specifies the user-defined information of a physical RF site.
Cabinet No.	DEVSITEINFO.C N	This parameter specifies the cabinet number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 15 .
Subrack No.	DEVSITEINFO.S RN	This parameter specifies the subrack number of an RF module, CXU, or RHUB. It can be set to a value ranging from 60 to 254 .
Slot No.	DEVSITEINFO.S N	This parameter specifies the slot number of an RF module, CXU, or RHUB. It can be set to a value ranging from 0 to 5 .

Parameter Name	Parameter ID	Setting Notes
Physical RF Site ID	DEVSITEINFO.R FPHYSITEID	This parameter specifies the ID of a physical RF site, which uniquely identifies a physical RF site within an NE. It can be set to a value ranging from 0 to 1023. After a physical RF site is configured, you need to set this parameter to associate an RF module, CXU, or RHUB with the physical RF site. The physical RF site ID set for the RF module, CXU, or RHUB must be consistent with the actual device deployment. Upon configuration of physical RF sites using the RFPHYSITE MO, the RF modules, CXUs, or RHUBs at the same site must be configured with the same physical RF site ID.

 NOTE

- A physical RF site can refer to the RF modules and their supporting devices deployed on a pole, tower, or rooftop site. It needs to be configured based on the actual situation. For details about how to configure a physical RF site, see *3900 & 5900 Series Base Station Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*.
- It is recommended that physical RF sites be configured in the scenario where multiple sites are configured in one BBU subrack.

Scenario-specific Data Preparation

For details about scenario-specific data to be prepared for the star topology, chain topology, ring topology, and load sharing topology, see the scenario-specific data preparation for each topology.

Scenario 1: dual-star topology

In the dual-star topology, set **Topo Type** to **CHAIN(CHAIN)** for each RAT. For details about other data to be prepared, see the data preparation for each RAT in the star topology.

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT (5G gNodeB, LTE eNodeB)	Set this parameter to CHAIN(CHAIN) .

Scenario 2: load sharing topology (multimode)

Parameter Name	Parameter ID	Setting Notes
Topo Type	RRUCHAIN.TT (5G gNodeB, LTE eNodeB)	Set this parameter to LOADBALANCE(LOADBALANCE) .

4.10.2.2 Using MML Commands

Before using MML commands, refer to [4.3.2 Impacts](#) and [4.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Using MML Commands

- For a separate-MPT base station, perform operations on each single-RAT base station separately.
 - For GBTs, see [4.5.2 Data Configuration](#).
 - For eGBTs, see [4.6.2 Data Configuration](#).
 - For NodeBs, see [4.7.2 Data Configuration](#).
 - For eNodeBs, see [4.8.2 Data Configuration](#).
 - For gNodeBs, see [4.9.2 Data Configuration](#).
- For a co-MPT base station, perform the following operations.

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring, and set **Topo Type** to **LOADBALANCE(LOADBALANCE)**.

Step 2 Configure breakpoints before adding an RRU/RFU/AAU.

Run the **MOD RRUCHAIN** command to configure breakpoints on the RRU chain/ring. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **0**.

Step 3 Run the **ADD RRU** command to add an RRU/RFU/AAU. In this step, set **Topo Position** to **TRUNK(TRUNK)**.

Step 4 Run the **MOD RRUCHAIN** command to remove the breakpoints configured in step 2. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

Step 5 (Optional) Run the **ADD RFPHYSITE** command to configure a physical RF site.

Step 6 (Optional) After configuring a physical RF site, run the **ADD DEVSITEINFO** command to associate an RRU, CXU, or RHUB with the physical RF site through the **DEVSITEINFO** subobject of the RRU, CXU, and RHUB.

----End

MML Command Examples

- Configure the RRU chain/ring.

```
//Adding an RRU chain
ADD RRUCHAIN: RCN=2, TT=LOADBALANCE, HCN=0, HSRN=0, HSN=3, HPN=3, TCN=0, TSRN=0,
TSN=3, TPN=4;

//Configuring breakpoints
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;

//Adding an RRU/RFU/AAU
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=GL, RN= "MRRU_1",
RXNUM=2, TXNUM=2;

//Removing the breakpoints
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

- (Optional) Configure a physical RF site named RFPHYSICALSITE1.

```
//Configuring a physical RF site
ADD RFPHYSITE: RFPHYSITEID=1, RFPHYSITENAME="RFPHYSICALSITE1";

//Associating RXUs with the physical RF site
//Associating three 2100 MHz RRUs with RFPHYSICALSITE1
ADD DEVSITEINFO: CN=0, SRN=150, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=151, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=152, SN=0, RFPHYSITEID=1;
//Associating three 3500 MHz AAUs with RFPHYSICALSITE1
ADD DEVSITEINFO: CN=0, SRN=180, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=181, SN=0, RFPHYSITEID=1;
ADD DEVSITEINFO: CN=0, SRN=182, SN=0, RFPHYSITEID=1;

//Checking the configurations consistency of physical RF sites
DSP MBTSRFPHYSITECFG;
```

4.10.2.3 Using the MAE-Deployment

- For detailed operations of configuring separate-MPT base stations in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: Separate-MPT Base Station Initial Configuration Guide (MAE-Deployment-based) > Creating Separate-MPT Base Stations in Batches*.
- For detailed operations of configuring co-MPT base stations in batches, see the following section in *3900 & 5900 Series Base Station Initial Configuration Guide of 3900 & 5900 Series Base Station Product Documentation: Co-MPT Base Station Initial Configuration Guide (MAE-Deployment-based) > Creating Co-MPT Base Stations in Batches (MOC Export)/Creating Co-MPT Base Stations in Batches (Customization Tool)*.

4.10.3 Activation Verification

None

4.10.4 Network Monitoring

None

4.10.5 Reconfiguration

This section mainly describes how to use MML commands to adjust an RF module topology, and add or remove an RF module. It also provides related precautions. During the adjustment, strictly follow the operation steps.

MML-based Adjustment

After the adjustment, run the **DSP CPRILBR** command to query the CPRI line rate of the chain/ring. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

Restructuring a Star Topology to a Load Sharing Topology

When the bandwidth of a single CPRI link for a star topology is insufficient, the star topology needs to be restructured to an intra-board or inter-board load sharing topology. The precautions during the restructuring are as follows:

- Ensure that the RRU in the star topology supports load sharing topology.
- Ensure that the configuration of the BBP to be used is complete, and CPRI ports for load sharing topology are available and idle.

Step 1 Run the **MOD RRUCHAIN** command to configure a breakpoint.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=0;
```

Step 2 Run the **MOD RRUCHAIN** command with **Topo Type** set to **LOADBALANCE(LOADBALANCE)**.

```
MOD RRUCHAIN: RCN=0, TT=LOADBALANCE;
```

Step 3 Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to **255**.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Restructuring a Star Topology to a Dual-Star Topology

In the dual-star topology, one RF module is connected to two BBPs/BRIs of different RATs. The BBPs/BRIs are separately connected to the specified RF module through the star topology.

NOTICE

In general, RF modules in a dual-star topology work in multiple modes. In capacity expansion scenarios from a single mode to multiple modes, change the working standard of the shared RF module to multimode.

Restructuring a Star Topology to CPRI MUX

For a co-MPT base station, no special configuration is required for the CPRI MUX topology. Ensure that the physical connection is correct and the working standard of the RF module is correct.

For a separate-MPT base station, if the star topology is changed to CPRI MUX, no special configuration is required for the converging party and only the RRU chain/ring of the converged party needs to be configured. When an RRU chain/ring is configured for the converged party, set **Access Type** to **PEERPORT(peer port)**. The chain/ring head cabinet number, chain/ring head subrack number, chain/ring head slot number, and chain/ring head optical port number must be the same as those of the port that connects to the optical fiber on the converging party side. Local slot number specifies the number of the slot where the BBP used for convergence is located at the local end. The configuration example of the converged party is as follows:

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain/ring. In this step, set **Access Type** to **PEERPORT**.

```
ADD RRUCHAIN: RCN=0, TT=CHAIN, AT= PEERPORT, BM=COLD, HSN=4, HPN=0, LSN=0;
```

Step 2 Run the **ADD RRU** command to add an RRU/RFU/AAU and set related information.

```
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=UL, RN= "MRRU_2", RXNUM=2, TXNUM=2;
```

 NOTE

CPRI refers to CPRI and eCPRI. If eCPRI networking is used, the protocol type in step 1 must be set to eCPRI.

----End

Restructuring CPRI MUX/eCPRI MUX to a Dual-Star Topology

For a separate-MPT base station, if CPRI MUX/eCPRI MUX is changed to a dual-star topology, no configuration modification is required for the original converging party and only the configuration of the original converged party needs to be modified. During the change, set **Access Type** to **LOCALPORT(local port)**. The chain/ring head cabinet number, chain/ring head subrack number, chain/ring head slot number, and chain/ring head optical port number must be the same as those of the port that connects to the optical fiber on the local side. The configuration example of the original converged party is as follows:

- Step 1** Run the **MOD RRCHAIN** command. In this step, set **Access Type** to **LOCALPORT**, change the port that connects to the optical fiber, and configure a breakpoint.

```
MOD RRCHAIN: RCN=0, AT=LOCALPORT, HCN=0, HSRN=0, HSN=3, HPN=3, BRKPOS1=0;
```

- Step 2** Run the **MOD RRCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

Restructuring a Dual-Star Topology to a Star Topology

When a dual-star topology is changed to a star topology, each RF module must connect to only one BBP. In general, RF modules in a dual-star topology work in multiple modes. In reconstruction scenarios from multiple modes to a single mode, the working standard of the RF modules must be changed to a single mode.

Restructuring CPRI MUX/eCPRI MUX to a Star Topology

In general, RF modules for CPRI MUX/eCPRI MUX work in multiple modes. In reconstruction scenarios from multiple modes to a single mode, the working standard of the RF modules must be changed to a single mode.

For a separate-MPT base station where CPRI MUX/eCPRI MUX is changed to a dual-star topology, if the converged party needs to be removed, only the RRU chain configuration of the converged party needs to be removed. No special operation is required for the converging party. If the converging party needs to be removed, the optical fiber must connect to the BBP of the original converged party instead and the configuration of the original converged party needs to be modified. During the change, set **Access Type** to **LOCALPORT(local port)**. The chain/ring head cabinet number, chain/ring head subrack number, chain/ring head slot number, and chain/ring head optical port number must be the same as those of the port that connects to the optical fiber on the local side. The configuration example of the original converged party is as follows:

- Step 1** Run the **MOD RRCHAIN** command. In this step, set **Access Type** to **LOCALPORT**, change the port that connects to the optical fiber, and configure a breakpoint.

```
MOD RRUCHAIN: RCN=0, AT=LOCALPORT, HCN=0, HSRN=0, HSN=3, HPN=3, BRKPOS1=0;
```

- Step 2** Run the **MOD RRUCHAIN** command to remove the breakpoint configured in step 1. In this step, set both **BreakPoint Position1** and **BreakPoint Position2** to 255.

```
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

----End

4.11 Fault Management

In a GBTS, run the **DSP BTSCPRILBR** command to query the CPRI line rate on the chain/ring in either of the following scenarios after the alarms/events are cleared. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

- The BSC LMT or MAE-Access reports ALM-26236 RRU Cascading Levels and Configuration Mismatch or ALM-26238 RRU Network Topology Type and Configuration Mismatch.
- Alarms/Events caused by hardware faults in the chain/ring are reported, for example, the optical module on the BBU transmission port or on the RF module is faulty.

In an eGBTS, NodeB, eNodeB, or gNodeB, run the **DSP CPRILBR** command to query the CPRI line rate on the chain/ring in either of the following scenarios after the alarms/events are cleared. If the CPRI line rate is inconsistent with the planned value, run the **STR CPRILBRNEG** command to perform CPRI line rate negotiation.

- The LMT or MAE-Access reports ALM-26236 RRU Cascading Levels and Configuration Mismatch or ALM-26238 RRU Network Topology Type and Configuration Mismatch.
- Alarms/Events caused by hardware faults in the chain/ring are reported, for example, the optical module on the BBU transmission port or on the RF module is faulty.

In the following scenarios, the negotiated rate may not be optimal. The optimal negotiated rate can be achieved through operations described below.

- In cascaded networking scenarios, if RRUs are powered off and reset level by level, the negotiated rate after the RRUs start may not be optimal. In this case, after all RRUs start, run the **DSP BTSCPRILBR** (for GBTS)/**DSP CPRILBR** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to check whether the line rate is consistent with the planned value. If it is not consistent with the planned value, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings. If the negotiated rate is still not optimal, power off and reset the BBP or run the **MOD RRUCHAIN** command to set a fixed rate.
- In the new deployment, capacity expansion, or reconstruction scenario, if the default port protocol of the BBP is eCPRI and CPRI-based cascading networking is used as the target networking, run the **DSP CPRIPORT** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command after the RRUs start to check whether the rate of the RRU at each level is consistent with the planned value. If not, run the **RST BRD** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to reset the BBP.

- If an SFP electrical module is installed in a remote port on the RF module at the lowest-level of the CPRI link, and the SFP electrical module is not connected, run the **DSP SFP** command to query dynamic information about the module on the remote port of the RF module. If an electrical module is used, remove the electrical module and run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate negotiations on specific rings/chains.
- If the RF module at the first level or optical fiber is hot swapped, run the **DSP BTSCPRILBR** (for GBTS)/**DSP CPRILBR** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to check whether the line rate is consistent with the planned value. If it is not consistent with the planned value, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings.
- After dynamic RF module configuration or networking adjustment, run the **DSP BTSCPRILBR** (for GBTS)/**DSP CPRILBR** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to check whether the line rate is consistent with the planned value. If it is not consistent with the planned value, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings.
- If the networking or number of cascading levels is inconsistent with the actual one, clear the alarm first. Then, run the **DSP BTSCPRILBR** (for GBTS)/**DSP CPRILBR** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to check whether the line rate is consistent with the planned value. If it is not consistent with the planned value, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings.
- If there is intermediate equipment, such as OTN or RTN, run the **MOD RRUCHAIN** command to set a fixed rate.
- If unauthorized optical modules or faulty optical modules are used, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings.
- If the rate of used optical modules is inconsistent with the planned value, run the **DSP SFP** command to check whether capabilities of optical modules on the BBP and RRU optical ports comply with the plan. If they do not, replace the optical modules. After the replacement, run the **DSP BTSCPRILBR** (for GBTS)/**DSP CPRILBR** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to check whether the rate complies with the planned rate. If it does not, run the **STR BTSCPRILBRNEG** (for GBTS)/**STR CPRILBRNEG** (for eGBTS/NodeB/eNodeB/gNodeB/multimode base station) command to initiate rate negotiations on specific chains/rings.

In addition, you can run the **DSP RRUCPRIFLTINFO** command to query the module status, certificate status, and optical module information on the RRU/AAU side for link fault locating and troubleshooting.

- Based on the preceding fault information, you can determine whether the fault is caused by the RRU/AAU module itself, optical module, or certificate, and rectify the fault accordingly.

- The **DSP RRUCPRIFLTINFO** command has the following hardware requirements:
 - RF modules must be 5000 series AAUs/RRUs. Book RRUs are not supported.
 - AAUs/RRUs must be connected to the same DC or AC power module. The power module or power distribution device that uses isolated DC return (DC-I) between power supply ports is not supported.

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

1. 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
2. Documents in *3900 & 5900 Series Base Station Product Documentation*:
 - *3900 & 5900 Series Base Station MML Command Reference*
 - *3900 & 5900 Series Base Station MO and Parameter Reference*
 - *3900 & 5900 Series Base Station Alarm Reference*
 - *3900 & 5900 Series Base Station Event Reference*
 - *3900 & 5900 Series Base Station Initial Configuration Guide*
 - *GBSS Reconfiguration Guide*
 - *RAN Reconfiguration Guide*
 - *eRAN Reconfiguration Guide*
 - *5G RAN Reconfiguration Guide*
3. Documents in *3900 & 5900 Series Base Station Product Documentation (Hardware Installation)*:
 - Installation guides for relevant base stations
4. Documents in *BSC6910 GSM Product Documentation*:
 - *BSC6910 GSM MML Command Reference*
 - *BSC6910 GSM MO and Parameter Reference*
 - *BSC6910 GSM Alarm Reference*
 - *BSC6910 GSM Event Reference*
5. Documents in *BSC6900 GSM Product Documentation*:
 - *BSC6900 GSM MML Command Reference*
 - *BSC6900 GSM MO and Parameter Reference*
 - *BSC6900 GSM Alarm Reference*
 - *BSC6900 GSM Event Reference*
6. Feature parameter description documents in *eRAN Feature Documentation* and *5G RAN Feature Documentation*:
 - *CPRI MUX*
 - *CPRI MUX (LampSite)*
 - *SRAN Networking and Evolution Overview*